HOR 211

Production Technology of Fruit and Plantation Crops

Complete Unit-wise Theory Study Material

B.Sc. (Hons) Agriculture

Third Semester • 2025 Syllabus

Tamil Nadu Agricultural University

Units I to V • Tables, worked examples and text diagrams • Exam-focus sections

Prepared as exam-oriented study material. Figures are drawn from standard published sources (NHB, PIB, APEDA, commodity boards, TNAU and ICAR material, and standard texts). Where a value varies, a standard range is given.

Preface and How to Use This Material

This book is a complete unit-wise theory study material for the course HOR 211, Production Technology of Fruit and Plantation Crops, for B.Sc. (Hons) Agriculture, Third Semester, 2025 syllabus, Tamil Nadu Agricultural University. It covers all five units of the syllabus in order, with one numbered chapter per unit and a separate section for every crop named in the syllabus.

How the material is organised

Each unit opens with the common techniques that apply to all the crops of that unit - climate and soil, propagation and rootstock, planting systems and density, training and pruning, water and nutrient management, weed management, growth regulation, disorders, and maturity and harvest. Each crop then follows the same structure throughout the book: origin and importance, area and production, climate and soil, varieties, propagation and rootstock, planting density and spacing, training and pruning, nutrient management, irrigation and fertigation, weed management, special horticultural practices and growth regulators, important disorders, pests and diseases, maturity indices, harvest and yield, post-harvest handling and value addition, and a short economics note. This consistency makes revision and comparison easy.

Study aids

- **Tables** compare facts - classification, nutritive value, area and production, spacing, rootstocks, maturity indices and the package of practices.
- **Worked examples** show the calculations asked in examinations - plant population, fertiliser conversion, high-density economics and cost and return.
- **Text diagrams** show the layouts and systems - planting systems, training systems, pruning, the coconut layout, multi-tier cropping and the fertigation and processing flows.
- **Exam focus** ends every unit with key points, short-answer questions with model answers and long-answer questions with outlines.
- **Appendices** bring together the consolidated tables, the objective question bank, solved numerical problems, post-harvest and orchard management notes, short notes, the variety list and model question papers.

Note on figures and sources

All statistics are drawn from the standard published sources named in the reference list - the National Horticulture Board and the Department of Agriculture and Farmers Welfare estimates, APEDA export data, the commodity boards (Coconut, Rubber, Tea, Coffee and Spices Boards and the Directorate of Cashewnut and Cocoa Development), TNAU and ICAR material, and the standard textbooks. Where a figure varies between sources or years, a standard range is given rather than false precision. Figures for productivity, yield and economics are indicative and vary with variety, soil, climate and market; the student should quote the source and year with any figure and confirm current crop-wise recommendations.

Syllabus at a glance

Unit-wise coverage of the syllabus

Unit	Crops and topics covered
Unit I	Production status of fruit and plantation crops: importance and scope, nutritional value, classification, area, production, productivity and export potential
Unit II	Tropical fruit crops - mango, banana, papaya, guava, sapota; sub-tropical - citrus (acid lime and sweet orange) and grape; common production techniques
Unit III	Sub-tropical - pineapple, pomegranate, jackfruit; temperate - apple, pear, peach, plum, strawberry; minor fruits - ber, custard apple, amla, dragon fruit; nut crops - almond, pistachio, walnut
Unit IV	Palms and plantation crops - coconut, arecanut, oil palm, palmyrah, rubber; importance and scope, commodity boards, multi-tier cropping and processing
Unit V	Beverage crops - tea, coffee, cocoa; and cashew - production, harvest, processing and value addition

Every crop named in the syllabus has its own section within its unit.

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Contents

Placeholder for table of contents **0**

UNIT I | Production Status of Fruit and Plantation Crops

This unit sets the economic and biological background for the whole course. It explains why fruit and plantation crops matter to India, what they contribute nutritionally, how the crops are classified, and where the country stands on area, production, productivity and exports.

1.1 Importance and scope of the fruit and plantation crop industry in India

India is the second largest producer of fruits and vegetables in the world, after China. Horticulture now contributes about one-third of the value of India's agricultural output from roughly 18 per cent of the cropped area, and fruit production alone crossed 112 million tonnes in 2023-24. Fruit growing earns several times the net return of cereal cultivation per hectare because the produce is high-value, labour-intensive and in rising domestic demand. The industry supports not just orchard owners but a long chain of nurserymen, harvest labour, commission agents, cold-chain operators, processors and exporters.

Fruit and plantation crops together account for a very large share of India's agricultural exports. They are the backbone of several hill and coastal economies: tea in the Nilgiris of Tamil Nadu, coffee in Yercaud and the Palani and Kodaikanal hills, rubber and arecanut in the Western Ghats, cashew along the Tamil Nadu coast, and coconut in the coastal belt of Kanyakumari and Thanjavur. For a student of agriculture the practical importance is that these crops are perennial, capital-intensive and technically demanding, so the correct choice of variety, rootstock, spacing and management decides profitability for the next twenty to sixty years.

The scope for growth is large. Per-capita consumption of fruit in India is well below the levels recommended by nutritionists, urban demand for quality fruit and processed products is rising, and organised retail, food processing and exports are expanding. At the same time, productivity in many crops is low because of old and seedling orchards, poor planting material, inadequate irrigation and weak post-harvest handling. Each of these gaps is an opportunity for trained horticulturists.

Economic and social importance

- **High value per unit area:** A well-managed mango or banana orchard can return several times the income of a cereal crop on the same land.
- **Employment:** Fruit and plantation crops are labour-intensive in planting, pruning, harvesting and processing, and so support rural employment through the year.
- **Foreign exchange:** Grapes, mango, banana, pomegranate, cashew, tea, coffee and spices earn substantial export revenue.
- **Raw material for industry:** Fruit pulp, juice, jam, squash, wine, coconut oil, coir, copra, rubber, tea and coffee all depend on these crops.
- **Nutritional security:** Fruit supplies vitamins, minerals and dietary fibre that are scarce in cereal-based diets.
- **Environmental role:** Perennial orchards and plantations check soil erosion, conserve moisture and act as carbon sinks, which matters on hill slopes and in the Western Ghats.

1.2 Nutritional value of fruit crops

Fruit is valued less for calories and more for protective nutrients. Fruit and vegetables are the main dietary sources of vitamin C, beta-carotene (provitamin A), several B vitamins, potassium and other minerals, and dietary fibre. Diets rich in fruit lower the risk of micronutrient deficiency, constipation and several non-communicable diseases. India's low per-capita fruit intake is therefore a public-health concern as much as an economic one.

The table below compares the approximate nutritive value of the main fruit crops of this course. Values are for 100 g of the edible portion and are drawn from standard Indian food-composition tables; they vary with variety, maturity and growing conditions, so they should be read as typical figures rather than fixed constants.

Table 1.1 Approximate nutritive value of major fruit crops (per 100 g edible portion)

Crop	Energy (kcal)	Protein (g)	Carbohydrate (g)	Fibre (g)	Vitamin C (mg)	Carotene / provitamin A (microgram)	Notable mineral
Mango	74	0.6	16.9	0.4	16	2743	Potassium
Banana	116	1.2	27.2	0.4	8	78	Potassium
Papaya	32	0.6	7.2	0.8	57	666	Potassium
Guava	51	0.9	11.2	5.2	212	31	Iron
Sweet orange	48	0.7	10.9	0.3	64	1104	Calcium
Acid lime	30	0.7	6.0	0.4	63	15	Calcium
Sapota	98	0.7	21.4	2.6	6	97	Iron
Grape	71	0.5	16.5	0.9	1	3	Potassium
Pineapple	46	0.5	10.8	1.2	39	18	Potassium
Pomegranate	83	1.6	14.5	2.2	16	0	Iron
Apple	59	0.2	13.8	1.0	2	7	Potassium
Aonla (amla)	58	0.5	13.7	3.4	600	9	Iron
Ber	70	1.2	17.0	0.6	76	17	Calcium
Custard apple	104	1.6	23.5	3.0	37	5	Iron
Strawberry	32	0.7	7.7	1.4	59	9	Iron
Coconut (kernel)	444	3.4	6.8	4.0	1	0	Fat 41 g, P, K
Almond (dried)	655	20.8	10.5	11.8	0	0	Fat 56 g, Ca, Fe
Walnut	687	15.0	12.1	4.6	6	0	Fat 64 g, Ca, Fe

Source: Standard Indian food-composition tables (Gopalan et al.), rounded. Values vary with variety and maturity.

Reading the table. Guava and aonla stand out for vitamin C, mango and sweet orange for provitamin A, banana for energy and potassium, and papaya as a low-calorie, high-vitamin-A fruit. Nuts such as almond and walnut are energy-dense because of their high fat content, and coconut is unusual among fruits in supplying fat.

1.3 Classification of fruit crops

Because fruit crops are grown from the tropics to the temperate hills, a single classification cannot serve every purpose. Horticulturists therefore use several complementary systems: by climatic adaptation, by botanical family, by fruit type (pomology), by life cycle and by ripening behaviour. An exam answer should be able to give any one system with examples.

Diagram 1.1 Classification of fruit crops - master tree

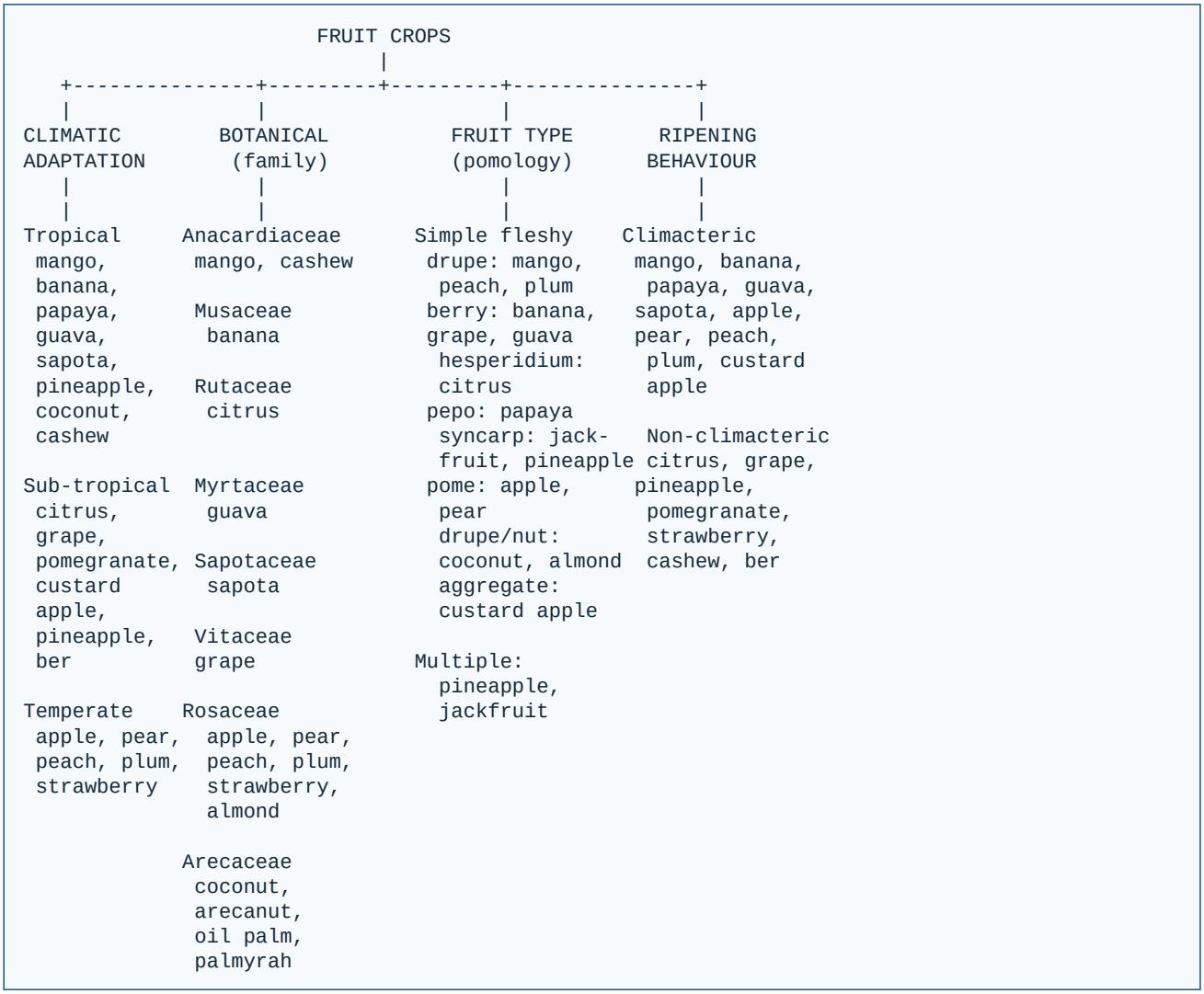


Figure 1.1 Master classification of fruit crops used in this course.

Table 1.2 Climatic classification of fruit crops with examples

Group	Climatic requirement	Representative crops	Main growing belts in India
Tropical	Warm, frost-free; mean temperature above about 21 deg C; high light	Mango, banana, papaya, guava, sapota, pineapple, coconut, arecanut, cashew, cocoa	Andhra Pradesh, Tamil Nadu, Maharashtra, Karnataka, Kerala, Gujarat
Sub-tropical	Distinct winter, mild frost, warm summer; 10-30 deg C	Citrus (acid lime, sweet orange, mandarin), grape, pomegranate, custard apple, ber, jackfruit	Maharashtra, Karnataka, Tamil Nadu, Punjab, Rajasthan, Telangana
Temperate	Cool winter with chilling, warm dry summer; below 7 deg C in winter	Apple, pear, peach, plum, strawberry, almond, walnut, pistachio	Himachal Pradesh, Uttarakhand, Jammu and Kashmir, Arunachal Pradesh; high hills of Tamil Nadu (limited)

Source: standard horticultural texts (Bose and Mitra; Chadha).

Table 1.3 Pomological (fruit-type) classification with examples

Fruit type	Botanical nature	Examples in this course
Drupe	Fleshy fruit with a single hard stone (endocarp)	Mango, peach, plum, coconut, cashew (nut is a drupe)
Berry	Fleshy fruit with many seeds, no hard stone	Banana, grape, guava, papaya (also classed as pepo)
Hesperidium	Berry with a leathery rind and oil glands	Acid lime, sweet orange, mandarin, lemon
Pome	Fleshy fruit from an inferior ovary, with a core	Apple, pear
Aggregate	Many small fruits fused into one	Custard apple, strawberry (achenes on a receptacle)
Multiple / syncarp	Many flowers fused into one fruit mass	Pineapple, jackfruit
Nut	Dry, hard, one-seeded fruit	Almond, walnut, pistachio, cashew (as nut)
Capsule / other	Dry dehiscent fruit	Tea, coffee, cocoa (seed crops grown as plantations)

Source: standard pomology texts.

Table 1.4 Ripening behaviour and its practical importance

Class	Definition	Crops	Practical consequence
Climacteric	Respiration and ethylene rise sharply at ripening; fruit ripens after harvest	Mango, banana, papaya, guava, sapota, apple, pear, peach, plum, custard apple, ber, pomegranate (weakly)	Harvested mature-green; can be ripened artificially (ethylene, calcium carbide banned); store and ship better
Non-climacteric	No respiration rise; fruit ripens only on the plant	Citrus, grape, pineapple, strawberry, cashew, cocoa	Must be harvested fully ripe; does not improve after picking

Source: standard post-harvest texts (Kader).

1.4 Area, production, productivity and export potential

The figures below are compiled from the Horticultural Statistics of the National Horticulture Board and the advance estimates of the Department of Agriculture and Farmers Welfare. They are given to the standard published precision, and the student should quote the source year with any figure, because estimates are revised each year.

Table 1.5 Area and production of the principal fruit crops of this course, India

Crop	Area (000 ha, 2022-23)	Production (000 t, 2023-24, 3rd advance estimate)
Mango	2346	22423
Banana	994	37807
Papaya	148	5304
Guava	352	5263
Sapota	73	878
Citrus - lime/lemon	312	3886
Citrus - mandarin	460	6150
Citrus - sweet orange (mosambi)	231	3814
Citrus - total	1100	14576
Grape	169	3911
Pineapple	108	1759
Pomegranate	258	2882
Jackfruit	190	3292
Apple	304	2699
Pear	35	277
Peach	16	111
Plum	18	70
Strawberry	2	17
Ber	52	526
Custard apple	52	513
Aonla (amla)	106	1330
Dragon fruit	5	55
Almond	10	12
Walnut	97	319
Total fruits (all India)	7025	112730

Source: NHB / DA&FW; horticultural estimates, 2022-23 final and 2023-24 third advance estimates. Rounded.

Productivity note. India's average fruit productivity works out to roughly 15 to 16 tonnes per hectare, but it ranges widely: banana and pineapple exceed 30 tonnes per hectare in

well-managed orchards, while mango and most temperate fruits average 8 to 12 tonnes per hectare. Closing this gap through improved varieties, high-density planting and irrigation is the single biggest opportunity in the sector.

Table 1.6 Area and production of plantation and beverage crops, India

Crop	Area	Production	Leading state(s)
Coconut	about 2.1-2.3 million ha	about 21,000-22,000 million nuts (2023-24)	Kerala, Karnataka, Tamil Nadu, Andhra Pradesh
Arecanut	949,000 ha (2023-24 final)	1,411,000 t (2023-24 final)	Karnataka, Kerala, Assam, Tamil Nadu
Oil palm	about 0.35-0.40 million ha	about 0.30-0.35 million t fresh fruit bunches	Andhra Pradesh, Telangana, Karnataka, Tamil Nadu (expanding)
Palmyrah	about 0.05 million ha (mostly Tamil Nadu)	not separately estimated nationally	Tamil Nadu, Andhra Pradesh
Rubber	about 0.85 million ha	857,000 t (2023-24)	Kerala, Tripura, Karnataka, Tamil Nadu (Kanyakumari)
Tea	about 0.62 million ha	1,382 million kg (2023-24)	Assam, West Bengal, Tamil Nadu (Nilgiris)
Coffee	about 0.47 million ha	about 374,000 t (2023-24)	Karnataka, Kerala, Tamil Nadu
Cocoa	109,000 ha	30,000 t	Kerala, Karnataka, Andhra Pradesh, Tamil Nadu
Cashew	1,195,000 ha	795,000 t raw cashew nut	Maharashtra, Andhra Pradesh, Odisha, Tamil Nadu

Source: NHB, Coconut Development Board, Rubber Board, Tea Board, Coffee Board and DCCD estimates, 2023-24. Figures rounded; area and production bases differ between crops.

Export potential

Exports are the fastest-growing end of the sector. India exported fresh fruits and vegetables worth about USD 1.8 billion (about Rs 15,000 crore) in 2023-24, and fresh fruits alone contributed about Rs 9,500 crore of that. India's share of the global horticulture market is still only about one per cent, so the export potential is far from exhausted. The main fresh-fruit earners are grapes, mango, pomegranate, banana and orange; the main markets are the Netherlands, the UAE, Iraq, Iran, Bangladesh and, for processed products, the USA.

Table 1.7 India's horticultural export performance (2023-24, selected products)

Product	Quantity (MT)	Value (Rs crore)	Value (USD million)	Main markets
Fresh fruits (all)	1,438,732	9,496	1,145	Netherlands, UAE, Iraq, Iran, Bangladesh
Fresh grapes (2023-24)	343,982	3,461	417	Netherlands, UK, Russia, UAE; Maharashtra contributes about 67 per cent
Cashew	65,801	2,809	339	USA, UAE, Japan, Netherlands
Processed fruits and juices	703,982	8,049	971	USA, Netherlands, Saudi Arabia, UAE, UK
Cocoa products	36,242	1,522	184	USA, Netherlands, UAE
Fresh vegetables (all)	2,686,175	7,378	891	Bangladesh, UAE, Malaysia, Sri Lanka, Nepal

Source: APEDA Annual Report 2023-24 and DGCI&S; data. Rounded.

Opportunities and constraints. The main opportunities are rising global demand for tropical fruit, the Gulf and South-East Asian markets close to India, growing processed-food demand, and GI-tagged products such as Alphonso mango and Nashik grapes. The main constraints are short shelf life, weak cold-chain and pack-house infrastructure, sanitary and phytosanitary barriers in the USA and Japan, and the fact that most exportable fruit is still produced by smallholders with variable quality.

Exam focus

Key points to remember

1. India is the second largest producer of fruits and vegetables in the world, after China; fruit production crossed 112 million tonnes in 2023-24.
2. India ranks first in the world in the production of banana, lime and lemon, and papaya.
3. Horticulture contributes about one-third of the value of agricultural output from about 18 per cent of the cropped area.
4. Fruit crops are classified by climatic adaptation, botanical family, fruit type (pomology), life cycle and ripening behaviour.
5. Guava and aonla are the richest common sources of vitamin C; mango and sweet orange are rich in provitamin A.
6. Climacteric fruit (mango, banana, papaya, guava, sapota, apple) ripen after harvest; non-climacteric fruit (citrus, grape, pineapple) do not.
7. Average Indian fruit productivity is about 15-16 tonnes per hectare, with wide variation between crops.
8. India exported fresh fruits and vegetables worth about USD 1.8 billion in 2023-24; grapes are the leading fresh-fruit earner.
9. Plantation crops - coconut, arecanut, rubber, tea, coffee, cocoa, cashew, oil palm and palmyrah - are the backbone of hill and coastal economies.

10. The five statutory commodity boards under the Department of Commerce are Tea, Coffee, Rubber, Spices and Tobacco Boards; coconut, cashew and cocoa are supported by the Coconut Development Board and the DCCD.

Short-answer questions with model answers

Q1. Why is horticulture important to India's farm economy?

Model answer. Horticulture contributes about one-third of the value of agricultural output from about 18 per cent of the cropped area. It is high-value, labour-intensive and export-oriented, supports processing industries, and supplies protective nutrients that cereal diets lack. India is the world's second largest producer of fruits and vegetables.

Q2. Distinguish climacteric from non-climacteric fruit with examples.

Model answer. Climacteric fruit show a sharp rise in respiration and ethylene at ripening and continue to ripen after harvest (mango, banana, papaya, guava, sapota, apple). Non-climacteric fruit show no such rise and ripen only on the plant (citrus, grape, pineapple, strawberry, cashew). Climacteric fruit are harvested mature-green and can be ripened artificially; non-climacteric fruit must be picked fully ripe.

Q3. Give the climatic classification of fruit crops with two examples each.

Model answer. Tropical: warm, frost-free, above about 21 deg C - mango, banana. Sub-tropical: distinct winter with mild frost - citrus, grape. Temperate: cool winter with chilling, warm summer - apple, pear.

Q4. Name four leading fruit crops of India by production and their approximate production.

Model answer. Banana about 37.8 million tonnes, mango about 22.4 million tonnes, citrus about 14.6 million tonnes and papaya about 5.3 million tonnes (2023-24 estimates).

Long-answer questions with model answer outlines

Q1. Describe the importance and scope of the fruit and plantation crop industry in India.

Outline. Define the sector (fruit + plantation + beverage crops). Economic importance: high value per hectare, employment, foreign exchange, industrial raw material, nutrition, environment. Statistical support: fruit production above 112 million tonnes, second largest producer in the world, exports about USD 1.8 billion fresh fruit and vegetables. Scope: rising demand, processing, exports, low productivity of old orchards, need for improved planting material. Mention specific belts - TN mango and banana, Nilgiris tea, Yercaud coffee, Western Ghats rubber and arecanut, coastal cashew and coconut.

Q2. Explain the classification of fruit crops with examples, and state the practical use of each system.

Outline. Climatic classification (tropical, sub-tropical, temperate) guides the choice of crop for a location. Botanical classification groups crops by family and so predicts pest, disease and cultural similarities. Pomological classification (drupe, berry, hesperidium, pome, aggregate, multiple, nut) helps in understanding fruit development and harvest. Ripening classification (climacteric vs non-climacteric) decides harvest maturity, ripening and storage. Life-cycle classification (herbaceous, woody, palms, vines) guides training and pruning. Give the master tree diagram and two examples for each class.

Q3. Discuss the nutritional importance of fruit crops and India's area, production and export position.

Outline. Nutrition: sources of vitamin C (guava, aonla, citrus), provitamin A (mango, papaya), potassium (banana), fibre and antioxidants; low per-capita intake is a public-health issue. Area and production: about 7.0-7.2 million ha and 110-113 million tonnes of fruit; crop-wise table; productivity about 15-16 tonnes per hectare. Exports: fresh fruit and vegetables about USD 1.8 billion, grapes the leading earner; main markets Netherlands and UAE; constraints of cold chain and SPS barriers. Conclude with the scope for improvement.

UNIT II | Production Techniques in Tropical and Sub-tropical Fruit Crops

This unit covers the tropical crops mango, banana, papaya, guava and sapota, and the sub-tropical crops acid lime, sweet orange and grape. Section 2.1 sets out the common techniques that apply to all of them - climate and soil, propagation and rootstock, planting systems and density, training and pruning, water and nutrient management, fertigation, weed management, growth regulation, disorders, maturity and harvest. Each crop section then applies and adds to this framework.

2.1 Common production techniques for tropical and sub-tropical fruit crops

2.1.1 Climate and soil requirements

Tropical fruit crops need a warm, frost-free climate with plenty of sunshine; sub-tropical crops need a distinct but mild winter. Soil for almost all fruit crops should be deep (at least 1 m), well drained, with a pH of 6.0 to 7.5, adequate organic matter and no hard pan or waterlogging. Waterlogging is the single most common cause of orchard failure in the heavy-rain coastal belts of Tamil Nadu, so drainage is designed before planting. Frost pockets are avoided for tropical crops; slopes and air-drained sites are preferred.

2.1.2 Varieties and planting material

Only recommended varieties of known parentage should be planted. Clonal planting material from a registered nursery is used for mango, guava, sapota, citrus and grape; tissue-cultured plantlets are used for banana; and gynodioecious or hybrid seed for papaya. Planting material should be true to type, free from disease, and raised on a suitable rootstock. The crop sections give the recommended varieties for Tamil Nadu and India, including TNAU and ICAR releases.

2.1.3 Propagation and use of rootstocks

Propagation may be sexual (seed) or asexual (vegetative). Asexual propagation - grafting, budding, layering, cutting and tissue culture - is used for most fruit crops because it fixes the characters of the parent and brings early and uniform bearing. The seedling used to carry the scion is the rootstock. A rootstock influences vigour, precocity, size, fruit quality, and tolerance of soil, drought, salinity and disease. Choosing the right rootstock is as important as choosing the right variety.

Table 2.1 Rootstock comparison for major tropical and sub-tropical fruit crops

Crop	Common rootstock	Method of propagation	Main advantage of the rootstock
Mango	Seedling of local / polyembryonic types (e.g. Olour, Bappakai)	Veneer / stone / epicotyl grafting on seedling	Vigorous, adapted to local soil; polyembryonic stock gives true-to-type uniformity
Banana	Sucker or tissue-cultured plantlet (no separate rootstock)	Sword sucker or tissue culture	Disease-free, uniform, high-yielding planting material
Papaya	Own-root seedling; gynodioecious or hybrid seed	Seed	Sex expression managed by selection of gynodioecious lines
Guava	Seedling of local guava (for grafting); own root for air-layers	Air-layering, stooling, grafting	Air-layers bear early and stay true to type; seedling stock gives vigour
Sapota	Seedling of sapota (Kalippatti / local)	Softwood grafting / air-layering	Grafted plants bear in 3-4 years; seedling takes 7-8 years
Citrus	Rough lemon, Rangpur lime, Trifoliate orange, Troyer citrange	Budding (T / shield) or grafting	Rough lemon: vigour and drought tolerance; Trifoliate: Phytophthora and cold tolerance; Troyer citrange: greening tolerance
Grape	Own root (cuttings); Dog Ridge / 110-R / 1103-P for salt and nematode	Hardwood cuttings; grafting on resistant stock	Resistant stocks allow planting on saline or nematode-infested soil

Source: standard horticultural texts and TNAU package of practices.

2.1.4 Planting density and systems of planting

Planting density is fixed by the spacing, which in turn depends on the vigour of the crop, the soil, the training system and the machinery to be used. The four classical systems of planting are square, rectangular, quincunx and hexagonal; on slopes, contour or terracing is used.

Diagram 2.1 Systems of planting - layouts

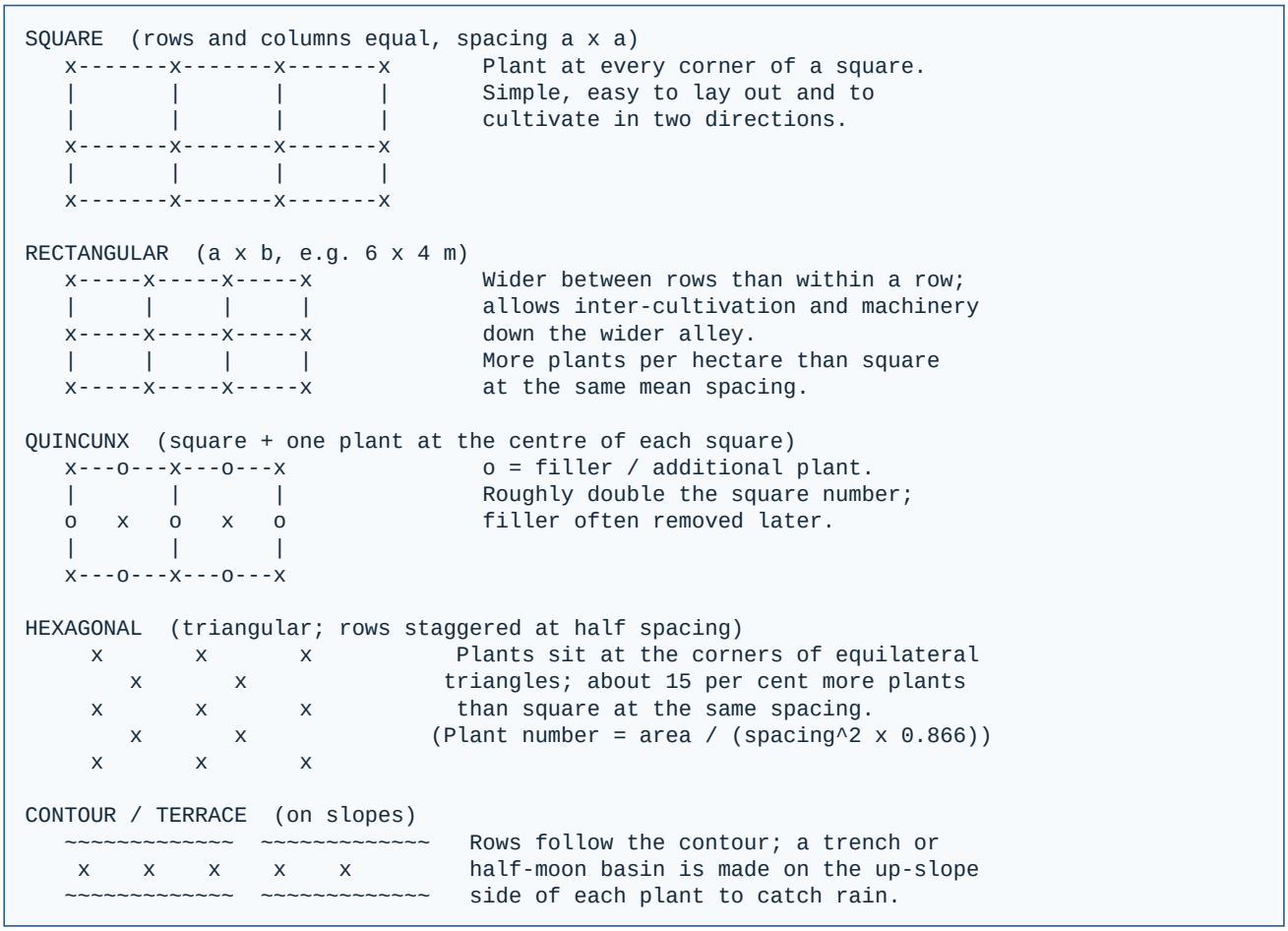


Figure 2.1 The four classical planting systems and the contour system for slopes.

Worked example 2.1 Plant population for a given spacing and area

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Given: 1 hectare = 10,000 m2. Spacing taken as 6 x 6 m for illustration.

SQUARE (6 x 6 m)
   Plants/ha = 10,000 / (6 x 6) = 10,000 / 36 = 277.8  ->  about 278 plants

RECTANGULAR (6 x 4 m)
   Plants/ha = 10,000 / (6 x 4) = 10,000 / 24 = 416.7  ->  about 416 plants

HEXAGONAL (6 m spacing)
   Plants/ha = 10,000 / (6 x 6 x 0.866) = 10,000 / 31.18 = 320.8
   ->  about 320 plants (about 15 per cent more than square)

QUINCUNX (6 x 6 m square plus a centre plant)
   Square grid gives 278 plants; one extra plant in each square
   roughly doubles it:  ->  about 555 plants (filler plants often
   removed after the main crop fills the canopy).

General rule:  Plants per hectare = 10,000 / (area occupied per plant in m2).
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The same method is used for any spacing or plot size; for a 0.5 ha plot, multiply by 0.5.

2.1.5 High density and ultra-high-density planting (HDP and UHDP)

High-density planting means planting more trees per hectare than the traditional spacing, to use land, light and labour more efficiently and to raise early and total yield. Ultra-high-density planting (UHDP) uses very close spacing with a small, controlled canopy. The techniques work only if the canopy is kept small by dwarfing rootstocks, training, pruning and growth regulators, and if water and nutrients are supplied precisely.

Table 2.2 Planting densities and expected performance (mango as model crop)

System	Spacing	Plants per ha	Years to full bearing	Typical yield (t/ha) at maturity
Conventional	10 x 10 m	100	10-12	8-10
High density	5 x 5 m	400	5-6	12-18
Ultra-high density	3 x 2 m or 2.5 x 2.5 m	1333-1667	3-4	18-25 (early; may decline later)

Indicative figures; performance depends on variety, rootstock, pruning and management.

Worked example 2.2 Economics of high-density planting - mango

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Assumptions for a one-hectare comparison (indicative, using round figures).
  Farm-gate price of fruit: Rs 40 per kg.

                            Conventional     High density
  Plants per hectare            100              400
  Cost of plants (@Rs 80)      Rs 8,000        Rs 32,000
  Cost of pits and planting    Rs 40,000       Rs 1,20,000
  Annual management cost       Rs 60,000       Rs 1,20,000
  Total cost to 6th year       Rs 4,08,000     Rs 8,32,000
  Yield at 6th year (t/ha)          8              15
  Gross return (15,000 kg x Rs 40 for HDP = Rs 6,00,000)
  Gross return (8,000 kg x Rs 40 for conventional = Rs 3,20,000)
  Net return at 6th year       -Rs 88,000      -Rs 2,32,000

Interpretation: HDP costs more up front, but it recovers the extra cost
from the 7th-8th year onward and gives a higher cumulative return over a
25-year orchard life. It suits small holdings and early-return varieties.
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The same framework is used for any crop; substitute the actual spacing, costs and price.

2.1.6 Cropping systems

Young orchards leave a great deal of vacant inter-row space. This is used by intercropping with short-duration crops such as pulses, groundnut, vegetables, banana or pineapple, which gives income before the main crop bears and also suppresses weeds. Care is taken to keep the intercrop away from the tree basin, not to shade the young tree, and not to grow another crop that competes for water or shares pests. Mixed planting of two or more fruit crops of different stature, and multi-tier cropping, are discussed in Unit IV.

2.1.7 After care

After planting, the young plant is shaded for a few weeks, staked if necessary, and watered regularly. The basin is mulched and kept free of weeds. For grafted plants the rootstock sprout below the graft union is removed. Tree guards protect against animals, and windbreaks protect against wind in the coastal belts.

2.1.8 Training and pruning

Training builds the framework of the young tree; pruning removes or shortens wood on the bearing tree. The aim is a strong, open framework that admits light to the centre, keeps the tree to a manageable height, removes dead, diseased and crowding branches, and regulates the balance between vegetative growth and fruiting. The main training systems are the central leader, the open centre (vase) and the modified leader.

Diagram 2.2 Training systems

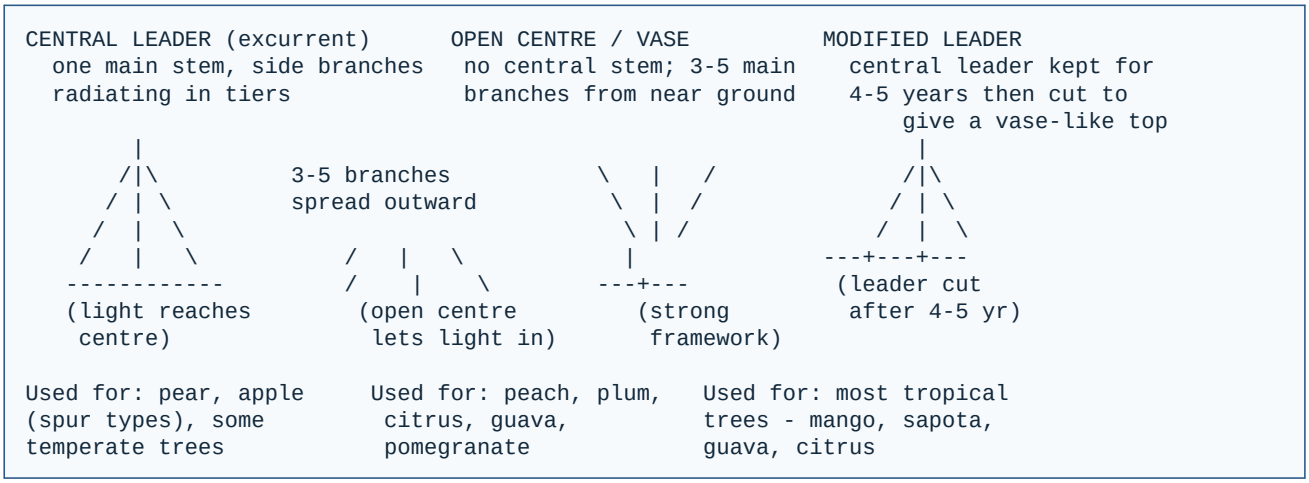


Figure 2.2 The three classical training systems and the crops they suit.

Diagram 2.3 Pruning of mango and grape

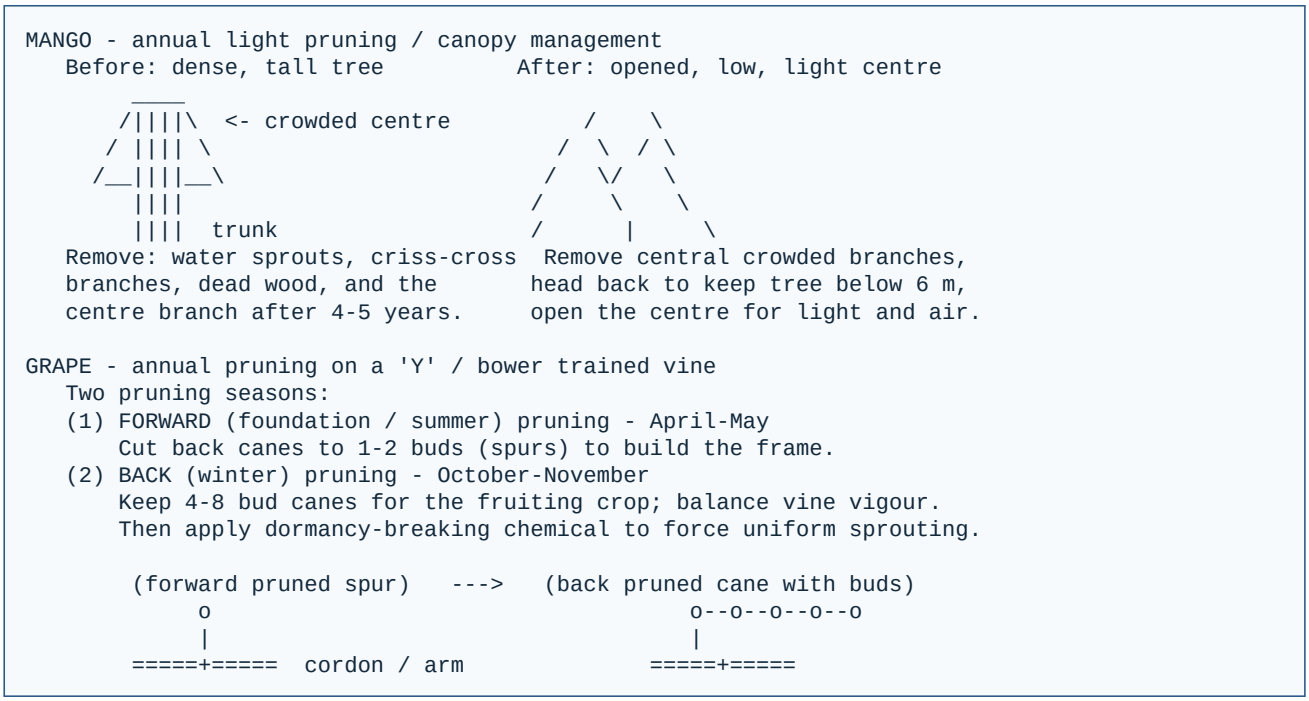


Figure 2.3 Pruning principles for mango (canopy opening) and grape (forward and back pruning).

2.1.9 Water management

Irrigation is essential and is scheduled by soil moisture, pan evaporation or tensiometer. The basin system of irrigation is traditional; drip irrigation is now standard because it saves water, keeps the soil evenly moist and allows fertigation. Critical stages when water shortage does the most damage are flowering, fruit set and fruit development. Excess water is as harmful as shortage: waterlogging causes root death, fruit drop and diseases such as Phytophthora in citrus. Drainage channels are essential in heavy-rain areas.

2.1.10 Nutrient management

Fruit crops need nitrogen for growth, phosphorus for root and flower development, potassium for fruit quality, and calcium, magnesium, sulphur and micronutrients (zinc, boron, iron, manganese) for balanced nutrition. Farmyard manure or compost is applied each year, and the fertiliser dose is split into two or three applications so that nutrients are available at the right growth stages. The dose rises with the age of the tree up to maturity, then stays constant. Foliar sprays correct micronutrient deficiencies quickly.

Worked example 2.3 Fertiliser dose per tree per year, split into applications

```
Given dose for a bearing mango tree: 500 g N, 250 g P2O5 and 500 g K2O per year.

Step 1 - convert nutrient to fertiliser (using the nutrient content of each fertiliser):
   Urea (46 per cent N):   500 / 0.46 = 1087 g  ->  1.09 kg urea
   SSP (16 per cent P2O5): 250 / 0.16 = 1562 g  ->  1.56 kg single superphosphate
   MOP (60 per cent K2O):  500 / 0.60 =  833 g  ->  0.83 kg muriate of potash

Step 2 - split the annual dose:
   Application 1 (June-July, after harvest / with the rains):
       half of the urea + all of the SSP + half of the MOP
       = 0.55 kg urea + 1.56 kg SSP + 0.42 kg MOP
   Application 2 (September-October, fruit development):
       the remaining half of the urea + remaining half of the MOP
       = 0.55 kg urea + 0.42 kg MOP

Step 3 - method: place in a ring 30-45 cm from the trunk, 15-20 cm deep,
cover with soil and irrigate. Add 25-50 kg farmyard manure per tree per year.
```

The same three steps convert any recommended dose into a split fertiliser schedule.

2.1.11 Weed management

Weeds compete for water and nutrients, especially in the young orchard, and harbour pests. Control by clean cultivation of the basin, mulching with organic matter or black polythene, growing a cover crop or green manure in the alleys, and, where necessary, selective herbicides. Glyphosate and paraquat are used as directed, taking care not to touch the crop. Mulching also conserves moisture and moderates soil temperature.

2.1.12 Fertigation

Fertigation is the application of water-soluble fertilisers through the drip irrigation system. It places nutrients in the root zone in small, frequent doses, so nutrient-use efficiency rises, labour falls and growth and yield improve. Only fully soluble fertilisers are used, the injection is followed by a clear-water flush to prevent clogging, and the schedule follows the crop's growth stages.

Diagram 2.4 Drip fertigation layout

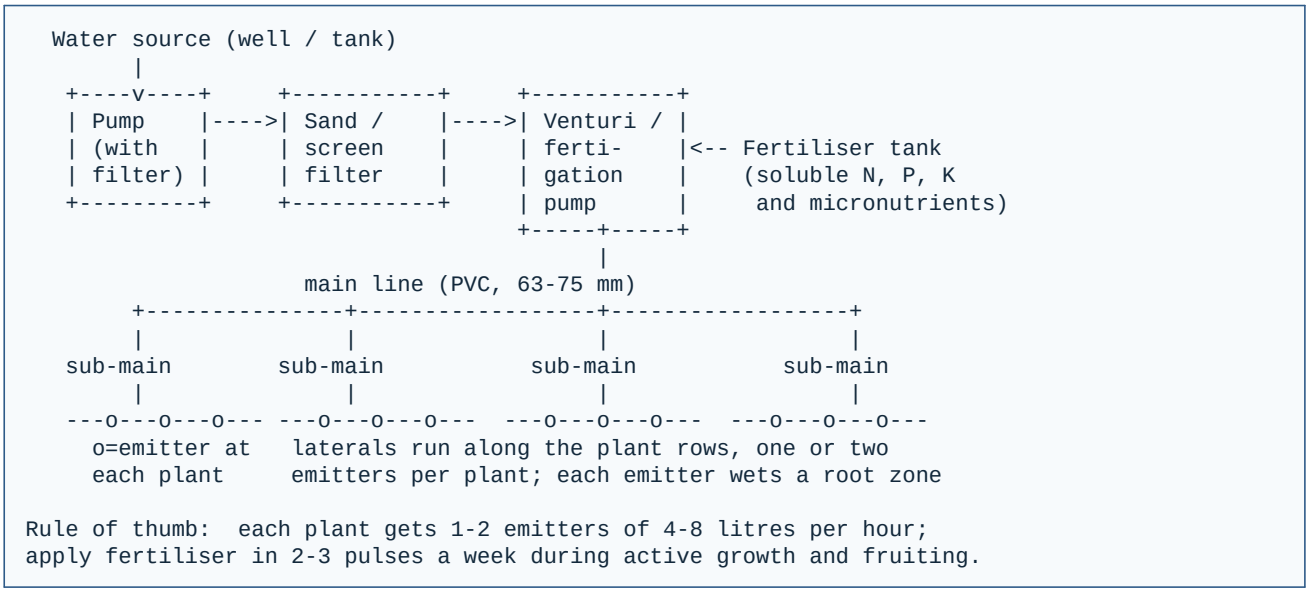


Figure 2.4 Layout of a drip-fertigation system.

2.1.13 Special horticultural techniques and growth regulators

Special techniques are used to control tree size, induce regular flowering, improve fruit set and improve fruit quality. These include bending and ringing of branches to induce flowering, scoring or girdling, smudging in mango, bahar treatment in citrus and grape, de-blossoming in some crops, and bagging of fruit to protect it from pests and sunburn. Growth regulators are used as foliar sprays or dips: auxins (NAA, 2,4-D) to check fruit drop, gibberellins (GA3) to increase fruit size and delay maturity, cytokinins to improve set, ethephon to hasten ripening and degreening, and paclobutrazol or CCC to dwarf trees and induce early flowering.

2.1.14 Important disorders

Disorders are functional problems caused by nutrient deficiency, imbalance or environment, not by a pathogen. Common ones are the fruit drop and spongy tissue of mango, the K and Mg deficiency and bunchy top-like rosetting of banana, the boron deficiency and leaf curl in papaya, the zinc and boron deficiency of guava, the leaf chlorosis of sapota on high-pH soils, and the granulation, mottling and fruit cracking of citrus. Management is by correcting the nutrient or water supply and choosing a tolerant rootstock or variety. The crop sections list the disorders of each crop.

2.1.15 Maturity indices and harvest

Fruit must be harvested at the correct maturity, because neither immature nor over-ripe fruit stores or sells well. Maturity is judged by a combination of physical and chemical indices. Some fruit (mango, banana) are picked mature but unripe and ripened later; others (grape, citrus) are picked when ready to eat. The table below summarises the main indices; the crop sections give the detail for each crop.

Table 2.3 Maturity indices of major tropical and sub-tropical fruit crops

Crop	Physical indices	Chemical / other indices	Approx. days from flowering
Mango	Shoulders fill out, fruit turns from green to light green-yellow, lenticels prominent	Specific gravity above 1.0-1.02, TSS 12-15 per cent, sap turns from milky to clear	90-140 (variety dependent)
Banana	Fingers plump, ridges become less angular, colour breaks from green to light green	Pulp-to-peel ratio, starch-to-sugar conversion, TSS 12-15 per cent	90-150
Papaya	Skin shows first yellow streaks, latex becomes watery	TSS 11-14 per cent	120-150
Guava	Skin turns light green to yellow, surface smooth	TSS 10-13 per cent, firmness	90-120
Sapota	Skin loses brown scurf, becomes smooth and light brown	TSS 18-22 per cent, less latex	120-150
Acid lime / sweet orange	Colour change to yellow / orange, surface smooth and glossy	TSS:acid ratio, juice content above 40-45 per cent	150-240
Grape	Berries attain full size and colour, easy separation from stalk	TSS 16-22 per cent (Brix), TSS:acid ratio	100-140

Source: standard post-harvest texts and TNAU package of practices. Days from flowering vary widely with variety and climate.

2.2 Mango (Mangifera indica)

Origin and importance

Mango is native to the Indo-Burma region and has been cultivated in India for over 4,000 years. It is the national fruit of India and is called the king of fruits. The ripe fruit is eaten fresh and is rich in provitamin A; the unripe fruit is used for pickle, chutney, amchur and panna. Mango is the largest fruit crop of India by area and the second largest by production.

Area and production

India grows mango on about 2.35 million hectares and produces about 22.4 million tonnes (2023-24 estimate), the largest in the world. Uttar Pradesh, Andhra Pradesh, Karnataka, Bihar, Gujarat, Maharashtra and Tamil Nadu are the leading states. In Tamil Nadu the main belts are Krishnagiri, Dharmapuri, Vellore, Salem and Theni, and the Periyakulam and Palani areas. India exports mango and mango pulp mainly to the UAE, UK, USA, Saudi Arabia and Bangladesh.

Climate and soil

Mango needs a warm, frost-free climate; it grows well from the tropics to about 30 deg N. A distinct dry spell before flowering helps profuse, uniform flowering, while rain or high humidity at flowering causes poor set and disease. Deep, well-drained alluvial or lateritic soil with pH 5.5 to 7.5 is ideal. The tree tolerates a range of soils but not waterlogging.

Varieties

Tamil Nadu and India grow both regular and irregular bearers. Important commercial varieties include Banganapalli, Neelam, Alphonso, Totapuri (Bangalora), Rumani, Mulgoa, Kalepad, Peter, Imam Pasand, Kesar, Dashehari and Langra. TNAU has released a series of varieties - CO 1 to CO 5 and PKM 1 and PKM 2 - selected for yield, fruit quality and regular bearing; the PKM selections from the Horticultural Research Station, Periyakulam, are notable for high yield and good quality. Neelam and its derivatives are valued for regular bearing, and Banganapalli for fruit quality.

Table 2.4 Important mango varieties and their salient features

Variety	Source / belt	Salient features
Banganapalli	Andhra Pradesh, Tamil Nadu	Large, sweet, fibreless; leading table and export variety of the south
Alphonso	Konkan, Maharashtra	Rich aroma, high quality; leading export variety
Neelam	Andhra Pradesh, Tamil Nadu	Regular bearer; good for processing and fresh use
Totapuri (Bangalora)	South India	Large fruit, good for pulp, juice and pickle
Rumani	Tamil Nadu	Regular bearer, good yield, mild flavour
Kalepad	Tamil Nadu	Local selection; good quality
TNAU CO 1 to CO 5	TNAU	Selections/hybrids for yield and quality; some are regular bearers
PKM 1, PKM 2	TNAU, Periyakulam	High-yielding clonal selections for Tamil Nadu
Kesar, Dashehari, Langra	North and west India	Kesar for aroma and colour; Dashehari and Langra for table quality

Source: TNAU package of practices and standard texts. Variety descriptions are indicative.

Propagation and rootstock

Mango is propagated by grafting on a seedling rootstock. The rootstock is raised from seed of a vigorous local or polyembryonic type such as Olour or Bappakai; polyembryonic seed gives true-to-type, uniform stock. The common grafting methods are veneer grafting, stone grafting (used in the Konkan for rapid, cheap planting) and epicotyl grafting. Stone grafting is done on 8-14 day old seedlings and gives a plant ready for planting in the same season. Air-layering is possible but is not commercial.

Planting density and spacing

The traditional spacing is 10 x 10 m (about 100 trees per hectare), giving large trees. High-density planting at 5 x 5 m (400 trees per hectare) and ultra-high-density planting at 3 x 2 m or 2.5 x 2.5 m (about 1,300-1,700 trees per hectare) are now recommended for early and higher returns, provided the canopy is kept small by pruning and, where needed, paclobutrazol. Pits of 1 x 1 x 1 m are dug, filled with topsoil, farmyard manure and a little rock phosphate, and planting is done at the onset of the monsoon.

Training and pruning

The young plant is trained to a single stem up to about 1 m, then headed to produce 3 to 5 well-spaced main branches - the modified-leader system. Regular light pruning after harvest removes water sprouts, criss-cross branches, dead wood and the crowded central branch, keeps the tree below about 6 m and opens the centre to light and air. This improves fruit colour and reduces disease, and is essential in high-density orchards.

Nutrient management

A bearing mango tree is given farmyard manure each year and a fertiliser dose rising with age to about 500 g N, 250 g P2O5 and 500 g K2O per tree per year at maturity, split into two or three applications (see Worked example 2.3). Micronutrient sprays of zinc sulphate, borax and ferrous sulphate correct the common deficiencies, especially on high-pH soils.

Irrigation and fertigation

Irrigation is given at 7-12 day intervals in the basin system or by drip. The critical stages are flowering, fruit set and fruit development; water is withheld for two to three months before flowering to induce the flowering flush, but stress during fruit development causes heavy drop. Fertigation through drip is increasingly used, applying the split dose in weekly pulses.

Weed management

The basin is kept weed-free by hand weeding or mulching, and the alleys are cultivated or kept under a cover crop. In the young orchard, intercrops such as pulses, groundnut or vegetables suppress weeds and add income.

Special horticultural practices and growth regulators

Smudging (burning leaves and dung under the canopy to raise smoke and ethylene) is used in some regions to induce off-season flowering. Paclobutrazol applied to the soil or as a foliar spray dwarfs the tree and induces early, regular flowering, which is central to high-density planting. NAA (10-20 ppm) is sprayed to check fruit drop, and GA3 is used to improve fruit size and delay maturity.

Important disorders

Mango suffers from alternate (irregular) bearing, in which a heavy crop is followed by a light one; from mango malformation, in which floral and vegetative buds become distorted; from fruit drop, especially the June drop; from spongy tissue (a physiological defect of Alphonso in which the pulp becomes soft and pale); and from nutrient disorders such as zinc, boron and iron deficiency. Alternate bearing is managed by thinning, pruning and paclobutrazol; spongy tissue by harvesting at correct maturity and by orchard management.

Pests and diseases

The main pests are the mango hopper, which causes flower and fruit drop, the mango fruit fly, the mealybug and the stone (nut) weevil. The main diseases are anthracnose, powdery mildew and bacterial canker. Management combines pruning to open the canopy, orchard sanitation, fruit bagging and timely sprays of the recommended insecticide and fungicide; the fruit fly is controlled by bait sprays and by collecting and destroying fallen fruit.

Maturity indices

Mango is harvested mature but unripe. The fruit is ready when the shoulders fill out, the colour lightens from dark green to light green or yellow, the lenticels become prominent, the specific gravity exceeds 1.0-1.02, the sap becomes less milky and total soluble solids reach about 12-15 per cent. In Tamil Nadu the main harvest is from April to June, earlier in the Periyakulam belt.

Harvest, yield and post-harvest

Fruit is harvested with a stalk of about 1 cm using a harvester or pole, never by shaking, to avoid bruising and latex burn. Yield ranges from about 8-10 tonnes per hectare in conventional orchards to 15-25 tonnes per hectare in well-managed high-density orchards. After harvest the fruit is de-sapped, washed, graded, and either marketed fresh or ripened. Value-added products are mango pulp, juice, squash, jam, pickle, chutney, amchur and dried slices.

Economics note

Mango is a long-term investment with a life of 25-40 years. A grafted plant begins to bear in 3-5 years and gives economic crops from the 6th-8th year. High-density planting brings earlier returns and higher yield per hectare, though at higher initial cost. Returns depend heavily on the variety (table versus processing), the season (early fruit fetches a premium) and the market. The crop is well suited to Tamil Nadu because early Periyakulam fruit reaches the market before the northern crop.

2.3 Banana (Musa species)

Origin and importance

Banana originated in South-East Asia and is now the largest fruit crop of India by production. It is a giant herb, not a tree, with a pseudostem formed of leaf sheaths. The fruit is eaten ripe and used for chips, puree and baby food; the male bud and the stem are used as vegetables. Banana is a staple energy source and a major export crop.

Area and production

India is the largest producer of banana in the world, with about 0.99 million hectares and about 37.8 million tonnes (2023-24 estimate). Tamil Nadu, Maharashtra, Gujarat, Andhra Pradesh, Karnataka and Kerala are the leading states. The main Tamil Nadu belts are Theni, Trichy, Karur, Coimbatore, Erode, Kanyakumari and the Cauvery delta. India exports banana mainly to the UAE, Saudi Arabia, Oman and, increasingly, to the Netherlands by sea.

Climate and soil

Banana needs a warm, humid climate with 75-85 per cent relative humidity and a temperature of 15-35 deg C; it is damaged by frost and by strong wind. It grows on a wide range of soils but needs a deep, rich, well-drained loam with high organic matter and pH 6.0-7.5. It is a heavy feeder and a heavy water user, and does not tolerate waterlogging or salinity.

Varieties

Commercial varieties include Poovan (Mysore), Nendran, Robusta and Grand Naine (Cavendish), Rasthali (Silk), Karpuravalli, Monthan, Ney Poovan, Red Banana, Pachanadan, Virupakshi and Kunnan. TNAU has released CO 1 and CO 2 and the PKM series (PKM 1 to PKM 5) with high yield and quality; tissue-cultured Grand Naine is widely planted because of its uniformity, disease-free planting material and high yield.

Table 2.5 Important banana varieties and their uses

Variety	Type / genome	Salient features
Grand Naine (Robusta)	Cavendish (AAA)	High yield, uniform, export quality; mostly tissue-cultured
Poovan (Mysore)	Mysore (AAB)	Sweet-acid flavour, dual purpose, good keeping quality
Nendran	Plantain (AAB)	Large fruit, used for chips and ripe use; Kerala and TN
Rasthali (Silk)	Silk (AAB)	Rich flavour; susceptible to bunchy top and Fusarium
Karpuravalli	Pisang Awak (ABB)	Hardy, used for fruit and stem vegetable
Monthan	ABB	Used mainly as vegetable and for chips
Red Banana	AAA (red)	Coloured, sweet; premium price
TNAU CO 1, CO 2; PKM 1 to PKM 5	Various	TNAU selections for high yield and quality in Tamil Nadu

Source: TNAU package of practices and standard texts.

Propagation and planting material

Banana is propagated vegetatively by sword suckers or, increasingly, by tissue-cultured plantlets. Sword suckers of 1.5-2 kg with a healthy corm are used; water suckers are avoided. Tissue-cultured plants are disease-free, uniform and high-yielding, and are hardened in a nursery before planting. Suckers are pared, treated with a fungicide and sometimes with a nematicide before planting.

Planting density and spacing

Spacing varies with variety and system. For tall Cavendish types 1.8 x 1.8 m (about 3,000 plants per hectare) is common; dwarf types are planted closer, and paired-row or double-row systems raise the population further. Pits of 45 x 45 x 45 cm are dug and filled with farmyard manure and a little fertiliser. Planting is done at the onset of the monsoon; in Tamil Nadu planting is staggered to catch favourable market windows.

Training and pruning

Banana needs desuckering, in which all suckers except one follower are removed to divert nutrients to the mother plant. The follower is retained to ratoon the crop. After harvest the pseudostem is cut and the follower is allowed to grow. Banana is not pruned in the tree sense, but the dry leaves are removed, the flower (male bud) is removed after the last hand to improve bunch size, and the bunch is propped or the plant staked to prevent lodging.

Nutrient management

Banana is a heavy feeder. A common dose for a high-yielding crop is about 200 g N, 50-100 g P2O5 and 200-300 g K2O per plant per year, applied in several splits, plus farmyard manure. Potassium is especially important for bunch weight and quality. The dose is applied in a ring or through drip as fertigation; calcium, magnesium and micronutrients are added as needed.

Irrigation and fertigation

Banana needs frequent, generous irrigation - the crop uses about 1,000-1,500 mm of water in a season. Drip irrigation with fertigation is standard and gives the best results; the crop is irrigated every 2-3 days in light soil. Water shortage reduces bunch size and causes premature ripening, while waterlogging causes root rot.

Weed management

The crop is kept weed-free by hand weeding, mulching with banana trash or black polythene, and intercropping with short-duration vegetables or pulses in the early months. Chemical weed control is used where labour is scarce.

Special horticultural practices and growth regulators

Important practices are desuckering, removal of the male bud (denavelling) to improve bunch weight, removal of the last small hands, and bunch covering with polythene to protect against sunburn and pests. Propping with bamboo prevents lodging of heavy bunches. Growth regulators are used less than in other crops, though GA3 and cytokinins are sometimes used to improve finger size.

Important disorders

Banana disorders include the rosetting and yellowing of potassium and magnesium deficiency, boron deficiency causing distorted fruit, and 'choking' in which the bunch fails to emerge properly. Bunchy top and Fusarium wilt are diseases, not disorders, but they are managed by using clean tissue-cultured planting material and resistant varieties.

Pests and diseases

The main pests are the pseudostem weevil and the rhizome weevil, which bore into the plant, and the aphid that transmits the bunchy top virus. The main diseases are Panama wilt (Fusarium), bunchy top and the Sigatoka leaf spot. Management uses clean tissue-cultured planting material, resistant varieties, good drainage and the removal and destruction of affected plants.

Maturity indices

The bunch is ready about 90-150 days after flowering depending on variety and season. Indices are the plumpness of fingers, loss of angularity of the ridges, colour change from dark green to light green, drying of the flower remnants, and total soluble solids of about 12-15 per cent in the mature fruit. Fruit is harvested while still green for distant markets and allowed to ripen in ripening chambers.

Harvest, yield and post-harvest

The bunch is harvested by cutting the peduncle, and the hands are separated, graded and packed. Average yield is 40-60 tonnes per hectare, and up to 70 tonnes with best management. Post-harvest operations are de-handing, cleaning, grading, packing in ventilated boxes and ripening with ethylene. Value-added products are banana chips, puree, pulp, flour, wine and vinegar, and fibre from the pseudostem.

Worked example 2.4 Cost and return for one hectare of banana (indicative)

```
Spacing 1.8 x 1.8 m  ->  plant population = 10,000 / 3.24 = about 3,086 plants.
Assume tissue-cultured plants at Rs 15 each.

  ITEM                                        AMOUNT (Rs)
  Cost of plants (3,086 x 15)                    46,290
  Land preparation, pits and planting            35,000
  Farmyard manure and fertilisers                60,000
  Drip and irrigation (share for the crop)       25,000
  Plant protection                                20,000
  Weeding and labour (desuckering, propping)     55,000
  Harvesting and packing                         40,000
  Total cost                                   2,81,290

  Yield: 55 t/ha = 55,000 kg.
  Gross return at Rs 15 per kg = 55,000 x 15 = Rs 8,25,000
  Net return = 8,25,000 - 2,81,290 = Rs 5,43,710
  Benefit-cost ratio = 8,25,000 / 2,81,290 = about 2.9

Note: a ratoon crop in the second year reduces cost (no new planting) and
improves the cumulative return. Actual figures depend on market price,
yield and the variety grown.
```

Indicative only; substitute local costs and prevailing market price.

2.4 Papaya (Carica papaya)

Origin and importance

Papaya is native to tropical America and is now grown throughout the tropics. It is the most productive of the fruit crops per unit area and per unit time, bearing fruit within a year of planting. The ripe fruit is eaten fresh and is rich in provitamin A and vitamin C; the unripe fruit is used as a vegetable, and the enzyme papain from the green fruit and latex is used in medicine, meat tenderising and the leather and textile industries.

Area and production

India is the largest producer of papaya in the world, with about 0.15 million hectares and about 5.3 million tonnes (2023-24 estimate). Andhra Pradesh, Gujarat, Maharashtra, Karnataka, Tamil Nadu, Madhya Pradesh and West Bengal are the leading states. In Tamil Nadu, Coimbatore, Erode, Salem, Dindigul and Trichy are important.

Climate and soil

Papaya needs a warm, humid climate, ideally 22-32 deg C, and is very sensitive to frost and to waterlogging. A dry spell or low temperature at flowering causes poor set and sex reversal. It grows on a wide range of soils but prefers a rich, well-drained loam with pH 6.0-7.0. Waterlogging for even a day can kill the plant, so raised beds and drainage are essential.

Varieties

Important varieties include Ranchi, Pusa Dwarf, Pusa Nanha, Red Lady (a hybrid), Coorg Honey Dew, and the TNAU varieties CO 1 to CO 7 and PKM 1 and PKM 2, selected for yield, fruit size and gynodioecy (production of only female and bisexual flowers, giving high marketable yield). Gynodioecious and hybrid varieties are preferred because every plant bears fruit and the crop is uniform.

Table 2.6 Important papaya varieties and their salient features

Variety	Source	Salient features
Ranchi	Bihar	Medium fruit, good quality, popular in the east
Pusa Dwarf / Pusa Nanha	IARI	Dwarf, high density planting, early bearing
Red Lady	Hybrid (imported/ICAR)	Gynodioecious, red flesh, high yield, export quality
Coorg Honey Dew	Karnataka	Sweet, good quality
TNAU CO 1 to CO 7	TNAU	Selections/hybrids for yield, fruit size and gynodioecy
PKM 1, PKM 2	TNAU, Periyakulam	High-yielding selections for Tamil Nadu

Source: TNAU package of practices and standard texts.

Propagation

Papaya is grown from seed. Seed is taken from a selected, healthy, high-yielding plant - preferably a bisexual (hermaphrodite) or gynodioecious parent - and sown in nursery beds or polybags. Seedlings of 15-20 cm height (about 6-8 weeks old) are transplanted. About 10 per cent extra seedlings are kept to replace weak plants. Being a seed-propagated crop, papaya does not use a rootstock.

Planting density and spacing

Common spacings are 1.8 x 1.8 m (about 3,000 plants per hectare), 2 x 2 m (2,500 plants per hectare) and, for dwarf varieties, closer spacing. A spacing of 2 x 2 m with one seedling per pit, on raised beds with good drainage, is standard in Tamil Nadu. Planting is done at the onset of the monsoon or with assured irrigation.

Training and pruning

Papaya is not pruned. Weak, diseased and male plants are removed, and the sex ratio is adjusted to about one male plant for every 10-15 female plants in dioecious types (in gynodioecious types all plants bear fruit). Side shoots are removed and the plant is staked if it leans.

Nutrient management

Papaya responds well to heavy manuring. A common dose is about 200-250 g N, 200-250 g P2O5 and 200-250 g K2O per plant per year, applied in monthly splits, plus about 10-20 kg farmyard manure. Boron and zinc sprays correct the common deficiencies. The crop is sensitive to fertiliser burn, so nutrients are applied in small, frequent doses.

Irrigation and fertigation

Papaya needs regular irrigation, especially at flowering and fruit development; drip irrigation with fertigation gives the best growth and yield. Water shortage causes flower drop and small fruit; waterlogging causes root and stem rot. Good drainage is essential during the monsoon.

Weed management

Weeds are controlled by hand weeding, mulching and, in the early stage, intercropping with short-duration crops. Mulching with organic matter conserves moisture and keeps the collar region dry.

Special horticultural practices and growth regulators

The main practices are the removal of male plants in dioecious varieties, the removal of side shoots, and, where fruit set is poor, hand pollination. Growth regulators are used to increase yield: GA3 (100-200 ppm) sprayed at flowering increases fruit set and size; NAA reduces flower drop. De-blossoming of the first flowers is sometimes practised to strengthen the plant.

Important disorders

Papaya disorders include 'little leaf' and 'leaf curl' caused by boron deficiency, fruit distortion, and the 'female sterility' of some seasons. The papaya ring spot virus is a serious disease, not a disorder, and is managed by growing tolerant varieties and controlling the aphid vector. Root and stem rot from waterlogging is a common cause of loss.

Pests and diseases

The main pests are the aphid that transmits the ring spot virus, the fruit fly and the mealybug. The main diseases are the papaya ring spot virus, which is the most serious, and root and stem rot from waterlogging. Management uses tolerant or resistant varieties, control of the aphid vector, raised beds with good drainage and orchard sanitation.

Maturity indices

Fruit is ready about 120-150 days after transplanting. The first indication is a yellow streak on the skin at the blossom end, accompanied by a change of latex from milky white to watery. Fruit for distant markets is harvested at this stage; fruit for local sale is left until the skin is half-yellow. TSS at harvest is about 11-14 per cent.

Harvest, yield and post-harvest

Fruit is harvested by hand or with a sharp knife, leaving a short stalk, and is handled carefully because the skin is thin. Yield is 40-75 tonnes per hectare in a year, and the crop is usually replanted after 2-3 years. Value-added products are papain, papaya pulp, juice, jam, tutti-frutti from the raw fruit, and dried and candied products.

Economics note

Papaya gives the quickest return of all fruit crops, bearing fruit within a year, but the orchard is short-lived (2-3 years) and disease pressure (ring spot virus) is high. Profit depends on a good market for the fruit and on the papain and processed-fruit markets. It is well suited to smallholdings and to intercropping with coconut and arecanut.

2.5 Guava (Psidium guajava)

Origin and importance

Guava is native to tropical America and is one of the hardiest and most nutritious fruit crops. It is called the poor man's apple because it is cheap, hardy and rich in vitamin C. The fruit is eaten fresh and used for jelly, jam, cheese, juice and nectar; it is also a good source of pectin.

Area and production

India is the largest producer of guava in the world, with about 0.35 million hectares and about 5.3 million tonnes (2023-24 estimate). Uttar Pradesh, Madhya Pradesh, Maharashtra, Andhra Pradesh, Karnataka, Tamil Nadu and Bihar are the leading states. In Tamil Nadu the crop is grown in Coimbatore, Dindigul, Madurai, Salem and Kanyakumari districts.

Climate and soil

Guava grows from the tropics to the sub-tropics and tolerates both drought and some frost, though it is damaged by severe frost. It grows on almost any soil, including poor and alkaline soils, but does best on a well-drained loam with pH 6.0-7.5. It is more tolerant of salinity and waterlogging than most fruit crops.

Varieties

Important varieties are Allahabad Safeda, Lucknow-49 (Sardar), Banarasi, Apple Colour, Chittidar, Arka Mridula, Arka Kiran and Arka Amulya. TNAU has released CO 1, CO 2, CO 3 and the PKM series, selected for yield, fruit size and quality. Lucknow-49 (Sardar) is the leading commercial variety because of its large, high-quality fruit.

Propagation

Guava is propagated by seed for rootstock and by air-layering, stooling and grafting for planting. Air-layering (gootee) is the common commercial method and gives true-to-type plants that bear in 2-3 years; it is done in the rainy season. Seedling plants are vigorous but variable and bear late. Grafting (wedge or patch) on seedling rootstock is also used.

Planting density and spacing

Traditional spacing is 6 x 6 m (about 278 plants per hectare); high-density planting at 3 x 3 m or 2 x 2 m is now recommended for early and higher yield. Pits of 45-60 cm cube are filled with farmyard manure and planted at the onset of the monsoon.

Training and pruning

Guava is trained to an open-centre or modified-leader system. The tree is kept low and open by removing the central stem and crowding branches. Regular pruning after harvest removes water sprouts and dead wood and encourages new shoots, on which the next crop is borne. In high-density planting, pruning is essential to control size. The crop can be made to bear two crops a year (bahar regulation) by pruning at the right time.

Nutrient management

A bearing tree is given about 250-600 g N, 150-300 g P2O5 and 250-600 g K2O per year depending on age, split into two applications, plus farmyard manure. Zinc and boron sprays correct the common deficiencies, and the crop responds well to organic manuring.

Irrigation and fertigation

Guava is fairly drought-tolerant but responds well to irrigation, especially at flowering and fruit development. Drip irrigation with fertigation gives earlier and higher yield. The crop is not irrigated during the rest period if bahar regulation is practised.

Weed management

Weeds are controlled by clean cultivation, mulching and intercropping in the young orchard. Guava can also be grown as a filler crop in mango and sapota orchards.

Special horticultural practices and growth regulators

Bahar regulation - the deliberate timing of flowering by pruning and water management - is the key technique, allowing the crop to be harvested in the most remunerative season. Paclobutrazol and ethephon are used to regulate flowering and to hasten fruit maturity. Bagging of fruit protects against fruit fly and birds.

Important disorders

Guava disorders include the 'hardness' of fruit from boron deficiency, fruit drop from poor nutrition or water stress, and zinc deficiency causing small leaves. Wilt (Fusarium) is a disease, not a disorder, but it is a major problem and is managed by using resistant rootstocks and clean planting material.

Pests and diseases

The main pests are the fruit fly, the tea mosquito bug and the mealybug. The main diseases are wilt (Fusarium), which is the most serious, and anthracnose. Management uses resistant rootstocks, clean planting material, fruit bagging and orchard sanitation.

Maturity indices

Fruit is ready about 90-120 days after flowering. The skin turns from dark green to light green or yellow, the surface becomes smooth, and the fruit softens slightly. TSS at harvest is about 10-13 per cent. Fruit for distant markets is harvested slightly before full ripeness.

Harvest, yield and post-harvest

Fruit is harvested by hand, and the crop is picked two or three times in a season. Yield is 20-40 tonnes per hectare in a well-managed orchard. Value-added products are guava jelly, jam, cheese, juice, nectar and toffee. Guava is a good source of pectin for the jelly industry.

Economics note

Guava is hardy, early-bearing, and can be grown on marginal land, so it is popular with smallholders. It gives two crops a year in many regions, and the fruit has a ready market fresh and for processing. Profit depends on controlling wilt and on bahar regulation to catch the best price.

2.6 Sapota / Sapodilla (Manilkara zapota)

Origin and importance

Sapota is native to Central America and the West Indies and was introduced to India several centuries ago. The fruit is sweet and eaten fresh; it is also used for milk shakes and preserves. The tree yields chicle (latex) used in chewing gum, and the wood is used in furniture. Sapota is valued for its high yield and its ability to grow in the dry tropics.

Area and production

India is the largest producer of sapota in the world, with about 73,000 hectares and about 0.88 million tonnes (2023-24 estimate). Gujarat, Maharashtra, Karnataka, Andhra Pradesh and Tamil Nadu are the leading states. In Tamil Nadu the main belts are Kanyakumari, Tirunelveli, Coimbatore and Dindigul.

Climate and soil

Sapota needs a warm, dry tropical climate and is sensitive to frost; it tolerates drought well. It grows on a range of soils, including poor and rocky soils, but does best on a deep, well-drained loam with pH 6.0-8.0. It is more tolerant of saline and dry soils than most fruit crops.

Varieties

Important varieties are Kalippatti (Kalipatti), Cricket Ball, Oval, Dwarapudi, Kirthabarthi, PKM 1, PKM 2, PKM 3, PKM 4 and PKM 5. The TNAU PKM selections from Periyakulam are high-yielding and are widely grown in Tamil Nadu; Kalippatti is prized for fruit quality, and PKM 3 for high yield.

Table 2.7 Important sapota varieties and their salient features

Variety	Source / belt	Salient features
Kalippatti	Maharashtra / Karnataka	Small, very sweet, high quality; leading table variety
Cricket Ball	Maharashtra	Large round fruit, good quality
Oval	Andhra Pradesh	Oval fruit, good yield
Dwarapudi	Andhra Pradesh	High yield, good quality
PKM 1 to PKM 5	TNAU, Periyakulam	High-yielding selections for Tamil Nadu; PKM 3 noted for yield

Source: TNAU package of practices and standard texts.

Propagation and rootstock

Sapota is propagated by grafting on a seedling rootstock (raised from seed of a vigorous local or Kalippatti type). Softwood grafting and air-layering are the common methods; grafting gives plants that bear in 3-4 years, whereas seedlings take 7-8 years. Air-layered plants are true to type and bear early but have a weaker root system.

Planting density and spacing

The usual spacing is 8 x 8 m (about 156 plants per hectare); closer spacing of 5 x 5 m or 6 x 6 m is used with pruning. Pits of 60 cm cube are filled with farmyard manure and planted at the onset of the monsoon.

Training and pruning

Sapota is trained to a single stem with 3-4 well-spaced main branches (modified leader). The tree is kept low and open by removing the central crowded branches after a few years. Regular pruning removes dead wood and water sprouts; the tree is prone to a heavy canopy, so thinning improves light and yield.

Nutrient management

A bearing tree is given about 500 g N, 250 g P2O5 and 500 g K2O per year at maturity, split into two applications, plus farmyard manure. Iron and zinc sprays correct the chlorosis common on high-pH soils.

Irrigation and fertigation

Sapota tolerates drought but responds to irrigation, especially at flowering and fruit development. Drip irrigation with fertigation is recommended. Excess water is avoided because the tree is sensitive to waterlogging.

Weed management

Weeds are controlled by clean cultivation and mulching. The tree is often grown as a filler or border crop in coconut gardens.

Special horticultural practices and growth regulators

The main practices are the removal of the terminal growing point to encourage branching and, where needed, the regulation of flowering. Growth regulators are not widely used; NAA is occasionally used to check fruit drop.

Important disorders

Sapota disorders include the leaf chlorosis of iron and zinc deficiency on high-pH soils, fruit drop, and the 'corky' texture of some fruit. The tree is prone to a heavy fruit set, which should be thinned to improve fruit size.

Pests and diseases

The main pests are the leaf webber, the bud borer and the chiku moth. The main diseases are sooty mould and leaf spot. Management uses pruning for an open canopy, timely sprays and orchard sanitation.

Maturity indices

Fruit is ready about 120-150 days after flowering. The brown scurf on the skin is shed and the skin becomes smooth and light brown, the fruit softens slightly and the latex content falls. TSS at harvest is about 18-22 per cent. Fruit for distant markets is harvested slightly firm.

Harvest, yield and post-harvest

Fruit is harvested by hand with a short stalk, taking care not to bruise it because latex stains the skin. Yield is 15-25 tonnes per hectare. Value-added products are sapota milk shake, preserve, jam and dried fruit. The fruit is a good source of energy and iron.

Economics note

Sapota is a hardy, long-lived tree (60-100 years) that yields well in dry areas where other fruit crops fail. It bears for many years with little care, and the fruit has a steady market. It is a good choice for rainfed and marginal lands of the southern districts.

2.7 Citrus - acid lime and sweet orange

The citrus group (family Rutaceae) includes acid lime, sweet orange, mandarin, lemon and grapefruit. This section covers acid lime (Citrus aurantifolia) and sweet orange (Citrus sinensis), the two most important in Tamil Nadu. Citrus fruit is rich in vitamin C and is used fresh and for juice, squash and the flavouring industry.

Origin and importance

Citrus originated in South-East Asia and China. India is a major producer and the world's leading producer of lime and lemon. Acid lime is grown for the fresh market and for juice; sweet orange (mosambi) is a popular fresh and juice fruit. The peel and essential oils are valuable by-products.

Area and production

India grows citrus on about 1.1 million hectares and produces about 14.6 million tonnes (2023-24 estimate), of which mandarin, sweet orange and lime/lemon are the main groups. Andhra Pradesh, Maharashtra, Madhya Pradesh, Gujarat, Rajasthan, Karnataka and Tamil Nadu are the leading states. In Tamil Nadu, acid lime is grown in Theni, Dindigul, Tirunelveli and Kanyakumari, and sweet orange in the Shevaroy and other hills.

Climate and soil

Citrus grows from the sub-tropics to the tropics; acid lime is more tropical and sweet orange more sub-tropical. A dry, sunny climate with moderate rainfall suits the crop; high humidity favours the diseases canker and gummosis. Deep, well-drained sandy loam or loam with pH 6.0-7.5 is ideal. Citrus is very sensitive to waterlogging and to salinity, so drainage and good water quality are essential.

Varieties

For acid lime the leading varieties are Kagzi lime, Pramalini, Vikram, Sai Sarbati, Balaji and Chakradhar, and the TNAU releases PKM 1 and PKM 2, selected for high yield and fruit quality. For sweet orange the leading varieties are Sathgudi, Batavian (Mosambi), Blood Red, Jaffa, Pineapple and Hamlin; Sathgudi is the main variety of the south.

Table 2.8 Important citrus varieties and their salient features

Crop	Variety	Salient features
Acid lime	Kagzi lime	Small, juicy, thin-skinned; leading commercial lime
Acid lime	PKM 1, PKM 2 (TNAU)	High yield, good quality; recommended for Tamil Nadu
Acid lime	Pramalini, Vikram, Sai Sarbati, Balaji	Improved selections for yield and juice content
Sweet orange	Sathgudi	Main southern variety; sweet, juicy, good colour
Sweet orange	Batavian (Mosambi)	Sweet, low acid; leading juice and table variety
Sweet orange	Blood Red	Red flesh; rich colour
Sweet orange	Jaffa, Hamlin, Pineapple	Early and mid-season types grown in the hills

Source: TNAU package of practices and standard texts.

Propagation and rootstock

Citrus is propagated by budding (T-budding or shield budding) on a seedling rootstock. Rough lemon is the common stock for vigour and drought tolerance; Rangpur lime for lime soils; Trifoliate orange for Phytophthora and cold tolerance; and Troyer citrange for tolerance to the greening disease. The rootstock strongly affects tree size, fruit quality and disease tolerance, so its choice is critical. Certified, disease-free budwood is used to avoid viruses.

Planting density and spacing

Acid lime is planted at 4 x 4 m or 5 x 5 m (about 400-625 plants per hectare); sweet orange at 6 x 6 m (about 278 plants per hectare). Pits of 60 cm cube are filled with farmyard manure and planted at the onset of the monsoon. High-density planting with pruning is used for early returns.

Training and pruning

Citrus is trained to an open-centre (vase) system to admit light and air and reduce disease. The young tree is headed at about 45-60 cm and 3-4 well-spaced branches are selected. Regular pruning removes water sprouts, dead wood, criss-cross branches and diseased wood. The tree is kept low for easy harvest. Bahar regulation (timing of flowering by withholding water and pruning) is practised in the Deccan for acid lime.

Nutrient management

A bearing citrus tree is given about 400-600 g N, 200-300 g P2O5 and 300-500 g K2O per year, split into two or three applications, plus farmyard manure. Zinc, iron and magnesium sprays correct the common deficiencies. Citrus is sensitive to excess nitrogen, which causes poor fruit quality.

Irrigation and fertigation

Citrus needs regular irrigation but is very sensitive to waterlogging; drip irrigation with fertigation is ideal. Water shortage causes fruit drop and small fruit, while excess water causes root rot and gummosis. The crop is irrigated every 5-10 days depending on soil and season.

Weed management

Weeds are controlled by clean cultivation of the basin, mulching and intercropping in the young orchard. The citrus basin should be kept free of weeds and the collar region dry to reduce gummosis.

Special horticultural practices and growth regulators

Bahar regulation is the key practice for acid lime: flowering is timed by withholding water and pruning so that the fruit is harvested in the most profitable season. Growth regulators - GA3 to improve fruit set and size, NAA to check fruit drop, and 2,4-D at low concentration to retain fruit - are used as directed. Fruit is sometimes bagged to protect against fruit fly and sunburn.

Important disorders

Citrus disorders include fruit cracking (from irregular water supply and boron deficiency), granulation or 'dry fruit' in which the juice sacs become hard and dry, mottling and deficiency of zinc, iron and magnesium, and 'fruit drop'. The serious diseases are citrus canker, gummosis (Phytophthora) and the greening (HLB) disease, managed by using tolerant rootstocks, clean budwood and vector control.

Pests and diseases

The main pests are the citrus leaf miner, the fruit fly, the aphid and mites. The main diseases are citrus canker, gummosis (Phytophthora) and the greening (HLB) disease. These are managed by using tolerant rootstocks such as Trifoliate orange and Troyer citrange, clean budwood, good drainage and vector control.

Maturity indices

Acid lime is ready about 150-180 days after flowering and sweet orange about 210-240 days. Fruit is harvested when the colour changes to yellow or orange, the surface is smooth and glossy, the juice content exceeds 40-45 per cent and the TSS:acid ratio is at the acceptable level for the variety. Colour is not always a reliable index in warm climates, so the TSS:acid ratio is used.

Harvest, yield and post-harvest

Fruit is harvested by clipping with a short stalk, never by pulling, to avoid tearing the skin. Yield is about 20-30 tonnes per hectare for acid lime and 15-25 tonnes per hectare for sweet orange. Post-harvest operations are washing, waxing, grading and packing; degreening with ethylene is used for early-season sweet orange. Value-added products are juice, squash, cordial, jam, marmalade, peel oil and pectin.

Economics note

Acid lime gives a steady income because it bears almost round the year in warm climates, and the fruit keeps well and has a strong market for juice. Sweet orange is more seasonal but fetches a good price. Profit depends on controlling canker and gummosis, on good water quality and on bahar regulation to catch the best price. Citrus is a long-term crop with an orchard life of 25-30 years.

2.8 Grape (Vitis vinifera and other species)

Origin and importance

Grape originated in the region between the Black Sea and the Caspian Sea and has been cultivated for thousands of years. It is grown for the table, for raisins and for wine. In India the crop is grown mainly for the table and for raisins, and the raisin industry is concentrated in Maharashtra. Grape is the leading fresh fruit export of India.

Area and production

India grows grapes on about 0.17 million hectares and produces about 3.9 million tonnes (2023-24 estimate). Maharashtra leads with about 67 per cent of the output, followed by Karnataka (about 28 per cent), Tamil Nadu and Mizoram. In Tamil Nadu the crop is grown in the Theni, Dindigul and Coimbatore districts and in the Cumbum valley. India exported about 344,000 tonnes of fresh grapes worth about Rs 3,460 crore in 2023-24.

Climate and soil

Grape needs a warm, dry climate with a long, sunny growing season and a dry period at ripening to prevent disease; rain at ripening causes berry splitting and rot. It is grown from the tropics to the sub-tropics. Deep, well-drained sandy loam with pH 6.5-8.0 is ideal; the vine is sensitive to waterlogging and to salinity. In the tropics the vine is pruned twice a year to give two crops.

Varieties

Important varieties are Thompson Seedless, Sonaka, Tas-A-Ganesh, Sharad Seedless, Manik Chaman, Flame Seedless, Anab-e-Shahi, Bangalore Blue, Gulabi, Panneer Drakshai, Pusa Navrang and Arka varieties. TNAU has released grape varieties including CO 1 and the PKM selections. Seedless varieties (Thompson Seedless and its selections) dominate the table and export trade; Bangalore Blue and Gulabi are grown for the local market.

Table 2.9 Important grape varieties and their salient features

Variety	Type	Salient features
Thompson Seedless	White seedless	Leading table and export variety; used for raisins
Sonaka, Tas-A-Ganesh, Manik Chaman	Selections of Thompson Seedless	Improved berry size and colour for export
Sharad Seedless	Black seedless	Bold black berries; good for export
Flame Seedless	Red seedless	Red colour; grown for export
Anab-e-Shahi	White seeded	Bold berries, good quality, local market
Bangalore Blue, Gulabi	Seeded	Hardy, local market and juice
Panneer Drakshai	Seeded	Tamil Nadu local type; hardy

Source: TNAU package of practices and standard texts.

Propagation and rootstock

Grape is propagated by hardwood cuttings taken from a healthy, high-yielding vine; the cutting is rooted in the nursery or planted directly. Grafted vines on resistant rootstocks (Dog Ridge, 110-R, 1103-P) are used on saline or nematode-infested soils. Rooted cuttings give a uniform vineyard and bear from the third year.

Planting density and spacing

Spacing depends on the training system. For the bower system the vines are planted 3 x 3 m or wider (about 1,000-1,600 vines per hectare); for the Y or telephone trellis, closer spacing of 2 x 2 m or 2 x 3 m is used. Pits are filled with farmyard manure and the vine is trained on a trellis or bower.

Training and pruning

Grape is trained on a bower, a kniffin, a telephone trellis or a Y trellis. The vine is pruned twice a year. In the tropics, forward (foundation) pruning is done after the harvest to build the frame, cutting the canes back to 1-2 buds; back (winter) pruning keeps 4-8 bud canes for the fruiting crop. Pruning balances the vine's vigour against the crop and is central to grape production. After pruning, a dormancy-breaking chemical (hydrogen cyanamide or a thiourea-benzyl adenine mixture) is applied to force uniform sprouting.

Nutrient management

A vineyard is given about 200-400 g N, 200-500 g P2O5 and 200-400 g K2O per vine per year in splits, plus farmyard manure. Potassium is important for berry size, colour and quality. Magnesium, zinc and boron sprays correct the common deficiencies. The dose is adjusted to the crop load and the growth stage.

Irrigation and fertigation

Grape is irrigated by drip, with the water withheld before pruning to induce dormancy and resumed after pruning. The crop is very sensitive to water stress and to excess water; drip fertigation gives precise control of both water and nutrients. Rain at flowering and ripening is the main risk, so the crop is timed to ripen in the dry season.

Weed management

Weeds are controlled by clean cultivation, mulching and the use of cover crops between rows. The vineyard floor is kept clean to reduce humidity and disease.

Special horticultural practices and growth regulators

Grape uses more growth regulators than any other fruit crop. GA3 is applied to elongate the bunch and increase berry size in seedless varieties; the application is timed to the flowering and berry stages. Cytokinins (like CPPU) improve berry size. Dormancy-breaking chemicals (hydrogen cyanamide) force uniform sprouting after pruning. Canopy management - shoot thinning, leaf removal and bunch thinning - improves light and air and berry quality. Girdling (ringing) is used to increase berry size and advance maturity.

Important disorders

Grape disorders include berry cracking (from rain or irregular irrigation), 'water berry' (soft, watery berries), berry shatter, and the nutrient disorders of magnesium, zinc and boron deficiency. The serious diseases are downy mildew, powdery mildew, anthracnose and the bacterial disease of Pierce's disease; these are managed by timely fungicide sprays and by a dry ripening season.

Pests and diseases

The main pests are the flea beetle, thrips, mealybug and mites. The main diseases are downy mildew, powdery mildew and anthracnose, managed by timely fungicide sprays and by a dry ripening season.

Maturity indices

Grape is ready about 100-140 days after pruning. The berries attain full size and colour, become easy to separate from the stalk, and the total soluble solids (Brix) reach about 16-22 per cent with the right TSS:acid ratio. Brix is the standard harvest index for the table and export trade. Berries are harvested when the Brix and colour are at the level required by the market.

Harvest, yield and post-harvest

Bunches are harvested by clipping with a short piece of cane, and are handled carefully because the berries bruise easily. Yield is about 20-30 tonnes per hectare in a well-managed vineyard, with some orchards taking two crops a year. Post-harvest operations are grading, bunch trimming, packing in ventilated boxes with SO2 pads (for export), pre-cooling and cold storage. Value-added products are raisins, wine, juice and grape seed oil.

Economics note

Grape is a high-investment, high-return crop; the trellis, drip system and the two-pruning schedule make it the most technically demanding fruit crop in this course. Returns depend on berry quality and on the export market, where colour and Brix decide price. Raisin production gives an alternative and often stable income in Maharashtra.

Exam focus

Key points to remember

1. Plant population per hectare = 10,000 / area occupied per plant in square metres.
2. Square gives the fewest plants; hexagonal gives about 15 per cent more; quincunx roughly doubles the square number; rectangular lies between square and hexagonal.
3. A rootstock controls vigour, precocity, size, fruit quality and tolerance of soil, drought, salinity and disease.
4. Training builds the young tree; pruning regulates the bearing tree. The three training systems are central leader, open centre (vase) and modified leader.
5. Grape is pruned twice a year - forward (foundation) pruning to build the frame and back (winter) pruning for the fruiting crop - followed by a dormancy-breaking chemical.
6. High-density planting raises yield per hectare but requires dwarfing, pruning and growth regulators (paclobutrazol in mango) to keep the canopy small.
7. Fertigation is the application of soluble fertilisers through drip; it raises nutrient-use efficiency and is followed by a clear-water flush.
8. TNAU varieties to remember: mango CO 1-CO 5 and PKM 1-2; banana CO 1-2 and PKM 1-5; papaya CO 1-7 and PKM 1-2; guava CO 1-3; sapota PKM 1-5; acid lime PKM 1-2; grape CO 1 and PKM selections.
9. Mango maturity: shoulders fill out, colour lightens, specific gravity above 1.0-1.02, TSS 12-15 per cent, sap less milky.
10. Grape maturity is judged mainly by Brix (16-22 per cent) and berry colour; the crop is timed to ripen in the dry season.

Short-answer questions with model answers

Q1. Calculate the number of mango plants in one hectare at 5 x 5 m spacing.

Model answer. Plants per hectare = 10,000 / (5 x 5) = 10,000 / 25 = 400 plants.

Q2. Why is a rootstock important in fruit crops?

Model answer. The rootstock influences tree vigour, size, precocity, fruit quality and the plant's tolerance of drought, salinity, waterlogging and soil-borne disease. For example, rough lemon gives vigour and drought tolerance in citrus, while Trifoliate orange gives tolerance to Phytophthora and cold.

Q3. Name two growth regulators used in grape and state their purpose.

Model answer. GA3 is used to elongate the bunch and increase berry size in seedless varieties, and hydrogen cyanamide is a dormancy-breaking chemical applied after pruning to force uniform sprouting.

Q4. Give the spacing and plant population for banana at 1.8 x 1.8 m.

Model answer. Plants per hectare = 10,000 / (1.8 x 1.8) = 10,000 / 3.24 = about 3,086 plants.

Long-answer questions with model answer outlines

Q1. Describe the systems of planting with diagrams and calculate the plant population for each at 6 x 6 m.

Outline. Describe square, rectangular, quincunx, hexagonal and contour systems with the layouts. Square 6x6 = 278; rectangular 6x4 = 416; hexagonal 6 m = about 320; quincunx 6x6 =

about 555. State the advantages and uses of each: square is simple; rectangular suits machinery; hexagonal uses land best; quincunx accommodates fillers; contour suits slopes.

Q2. Explain the production technology of mango from planting to harvest.

Outline. Climate and soil; varieties (Banganapalli, Neelam, Alphonso, Totapuri, TNAU CO and PKM series); propagation by veneer or stone grafting on polyembryonic rootstock; spacing 10x10 m or high-density 5x5 m; training (modified leader) and annual light pruning; nutrient dose 500 g N, 250 g P2O5, 500 g K2O split into two applications; irrigation and withholding before flowering; paclobutrazol for regular flowering; NAA to check drop; disorders (alternate bearing, malformation, spongy tissue); maturity indices; harvest and yield 8-25 t/ha; value addition (pulp, pickle).

Q3. Compare the cultivation of grape and banana, bringing out the main differences in propagation, pruning and harvest.

Outline. Grape: propagated by hardwood cuttings, trained on a bower or trellis, pruned twice a year with a dormancy-breaking spray, harvested by Brix, high investment and export-oriented, 20-30 t/ha. Banana: propagated by suckers or tissue culture, desuckered rather than pruned, harvested on finger plumpness and colour, 40-60 t/ha, ripened in chambers. Bring out differences in climate, spacing and economics.

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UNIT III | Sub-tropical and Temperate Fruit Crops, Minor Fruits and Nuts

This unit covers the sub-tropical crops pineapple, pomegranate and jackfruit; the temperate crops apple, pear, peach, plum and strawberry; the minor fruits ber, custard apple, amla and dragon fruit; and the nut crops almond, pistachio and walnut. Section 3.1 sets out the techniques that distinguish these crops from the tropical crops of Unit II, after which each crop is treated in the same structure used throughout this book.

3.1 Common techniques for sub-tropical, temperate and minor fruit crops

The general framework of Unit II - climate and soil, propagation and rootstock, planting systems, training and pruning, water and nutrient management, weed management, growth regulation, disorders, maturity and harvest - applies here as well. Three features make the crops of this unit different. First, temperate crops need a period of winter cold, called the chilling requirement, without which they do not break dormancy and flower properly. Second, deciduous crops are pruned in the dormant season, not after harvest. Third, many minor fruits and nuts are hardy, rainfed or dryland crops with a lower level of management.

Table 3.1 Chilling requirement of temperate and nut crops

Crop	Approximate chilling requirement (hours below 7 deg C)	Low-chill option for warm areas
Apple	1000-1600 (Royal Delicious about 1200)	Low-chill types: Anna, Dorsett Golden, Vered, Michael
Pear	600-1500 depending on type	Patharnakh, Gola and Le Conte are low-chill sub-tropical pears
Peach	300-1200	Flordasun, Flordaprince, Prabhat, Partap (150-300 h)
Plum	700-1000	Satluj Purple, Kala Amritsari are low-chill
Almond	200-700	Most varieties need a mild winter only
Walnut	400-1500	Some selections tolerate mild winters
Pistachio	600-1000	Kerman needs a definite winter
Strawberry	Very low; day-neutral and low-chill types grow in the plains	Chandler, Winter Dawn, Sweet Charlie

Source: standard temperate-fruit texts. Figures are indicative and vary with variety and locality.

Table 3.2 Training and pruning systems for the crops of this unit

Crop	Training system	Pruning practice
Pineapple	No training; suckers and slips form the crop	Removal of slips and suckers to divert food to the fruit
Pomegranate	Open centre / multi-stem bush	Annual removal of water sprouts, dead wood and suckers; thinning of crowded shoots
Jackfruit	Modified leader	Light pruning of dead wood and crowded branches; tree kept low
Apple	Central leader or modified central leader	Dormant pruning to open the centre and remove crowding; renewal of fruiting wood
Pear	Central leader (European) or open centre	Dormant pruning; spur thinning
Peach	Open centre (vase)	Severe dormant pruning; peach bears on one-year-old wood, so renewal is essential
Plum	Open centre or modified leader	Dormant pruning; thinning of overcrowded branches
Strawberry	No training; runner removal and mulching	Removal of runners and old leaves; renovation after harvest
Ber	Open centre / modified leader	Annual heading back to force new fruiting shoots; very responsive to pruning
Custard apple	Modified leader	Light pruning after harvest; tipping to encourage branching
Amla	Modified leader	Removal of water sprouts and dead wood; heading back of long shoots
Dragon fruit	Trained on a concrete post with a ring (trellis)	Removal of excess and downward shoots to keep the canopy manageable
Almond	Open centre / modified leader	Dormant pruning; removal of crowding and dead wood
Walnut	Central leader (modified)	Light dormant pruning; walnut bleeds, so heavy pruning is avoided

Source: standard texts and TNAU/ICAR material.

Diagram 3.1 Apple training (central leader) and strawberry runner propagation

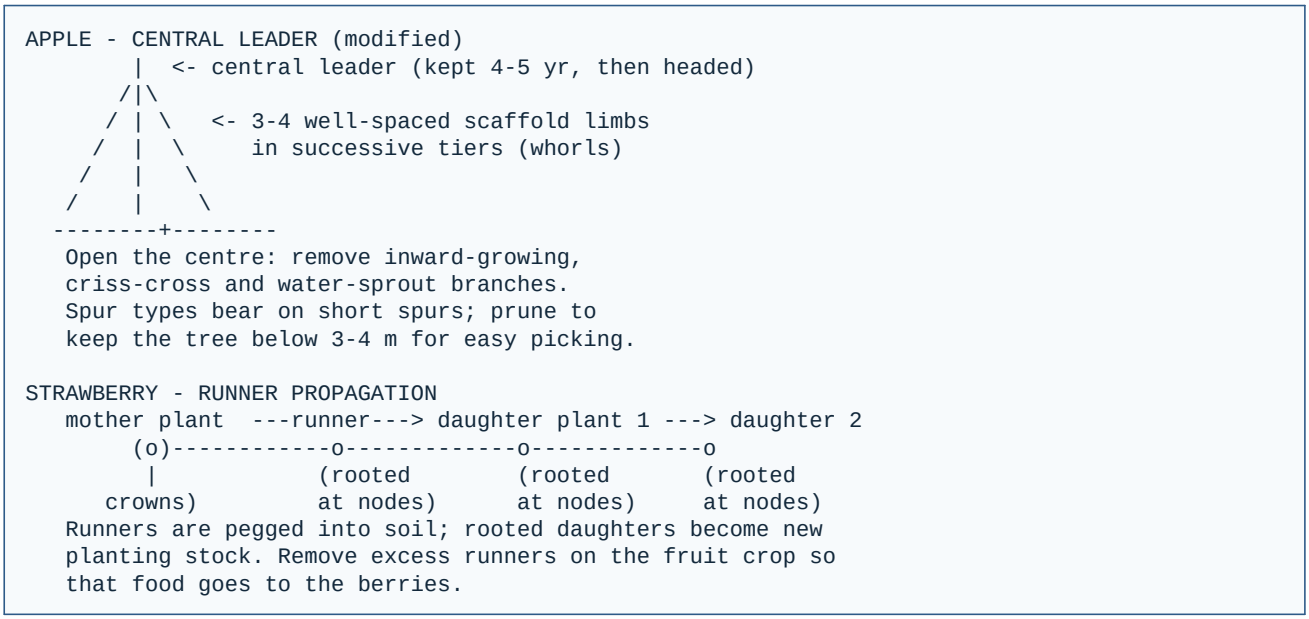


Figure 3.1 Apple central-leader training and strawberry runner propagation.

3.2 Pineapple (Ananas comosus)

Origin and importance

Pineapple is native to tropical South America and is grown throughout the tropics. It is a herbaceous perennial with a rosette of spiny or smooth leaves and a composite (multiple) fruit formed by the fusion of many berries. The fruit is eaten fresh and canned, and the juice is a popular beverage; the fibre yields a soft cloth, and the plant yields the enzyme bromelain.

Area and production

India grows pineapple on about 0.11 million hectares and produces about 1.76 million tonnes (2023-24 estimate). Assam, West Bengal, Tripura, Kerala, Karnataka and Meghalaya are the leading states. In Tamil Nadu the crop is grown in the hills of Kanyakumari, the Nilgiris and the Shevaroys.

Climate and soil

Pineapple needs a warm, humid climate, ideally 22-32 deg C, and tolerates light shade. It is sensitive to frost and to waterlogging. A well-drained sandy loam rich in organic matter with pH 5.5-6.5 is ideal; the crop is slightly acid-loving. Good drainage is essential because the plant rots in waterlogged soil.

Varieties

Important varieties are Kew (the main canning type), Queen, Mauritius, Jaldhup, Lakhat, Amritha and the TNAU selections. Kew is the leading commercial and canning variety; Queen is sweet and used fresh; Amritha is a TNAU release with good fruit quality. The smooth-leaved types are preferred where leaf handling is a problem.

Propagation and planting material

Pineapple is propagated vegetatively from suckers, slips, crowns and stumps. Suckers are the best planting material; slips and crowns are also used. Planting material is treated with a fungicide and dried before planting. The crop is planted in a double-row or paired-row system for easy harvest and good light use.

Planting density and spacing

Spacing depends on the system. A common double-row system uses rows 60-90 cm apart with plants 30-45 cm apart, giving about 40,000-55,000 plants per hectare. Closer spacing gives smaller fruit; wider spacing gives larger fruit. The crop is planted at the onset of the monsoon on raised beds.

Training, pruning and special practices

There is no training. The important practices are the removal of slips and suckers (de-suckering) so that food is diverted to the fruit, and the removal of the crown or the application of growth regulators to adjust fruit size. Flower induction is done with ethephon or calcium carbide (though carbide is discouraged) to synchronise the crop and to induce off-season flowering.

Nutrient, water and weed management

Pineapple is a moderate feeder; a dose of about 8-12 g N, 3-4 g P_2O_5 and 8-12 g K_2O per plant per year is applied in splits, plus organic matter. Potassium improves fruit quality. The crop tolerates short dry spells but needs irrigation for good fruit size; drip irrigation and fertigation are used. Weeds are controlled by mulching with black polythene or organic matter and by hand weeding.

Important disorders

Pineapple disorders include 'multiple crown' and 'fasciation' from excess nitrogen or unbalanced nutrition, fruitlet core rot and the 'sunburn' of exposed fruit. 'Black heart' occurs when fruit is stored at low temperature. These are managed by balanced nutrition, shade and correct storage.

Maturity indices, harvest and yield

The fruit is ready about 15-20 months after planting (12-18 months after flowering). Indices are the change of the shell colour from green to yellow at the base, the flattening of the 'eyes', a hollow sound when tapped and total soluble solids of about 12-15 per cent. Yield is 40-60 tonnes per hectare. Post-harvest, the fruit is graded and packed; value-added products are canned slices, juice, squash, jam and bromelain.

Economics note

Pineapple is a relatively low-cost crop that gives a good return, and it can be grown as an intercrop in coconut and rubber plantations in the early years. The main risks are the long crop duration, the need for good drainage and the market for the fresh fruit, which is offset by the strong canning and juice market.

3.3 Pomegranate (Punica granatum)

Origin and importance

Pomegranate is native to Iran and the region around the Himalayas and is one of the oldest cultivated fruits. It is a hardy, drought-tolerant shrub or small tree. The fruit is eaten fresh and the arils (seeds with juicy coat) are used in juice, syrup and the flavouring industry; the rind and bark yield tannins and the fruit is valued in traditional medicine. Pomegranate is a major export fruit of India.

Area and production

India grows pomegranate on about 0.26 million hectares and produces about 2.9 million tonnes (2023-24 estimate). Maharashtra, Karnataka, Gujarat, Andhra Pradesh, Telangana, Rajasthan and Tamil Nadu are the leading states. In Tamil Nadu the crop is grown in Dindigul, Theni, Coimbatore and Salem districts, and it is an important export crop to the Gulf and Europe.

Climate and soil

Pomegranate grows from the tropics to the sub-tropics and tolerates drought, heat and mild frost. It fruits best in a dry climate with a hot summer and cool winter; rain at flowering and ripening causes fruit cracking and disease. It grows on a wide range of soils but prefers a deep, well-drained loam with pH 6.5-7.5. It tolerates slightly alkaline and saline soils better than most fruit crops.

Varieties

Important varieties are Ganesh, Bhagwa, Mridula, Ruby, Arakta, Kesar, Phule Arakta, Phule Bhagwa and the TNAU selections (CO 1 and others). Ganesh is a leading soft-seeded variety, and Bhagwa is prized for its deep red, soft arils and high export demand. Soft-seeded varieties fetch a premium.

Table 3.3 Important pomegranate varieties and their salient features

Variety	Source / belt	Salient features
Ganesh	Maharashtra (MPKV)	Soft-seeded, high yield, good quality; leading variety
Bhagwa	Maharashtra	Deep red arils, soft seed, high export demand
Mridula, Ruby, Arakta	Maharashtra	Improved selections for colour and yield
Kesar	Gujarat	Distinct flavour; good for juice
TNAU CO 1 and selections	TNAU	Selections for Tamil Nadu conditions

Source: standard texts and TNAU/ICAR material.

Propagation and rootstock

Pomegranate is propagated by hardwood cuttings of 20-30 cm length taken from a healthy, high-yielding mother plant, planted directly or rooted in the nursery. Air-layering and grafting are also used. Cuttings are the standard method because they are cheap, easy and true to type. Hardened, disease-free cuttings from a certified mother plant are used to avoid bacterial blight.

Planting density and spacing

The usual spacing is 3 x 3 m or 4 x 4 m (about 625-1,100 plants per hectare), with high-density planting at 2.5 x 3 m or closer under intensive management. The crop is trained as a multi-stem bush, so the plant is headed low and 3-4 stems are allowed to develop.

Training and pruning

Pomegranate is trained to a multi-stem bush or open centre. Regular pruning removes water sprouts, root suckers, dead wood and crowded branches, and the plant is kept low and open. Since the crop bears on new shoots, pruning encourages fresh fruiting wood. Bahar regulation (timing of flowering by water and pruning) is practised to catch the best market, especially the export window.

Nutrient management

A bearing plant is given about 200-400 g N, 100-200 g P2O5 and 200-400 g K2O per year in splits, plus farmyard manure. Potassium improves aril colour and quality. Zinc and boron sprays correct the common deficiencies. Balanced nutrition reduces fruit cracking.

Irrigation and fertigation

Pomegranate is drought-tolerant but needs irrigation at flowering and fruit development for good yield; drip fertigation gives the best control and reduces cracking. Irregular or heavy irrigation after a dry spell causes fruit cracking, so the water supply must be steady. The crop is very sensitive to waterlogging.

Weed management

Weeds are controlled by clean cultivation, mulching and intercropping in the young orchard. Mulching also moderates soil moisture and reduces cracking.

Special horticultural practices and growth regulators

Bahar regulation is the key technique, allowing flowering to be timed to the export season. Growth regulators - NAA and GA3 - are used to improve fruit set and size, and ethephon to hasten maturity. Fruit bagging protects against fruit borer and sunburn. Pruning and water management together control the bahar.

Important disorders

Pomegranate disorders include fruit cracking (from irregular water and boron deficiency), the 'internal breakdown' of arils, and sunburn. Bacterial blight (Xanthomonas) is a serious disease, not a disorder, and is managed by using clean planting material and resistant varieties such as Bhagwa. Fruit borer is a major pest controlled by bagging and timely sprays.

Maturity indices, harvest and yield

The fruit is ready about 120-150 days after flowering. Indices are the change of the rind colour from green to the characteristic colour (red or yellow) and the metallic sound when tapped. Yield is 15-25 tonnes per hectare. Post-harvest, the fruit is graded and packed for export; value-added products are juice, syrup, wine, anardana (dried aril) and rind tannin. Pomegranate is a leading fresh export fruit to the Gulf and Europe.

Economics note

Pomegranate is a high-value crop with strong export demand, and the soft-seeded Bhagwa variety earns a premium. It gives returns from the third year and has a long life. The main risks are bacterial blight and fruit cracking, both of which are managed by clean planting material and steady irrigation.

3.4 Jackfruit (Artocarpus heterophyllus)

Origin and importance

Jackfruit is native to the Western Ghats of India and is the largest tree-borne fruit in the world, often weighing 10-25 kg. The ripe fruit is eaten fresh and used for jam and preserves; the tender unripe fruit is used as a vegetable; the seeds are eaten roasted; and the wood is valued for furniture. Jackfruit is being promoted as a nutritious and climate-resilient crop.

Area and production

India grows jackfruit on about 0.19 million hectares and produces about 3.3 million tonnes (2023-24 estimate). Kerala, Karnataka, Tamil Nadu, West Bengal, Bihar and Maharashtra are the leading states. In Tamil Nadu it is grown in the Western Ghats districts, Coimbatore, Kanyakumari and the Nilgiris.

Climate and soil

Jackfruit is a tropical tree and grows well in warm, humid conditions with 25-35 deg C; it tolerates light frost only as a mature tree. It grows on a range of soils but prefers a deep, well-drained loam with pH 6.0-7.5. Young trees need protection from frost and drought.

Varieties

Important types are the local selections, and named varieties include Singapore Jack, Black Gold, Coorg Honey Dew, Muttom Varikka, Palur, and the TNAU selections. Varieties are grouped as 'koozha' (soft flesh) and 'varikka' (firm flesh); the firm-fleshed varikka types are preferred for the table and export.

Propagation and planting

Jackfruit is propagated by seed for rootstock and by grafting (softwood or epicotyl), air-layering and budding for planting. Grafted plants are true to type and bear in 3-4 years, whereas seedlings are variable and bear in 7-8 years. Planting is at 8 x 8 m (about 156 plants per hectare) or wider for the large-growing types, in pits filled with farmyard manure.

Training, pruning and management

The tree is trained to a modified-leader system and kept low for easy harvest. Light pruning removes dead wood and crowded branches. The tree needs little care once established. Irrigation at flowering and fruit development improves fruit set and size. Manuring is with farmyard manure and a moderate fertiliser dose. Fruit is sometimes bagged to protect it from fruit borer and bats.

Disorders, maturity and harvest

Disorders include poor fruit set, fruit drop and the 'choking' of young plants. Fruit is ready about 120-180 days after flowering. Indices are the change of the spine colour from green to yellow-brown, a strong aroma and a hollow sound when tapped. Yield is 15-30 tonnes per hectare for good trees. Value-added products are chips (kerala), canned young fruit, jam, and jackfruit flour from the seeds.

Economics note

Jackfruit is a hardy, low-input tree that gives a large crop with little care, and the young-fruit and chips markets are growing. It is a good choice for homesteads and for intercropping in coconut and arecanut gardens. The main constraint is the bulky fruit, which is difficult to transport and market.

3.5 Apple (Malus domestica)

Origin and importance

Apple is native to Central Asia and the Caucasus and is the leading temperate fruit of the world and of the Indian hills. The fruit is eaten fresh and used for juice, jam and cider; India is a large producer but also imports apples because hill production is seasonal. Apple growing supports the economy of Himachal Pradesh, Uttarakhand, Jammu and Kashmir and Arunachal Pradesh.

Area and production

India grows apple on about 0.30 million hectares and produces about 2.7 million tonnes (2023-24 estimate). Jammu and Kashmir, Himachal Pradesh, Uttarakhand and Arunachal Pradesh are the leading states. Apple is a minor crop in the Tamil Nadu hills, where a few low-chill types are tried.

Climate and soil

Apple needs a cool temperate climate with a distinct winter chilling of about 1000-1600 hours below 7 deg C, a mild summer and a dry, sunny ripening period. It grows best at 1,500-2,700 m elevation. A deep, well-drained loam with pH 5.5-6.5 and good organic matter is ideal. The tree is sensitive to waterlogging and to late frost.

Varieties

The leading varieties are Royal Delicious, Red Delicious, Golden Delicious, McIntosh, Granny Smith, Ambri (a Kashmiri variety), Scarlet Spur, Vance Delicious, Top Red and Oregon Spur. Low-chill varieties such as Anna, Dorsett Golden and Vered allow apple growing in warmer areas. The delicious group dominates the Indian market.

Propagation and rootstock

Apple is propagated by grafting or budding on a clonal or seedling rootstock. Dwarfing rootstocks such as M9, M7, MM106 and MM111 control tree size and allow high-density planting; seedling stocks give vigorous standard trees. The rootstock strongly affects tree size, precocity and anchorage, so the choice depends on the planting density and soil.

Planting density and spacing

Standard trees are planted at 6 x 6 m (about 278 plants per hectare) or wider; high-density planting on dwarfing rootstocks uses 3 x 2 m or 4 x 2 m (about 1,250-1,600 plants per hectare) with a trellis and training to a spindle or vertical-axis system. Pits are filled with farmyard manure and the plant is set at the same depth as in the nursery.

Training and pruning

Apple is trained to a central leader or modified central leader. Dormant pruning opens the centre, removes crowding and dead wood and renews fruiting wood; the tree is kept low for easy picking. Since apple bears mostly on spurs (short fruiting shoots) and on one-year wood depending on the variety, pruning is adjusted to the bearing habit. Fruit thinning improves size and reduces biennial bearing.

Nutrient, water and weed management

A bearing apple tree is given about 500-700 g N, 250-350 g P2O5 and 500-700 g K2O per year, plus farmyard manure, adjusted to tree age and crop load. Calcium sprays reduce bitter pit and improve storage. Irrigation is by drip or basin, especially at fruit development; water shortage reduces fruit size. Weeds are controlled by clean cultivation and mulching.

Growth regulators, disorders and maturity

Growth regulators - GA3 for fruit size, ethephon for ripening and colour, and growth retardants (daminozide, paclobutrazol) to control vigour - are used in intensive orchards. Disorders include bitter pit (calcium deficiency), cork spot, water core and biennial bearing. Fruit is harvested on the basis of days from full bloom, starch-iodine index, firmness, colour and TSS. Yield is 15-30 tonnes per hectare depending on density.

Post-harvest and economics

Apple is stored in cold stores and controlled-atmosphere stores for many months; value-added products are juice, concentrate, jam, sauce and vinegar. Apple is a high-investment crop with a good return in the hills, but the Indian crop is seasonal and competes with imports. High-density orchards on dwarfing rootstocks have transformed the economics of apple growing in Himachal Pradesh.

3.6 Pear (Pyrus species)

Origin and importance

Pear is native to Europe and Asia and is grown in both temperate and sub-tropical regions. The fruit is eaten fresh and used for juice, jam and preserve; it is a good source of dietary fibre and vitamin C. Pears are grouped as European (soft, buttery, bell-shaped) and Oriental or Asian (crisp, round) types, and the two groups differ in climate needs and in the market they serve.

Area and production

India grows pear on about 35,000 hectares and produces about 0.28 million tonnes (2023-24 estimate), mainly in Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Punjab and Arunachal Pradesh. Sub-tropical pears are also grown in the plains of Punjab and Haryana, and a few low-chill types are tried in the Tamil Nadu hills.

Climate and soil

European pears need a temperate climate with a winter chilling of about 600-1,500 hours, while Oriental and low-chill types such as Patharnakh, Gola and Le Conte grow in the sub-tropics. Pear grows on a range of soils but prefers a deep, well-drained loam with pH 6.0-7.5. It tolerates heavier and moister soil than apple but is still sensitive to waterlogging and to late frost at flowering.

Varieties

European varieties include Bartlett (Williams), Flemish Beauty, Bosc, Anjou and Conference; sub-tropical varieties include Patharnakh, Gola, Le Conte, Kieffer, Punjab Beauty and Punjab Nectar. Patharnakh and Gola are the leading sub-tropical pears of north India, and Bartlett is the leading European variety for processing.

Propagation and rootstock

Pear is propagated by grafting or budding on a rootstock. Seedling pear stocks are used for vigour; quince rootstock (Quince A) dwarfs the tree and gives early bearing; and Kainth (Pyrus pashia) is a vigorous, hardy stock for the hills. Rootstock choice controls vigour, precocity and, in some cases, compatibility, so it is matched to the planting density and soil.

Planting density and spacing

European pears are planted at 6 x 6 m or wider (about 278 plants per hectare) and sub-tropical pears at 5 x 5 m to 7 x 7 m (about 204-400 plants per hectare). Dwarfing rootstocks allow closer spacing and high-density planting. Pits are filled with farmyard manure and the plant is set at the nursery depth.

Training and pruning

Pear is trained to a central leader (European types) or an open centre, and pruned in the dormant season. Pruning removes crowding and dead wood, renews fruiting spurs and keeps the tree low for picking. Pear bears mostly on spurs, so spur thinning and careful renewal are important; some varieties are self-incompatible and need a polliniser.

Nutrient, water and weed management

A bearing tree is given about 400-600 g N, 200-300 g P_2O_5 and 400-600 g K_2O per year in splits, plus farmyard manure. Irrigation is by basin or drip, especially at fruit development, and water shortage reduces fruit size. Weeds are controlled by clean cultivation and mulching. Fruit is thinned to improve size.

Disorders, maturity and harvest

Disorders include the 'cork spot' and 'bitter pit' of calcium deficiency, and 'water core'. Fruit is harvested before it ripens fully, when the skin colour changes and the fruit separates easily from the spur, and is ripened off the tree. Yield is 15-25 tonnes per hectare. Value-added products are juice, nectar, jam, preserve and canned halves.

Economics note

Pear gives a good return in the hills and, with sub-tropical varieties, in the plains. It is a long-lived tree with a steady market, and the processing market for juice and nectar adds value. The main risks are late frost, fire blight and the need for pollinators.

3.7 Peach (Prunus persica)

Origin and importance

Peach is native to China and is a favourite temperate stone fruit. The fruit is eaten fresh and used for canning, juice and jam; the kernel yields oil. Peach trees are short-lived but bear early and heavily, and low-chill varieties grow in the sub-tropics and even in the warmer hills of the south, which makes peach an important early-return temperate fruit.

Area and production

India grows peach on about 16,000 hectares and produces about 0.11 million tonnes (2023-24 estimate), mainly in Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Punjab and Arunachal Pradesh. Low-chill varieties are also grown in the Deccan plateau and in the north-eastern hills.

Climate and soil

Peach needs a distinct winter chilling (300-1,200 hours, much less for low-chill types) and a warm, dry summer. It grows on a range of soils but prefers a light, well-drained sandy loam with pH 6.0-7.0; it is sensitive to waterlogging and to heavy clay soil. Late frost at flowering damages the crop, and rain at ripening causes fruit rot.

Varieties

Important varieties are Flordasun, Flordaprince, Shan-e-Punjab, Earli Grande, Elberta, July Elberta, Redhaven, Sharbati, Prabhat and Partap. The low-chill Florida and Indian varieties (Flordasun, Prabhat, Partap) are grown in the sub-tropics, while Elberta and Redhaven are grown in the temperate hills.

Propagation and rootstock

Peach is propagated by budding or grafting on a seedling rootstock of peach, or on wild peach, almond or plum stocks. The rootstock controls vigour, soil tolerance and sometimes resistance to nematodes. Budding is done in the nursery and the plant is ready in a year.

Planting density and spacing

Peach is planted at 4 x 4 m to 6 x 6 m (about 278-625 plants per hectare) and trained to an open-centre (vase) system, because peach needs full light for fruit colour and wood renewal. Pits are filled with farmyard manure and the plant is set at the nursery depth.

Training and pruning

Peach is trained to an open centre and pruned severely every year in the dormant season. Since peach bears on one-year-old wood, a part of the previous year's shoots is retained for the crop and the rest is removed to force new growth; without this renewal the fruiting wood moves outwards and upwards and the crop becomes small and poorly coloured. Fruit is thinned to improve size.

Nutrient, water and weed management

A bearing tree is given about 400-600 g N, 200-300 g P_2O_5 and 400-600 g K_2O per year in splits, plus farmyard manure; peach needs good nitrogen for the vigorous new growth that carries the next crop. Irrigation is by basin or drip, especially at fruit development. Weeds are controlled by clean cultivation and mulching.

Disorders, maturity and harvest

Disorders include the 'split pit' and 'gumming' from water or nutrient stress. Fruit is harvested when the skin colour develops, the fruit softens slightly and the TSS is at the required level. Yield is 10-20 tonnes per hectare. Value-added products are canned halves, juice, jam, nectar and dried fruit.

Economics note

Peach is a short-lived but early-bearing and high-return crop, well suited to smallholdings and to the early market. The main risks are late frost, fruit fly and the short storage life of the fruit, which makes prompt marketing or processing essential.

3.8 Plum (Prunus species)

Origin and importance

Plum is native to Europe and Asia and is grown in both temperate and sub-tropical regions. The fruit is eaten fresh and used for jam, jelly, sauce and the dried product (prune); it is a hardy stone fruit that bears early and gives a good return.

Area and production

India grows plum on about 18,000 hectares and produces about 70,000 tonnes (2023-24 estimate), mainly in Himachal Pradesh, Jammu and Kashmir, Uttarakhand, Punjab and the north-eastern hills.

Climate and soil

Plum needs a winter chilling of about 700-1,000 hours, less for the Japanese and low-chill varieties. It grows on a range of soils but prefers a deep, well-drained loam with pH 6.0-7.5 and is more tolerant of heavy and moist soil than peach.

Varieties

Important varieties are Santa Rosa, Satluj Purple, Kala Amritsari, Kelsey, Frontier, Methley, Alucha Yellow and Titron. Santa Rosa is the leading commercial variety; Satluj Purple and Kala Amritsari are low-chill types for the sub-tropics.

Propagation, planting and training

Plum is propagated by budding or grafting on a seedling rootstock of plum, peach or wild plum. It is planted at 5 x 5 m to 6 x 6 m (about 278-400 plants per hectare) and trained to an open centre or modified leader. Dormant pruning removes crowding and renews fruiting wood, since plum bears on spurs and on one-year wood. Most varieties need a polliniser.

Management, harvest and economics

The nutrient and water management follows the general temperate-fruit pattern; a bearing tree gets about 400-600 g N, 200-300 g P_2O_5 and 400-600 g K_2O per year in splits. Fruit is harvested when the skin colour and bloom develop and the fruit softens slightly; yield is 10-20 tonnes per hectare. Value-added products are jam, jelly, sauce, squash and dried prunes. Plum is a hardy, early-bearing crop with a steady market and a good processing outlet.

3.9 Strawberry (Fragaria x ananassa)

Origin and importance

Strawberry is native to Europe and the Americas and is the most important small fruit of the world. It is a low-growing herbaceous perennial; the 'fruit' is an aggregate of many small achenes on a fleshy receptacle. The fruit is eaten fresh and used for jam, ice cream and flavouring. It is rich in vitamin C and antioxidants, and it gives a very high return per unit area.

Area and production

India grows strawberry on a small area (about 2,000 hectares) with production of about 17,000-20,000 tonnes (2023-24 estimate). Maharashtra, Himachal Pradesh, Punjab, Haryana, Delhi, West Bengal and Uttarakhand are the leading states. In Tamil Nadu the crop is grown in the Nilgiris and on a small scale in the plains with low-chill varieties.

Climate and soil

Strawberry prefers a cool climate (15-25 deg C) and is grown in the hills in summer and in the plains in winter. It is sensitive to high temperature and to heavy rain. A well-drained sandy loam rich in organic matter with pH 5.5-6.5 is ideal. Raised beds with good drainage are essential because the plant rots in waterlogged soil.

Varieties

Important varieties are Chandler, Tioga, Selva, Sweet Charlie, Winter Dawn, Camarosa, Festival, Nabila and Douglas. Chandler and Winter Dawn are widely grown for their yield and fruit quality; Sweet Charlie and Winter Dawn are low-chill types suited to the plains.

Propagation and planting

Strawberry is propagated by runners (daughter plants) produced from a healthy mother plant; the runners are pegged into the soil and the rooted daughters are used as planting stock. Micropropagation is used to produce disease-free mother plants. Planting is done on raised beds covered with black or silver-black polythene mulch, which conserves moisture, controls weeds and keeps the fruit clean.

Planting density, training and management

Plants are set in double or four-row beds at 30 x 30 cm or 30 x 20 cm, giving about 50,000-100,000 plants per hectare. There is no training; the important practices are the removal of excess runners (so that food goes to the berries) and the removal of old leaves. A nutrient dose of about 5-10 g N, 3-5 g P2O5 and 5-10 g K2O per plant per season, plus organic matter, is applied through the bed. Irrigation is by drip, which also keeps the foliage dry and reduces disease.

Disorders, maturity and harvest

Disorders include boron deficiency (fruit distortion), calcium deficiency and 'albinism'. Fruit is ready about 30-40 days after flowering; the berry is harvested when it is fully coloured and firm. Yield is 20-30 tonnes per hectare. Post-harvest, the fruit is very perishable and is cooled, graded and packed immediately; value-added products are jam, squash, ice cream and freeze-dried fruit. Strawberry gives one of the highest returns per hectare of any fruit crop, but it is labour-intensive and needs a cold chain.

Worked example 3.1 Plant population and nutrient requirement for strawberry

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Bed system: 4 rows per bed, plants 30 cm apart within the row and 30 cm
between rows; beds 1 m wide with 0.5 m path.

Step 1 - plants per square metre of bed area:
   1 plant per (0.30 x 0.30) = 1 plant per 0.09 m2 = about 11 plants/m2 of bed.

Step 2 - for 1 hectare with 60 per cent of the area in beds (6,000 m2 bed):
   6,000 x 11 = 66,000 plants per hectare (about 60,000-70,000).

Step 3 - nutrient requirement: at 8 g N, 4 g P2O5 and 8 g K2O per plant:
   Urea  (46 per cent N)  = 8 / 0.46 = 17.4 g per plant
   SSP   (16 per cent P2O5) = 4 / 0.16 = 25.0 g per plant
   MOP   (60 per cent K2O) = 8 / 0.60 = 13.3 g per plant
   For 66,000 plants: 1,148 kg urea, 1,650 kg SSP and 878 kg MOP per hectare.
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Adjust the bed proportion and the dose to local recommendations.

3.10 Ber / Indian jujube (Ziziphus mauritiana)

Origin and importance

Ber is native to India and is one of the most important minor fruits of the dry regions. It is a hardy, drought-tolerant tree that fruits in winter when few other fruits are available. The fruit is eaten fresh and used for preserve and candy, and it is rich in vitamin C and minerals. Ber is often called the poor man's fruit because it grows on poor land with little care.

Area and production

India grows ber on about 52,000 hectares and produces about 0.53 million tonnes (2023-24 estimate). Madhya Pradesh, Maharashtra, Rajasthan, Gujarat, Punjab, Haryana and Uttar Pradesh are the leading states. In Tamil Nadu it is grown in the dry tracts of the southern districts.

Climate and soil

Ber is a tree of the dry tropics and sub-tropics; it tolerates drought, heat and mild frost and grows on poor, sandy, saline and alkaline soils with pH 6.0-9.0. It fruits best in a dry climate with a cool winter. It is one of the hardiest fruit trees and needs very little water.

Varieties

Important varieties are Umran, Gola, Seb, Kaithali, Katha, Mehrun, ZG-2, Goma Kirti, Banarasi Karaka and Illaichi. Umran is the leading commercial variety; Gola and Illaichi are early and sweet; Goma Kirti is a high-yielding release.

Propagation and rootstock

Ber is propagated by budding (T or shield) on a seedling rootstock of the wild ber (Ziziphus rotundifolia), which gives a hardy, drought-tolerant plant. It is planted at 6 x 6 m or 5 x 5 m (about 278-400 plants per hectare) in pits filled with farmyard manure. Grafted plants bear in 2-3 years.

Training and pruning

Ber is trained to an open centre or modified leader. It is very responsive to pruning: annual heading back of the previous year's growth in the dormant season forces new fruiting shoots and gives a heavy, uniform crop. Without annual pruning the tree becomes tall and the crop small and uneven.

Nutrient, water and weed management

The nutrient dose is moderate; farmyard manure and about 200-300 g N, 100-200 g P2O5 and 200-300 g K2O per tree are applied. Irrigation improves fruit size, but the crop tolerates dryland conditions and is largely rainfed. Weeds are controlled by clean cultivation and mulching.

Disorders, maturity and harvest

The main disorder is the fruit drop from water or nutrient stress. Fruit is harvested when the skin turns from green to golden-yellow and the TSS rises; yield is 15-25 tonnes per hectare. Value-added products are candy, preserve, murabba and dried ber.

Economics note

Ber is a low-cost, hardy crop that gives fruit in the lean winter season, when prices are high. It suits the drylands of the Deccan and Rajasthan and needs little water, which makes it valuable in rainfed areas.

3.11 Custard apple / Sitaphal (Annona squamosa)

Origin and importance

Custard apple is native to tropical America and is grown throughout the tropics and sub-tropics. The fruit is an aggregate of many segments and is sweet and aromatic; it is eaten fresh and used for ice cream and flavouring. It is a hardy, drought-tolerant tree that fruits in the dry season.

Area and production

India grows custard apple on about 52,000 hectares and produces about 0.51 million tonnes (2023-24 estimate). Maharashtra, Gujarat, Andhra Pradesh, Madhya Pradesh, Karnataka and Tamil Nadu are the leading states. In Tamil Nadu it is grown in the dry tracts of the southern districts.

Climate and soil

Custard apple is a crop of the dry tropics; it tolerates drought, heat and mild frost and grows on poor, sandy and stony soils with pH 6.0-8.0. It is damaged by waterlogging and by frost at the seedling stage, and a dry spell at flowering improves fruit set.

Varieties

Important varieties are Balanagar, Mammoth, Washington, Red Sitaphal, Purple, Arka Sahan and local types. Balanagar and Mammoth are leading commercial varieties; Arka Sahan is a hybrid with good fruit size.

Propagation, planting and training

Custard apple is propagated by grafting (wedge or softwood) on a seedling rootstock of the same species or a related Annona; budding and air-layering are also used. It is planted at 5 x 5 m or 4 x 4 m (about 400-625 plants per hectare) in pits filled with farmyard manure. The tree is trained to a modified leader and kept low; light pruning after harvest and tipping of shoots encourages branching and fruiting. Grafted plants bear in 3-4 years.

Management, harvest and economics

The nutrient dose is moderate and the crop is largely rainfed, though irrigation improves yield. Fruit is harvested when the colour changes from green to light green or yellow, the segments widen and the fruit softens slightly; yield is 8-15 tonnes per hectare. Value-added products are pulp, ice cream, nectar and custard powder. Custard apple is a hardy dryland crop with a strong seasonal market and good processing potential.

3.12 Amla / Aonla / Indian gooseberry (Phyllanthus emblica)

Origin and importance

Amla is native to India and is one of the richest natural sources of vitamin C. The fruit is used in Ayurveda and in the preparation of chyawanprash, and in hair oil and shampoo; it is also used for preserve, candy and dried powder. It is a hardy, drought-tolerant tree with a large and growing market for processed products.

Area and production

India grows amla on about 0.11 million hectares and produces about 1.33 million tonnes (2023-24 estimate). Uttar Pradesh, Rajasthan, Madhya Pradesh, Gujarat, Maharashtra and Tamil Nadu are the leading states.

Climate and soil

Amla is a crop of the dry tropics and sub-tropics; it tolerates drought, heat, salinity and poor soil and grows at pH 6.5-9.0. It is damaged by waterlogging and by frost at the seedling stage, and it fruits well in a dry climate.

Varieties

Important varieties are NA-7, NA-10, Chakaiya, Kanchan, Krishna, BSR-1, Anand-1, Anand-2 and the TNAU selections. NA-7 and NA-10 are high-yielding and are the leading commercial varieties; Chakaiya is valued for processing.

Propagation, planting and training

Amla is propagated by budding (patch or shield) on a seedling rootstock, or by softwood grafting. It is planted at 8 x 8 m (about 156 plants per hectare) or closer for high-density planting, in pits filled with farmyard manure. The tree is trained to a modified leader and kept low; pruning removes water sprouts, dead wood and crowding, and heading back encourages new fruiting wood. Grafted plants bear in 3-5 years.

Management, harvest and economics

The crop is largely rainfed but irrigation improves fruit size. Manuring is with farmyard manure and a moderate fertiliser dose. Fruit is harvested when the colour changes from green to light green or yellow and the fruit is full-sized; yield is 15-25 tonnes per hectare. Value-added products are preserve, candy, murabba, dried powder, juice and hair oil. Amla has a strong and growing market for processed and nutraceutical products, which makes it a good choice for dryland orchards.

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3.13 Dragon fruit / Kamalam (Hylocereus species)

Origin and importance

Dragon fruit is native to Central and South America and has become popular in India in the last decade. It is a climbing cactus with a large, colourful fruit and is valued for its antioxidants, vitamin C and dietary fibre. The crop is hardy, needs little water, gives a high return and has been renamed Kamalam in India; it is an attractive new crop for the dry and coastal districts.

Area and production

Dragon fruit is a new crop in India, with about 5,000 hectares and about 55,000 tonnes (2023-24 estimate), and it is expanding rapidly. Karnataka, Kerala, Tamil Nadu, Andhra Pradesh, Maharashtra and Gujarat are the leading states. In Tamil Nadu it is grown in the dry and coastal districts.

Climate and soil

Dragon fruit is a tropical to sub-tropical crop; it tolerates heat and short dry spells but is damaged by frost and by waterlogging. It grows on a wide range of soils, including poor and sandy soils, but prefers a well-drained soil with pH 6.0-7.5. It needs full sunlight and a support to climb.

Species and varieties

The main species are Hylocereus undatus (white flesh, red skin), Hylocereus polyrhizus (red flesh) and Selenicereus megalanthus (yellow skin, white flesh). Named selections and local types are grown; the red-fleshed and yellow types fetch a premium. There is no standard TNAU variety, and planting material is propagated from selected mother plants.

Propagation, planting and training

Dragon fruit is propagated by stem cuttings of 30-50 cm taken from a healthy mother plant and rooted in the nursery. The plant is trained on a concrete post with a ring at the top (a trellis or T-post), at a spacing of 2 x 2 m or 3 x 2 m (about 1,600-2,500 plants per hectare). Rooted cuttings bear in 1-2 years. The shoots climb the post and spread over the ring; excess and downward-growing shoots are removed to keep the canopy manageable and to expose the fruit to light.

Management, harvest and economics

The crop needs little water but responds to irrigation at flowering and fruiting; it is sensitive to waterlogging. Manuring is with farmyard manure and a light fertiliser dose. Fruit is harvested when the skin colour is fully red or yellow and the fruit softens slightly; yield is 20-30 tonnes per hectare. Value-added products are juice, jam, wine, dried fruit and the natural red colour from the flesh. Dragon fruit gives a high return per hectare and suits smallholders, but the market for the fresh fruit is still developing.

3.14 Almond (Prunus dulcis)

Origin and importance

Almond is native to Central and South-West Asia and is grown in the temperate hills. The 'nut' is the kernel of a drupe; it is a rich source of protein, fat, vitamin E and minerals, and is eaten raw, roasted and as a flavouring. India is a large consumer but a small producer and imports most of its almonds, so the crop has a large import-substitution potential.

Area and production

India grows almond on about 10,000 hectares and produces about 12,000 tonnes (2023-24 estimate), mainly in Jammu and Kashmir, Himachal Pradesh and Uttarakhand. Production is far below demand.

Climate and soil

Almond needs a temperate climate with a mild winter (chilling 200-700 hours), a warm spring and a dry summer; late frost at flowering is the main risk because almond flowers very early. It grows on a range of soils but prefers a deep, well-drained loam with pH 6.0-8.0 and is drought-tolerant once established.

Varieties and rootstock

Important varieties are Nonpareil, Ne Plus Ultra, Mission, Drake, IXL, Waris, Merced and Parbat. Nonpareil and Mission are the leading California-type varieties; IXL, Waris and Parbat are grown in Kashmir. Almond is propagated by budding or grafting on a seedling rootstock of almond, peach or plum; peach rootstock gives vigour and adapts to a range of soils.

Planting, training and management

Almond is planted at 6 x 6 m (about 278 plants per hectare) and trained to an open centre. Since most varieties are self-incompatible, polliniser varieties are planted in every third row. Dormant pruning removes crowding and dead wood. The crop is largely rainfed in the hills but needs irrigation at kernel development for good size. Harvest is judged by the splitting of the hull and the drying of the shell; yield is 1-2 tonnes of kernel per hectare. Value-added products are the kernel, oil and almond butter.

3.15 Pistachio (Pistacia vera)

Origin and importance

Pistachio is native to Central and West Asia and is a minor crop in India, where the climate of the few suitable areas is marginal. The kernel is a popular dry fruit, rich in fat, protein and antioxidants. India is a very large consumer and imports almost all of its pistachio, so any domestic production has a ready market.

Area and production

Pistachio is grown on a very small area in India, in the cold desert tracts of Jammu and Kashmir, Himachal Pradesh and Ladakh; reliable all-India area and production figures are not separately reported and the crop is grown mainly on an experimental scale.

Climate, soil and varieties

Pistachio needs a climate with a cold winter (chilling 600-1,000 hours), a long hot dry summer and low humidity; it does not tolerate humidity or heavy rain at ripening. It is very drought- and salt-tolerant and grows on poor, stony, calcareous soils with pH 7.0-8.5. Kerman is the leading variety, with Peters as a polliniser; Chico and other selections are also grown. It is propagated by budding or grafting on a rootstock of Pistacia integerrima or Pistacia atlantica, which gives drought and salt tolerance.

Planting, harvest and economics

Pistachio is planted at 6 x 6 m or wider and trained to an open centre or modified leader. It is dioecious, so polliniser trees are planted. The crop is largely rainfed but irrigation improves yield. Harvest is judged by the splitting of the shell and the change of hull colour; the nuts are hulled, dried and roasted. Yield is low (0.5-1.5 tonnes per hectare) and the crop takes several years to bear, which is why it is grown only on a small scale in India. Its main value is as a high-priced dry fruit with a large and assured domestic demand.

3.16 Walnut (Juglans regia)

Origin and importance

Walnut is native to Central Asia and is grown in the temperate hills. The kernel is rich in fat (especially omega-3 fatty acids), protein and antioxidants and is eaten raw and used in confectionery; the shell and wood are also useful. India is a significant producer in the Himalayan states and a large consumer.

Area and production

India grows walnut on about 97,000 hectares and produces about 0.32 million tonnes (2023-24 estimate), mainly in Jammu and Kashmir, Himachal Pradesh, Uttarakhand and Arunachal Pradesh.

Climate, soil and varieties

Walnut needs a temperate climate with a definite winter chilling (400-1,500 hours), a warm summer and a dry ripening period; late frost at flowering damages the crop. It grows on a range of soils but prefers a deep, well-drained loam with pH 6.0-7.5 and good organic matter, and it is sensitive to waterlogging. Important varieties are Lake English, Opex Culch, Sulaiman, Gobind, Hamdan, JKSK, and the California types Payne, Franquette, Eureka, Hartley, Chandler and Howard. Franquette and Payne are popular for their late flowering (which avoids frost) and good quality.

Propagation, planting and management

Walnut is propagated by budding or grafting on a seedling rootstock; walnut is difficult to graft because it bleeds, so grafting is done carefully at the right season. It is planted at 8 x 8 m or wider (about 156 plants per hectare) and trained to a modified central leader. Pruning is light, because heavy pruning causes bleeding. The crop is largely rainfed but irrigation at kernel development improves size and quality. Harvest is judged by the splitting of the green husk and the drying of the shell; yield is 2-4 tonnes per hectare. Value-added products are the kernel, walnut oil and shell products.

Exam focus

Key points to remember

1. Temperate fruit crops need a period of winter cold - the chilling requirement - to break dormancy and flower; low-chill varieties extend these crops to warm areas.
2. Apple is trained to a central leader; peach and plum to an open centre; deciduous crops are pruned in the dormant season.
3. Peach bears on one-year-old wood, so severe annual dormant pruning is essential to renew fruiting wood.
4. High-density planting on dwarfing rootstocks (M9, MM106 for apple) has transformed the economics of temperate orchards.
5. Pomegranate is trained as a multi-stem bush; bahar regulation is used to time the crop to the export window; Bhagwa is the leading soft-seeded export variety.
6. Strawberry is propagated by runners and grown on polythene-mulched raised beds; the berry is an aggregate of achenes on a fleshy receptacle.
7. Ber and amla are hardy dryland crops; ber is very responsive to annual heading-back pruning.
8. Dragon fruit is a climbing cactus trained on a post and ring; the red-fleshed and yellow types fetch a premium.
9. Most almond varieties are self-incompatible and need polliniser varieties; almond flowers very early and is at risk from late frost.
10. India is a small producer but a very large consumer of almond and pistachio, so both crops have import-substitution potential.

Short-answer questions with model answers

Q1. What is chilling requirement and why does it matter?

Model answer. Chilling requirement is the number of hours below about 7 deg C that a temperate fruit tree needs to break dormancy and flower normally. If it is not satisfied, the tree flowers erratically or not at all. It decides the elevation and region suitable for a crop, and low-chill varieties allow apple, peach and plum to be grown in warmer areas.

Q2. Why is peach pruned severely every year?

Model answer. Peach bears its fruit on one-year-old wood. Unless a part of the previous year's shoots is removed and new growth is forced every year, the fruiting wood moves outwards and upwards and the crop becomes small and poorly coloured. Hence severe dormant pruning renews the fruiting wood and keeps the tree open to light.

Q3. Name two leading varieties each of pomegranate and strawberry.

Model answer. Pomegranate: Ganesh (soft-seeded, high yield) and Bhagwa (deep red arils, export quality). Strawberry: Chandler (high yield) and Winter Dawn (low-chill, suited to the plains).

Q4. How is dragon fruit trained, and at what spacing is it planted?

Model answer. Dragon fruit is trained on a concrete post with a ring at the top; the shoots climb the post and spread over the ring. It is planted at 2 x 2 m or 3 x 2 m (about 1,600-2,500 plants per hectare).

Long-answer questions with model answer outlines

Q1. Compare apple and peach cultivation with respect to climate, training, pruning and harvest.

Outline. Climate: both temperate with chilling requirement, apple 1000-1600 h, peach 300-1200 h (low-chill types less); both need a dry ripening period and are damaged by late frost. Training: apple to a central leader, peach to an open centre. Pruning: apple on spurs and one-year wood, moderate dormant pruning; peach bears on one-year wood and needs severe annual dormant pruning. Rootstock: apple on M9/MM106 clonal stocks, peach on seedling peach/almond. Harvest: apple by days from bloom, starch index and firmness; peach by skin colour, firmness and TSS. Yield and economics: apple 15-30 t/ha, peach 10-20 t/ha.

Q2. Describe the production technology of pomegranate with emphasis on bahar regulation and export quality.

Outline. Climate and soil; varieties (Ganesh, Bhagwa); propagation by hardwood cuttings from a certified mother plant; spacing 3x3 to 4x4 m; training to a multi-stem bush; annual pruning; nutrient dose with potassium for aril colour; steady drip irrigation to prevent cracking; bahar regulation by water and pruning to time the crop to the export window; NAA and GA3 for set and size; fruit bagging; disorders (cracking, internal breakdown); disease (bacterial blight) and pest (fruit borer); maturity by rind colour and sound; harvest, grading and export packing; value addition (juice, anardana).

Q3. Write short notes on (a) minor fruits of the dry regions and (b) nut crops of India.

Outline. (a) Ber, custard apple, amla and dragon fruit are hardy, drought-tolerant crops of the dry tropics and sub-tropics. They need little water, tolerate poor and saline soils, and give fruit in lean seasons (ber in winter). They are propagated by budding or grafting and respond to pruning. Value addition - candy, preserve, powder, juice - is important because the fresh market is limited. (b) Almond, pistachio and walnut are temperate nut crops. India is a small producer but a large consumer and imports most of its requirement. All three need winter chilling and a dry ripening period, are propagated by budding or grafting on seedling stocks, and are largely rainfed. Most almond varieties are self-incompatible and need pollinators. They have large import-substitution potential.

UNIT IV | Production Techniques in Palms and Plantation Crops

This unit covers the palms coconut, arecanut, oil palm and palmyrah, and the plantation crop rubber. It opens with the importance, scope and export position of the plantation crop industry and the commodity boards that support it, then sets out the common techniques - propagation, nursery management, planting systems, intercropping, multi-tier cropping and mulching - before treating each crop in the standard structure.

4.1 Importance and scope of the plantation crop industry in India

Plantation crops are perennial crops grown on a plantation scale and processed before sale. In India the group includes coconut, arecanut, oil palm, palmyrah, rubber, tea, coffee, cocoa, cashew and spices. They are grown mainly in the coastal and hill belts of the south and the north-east, provide year-round employment, support a large processing industry and earn substantial foreign exchange. Coconut, arecanut and rubber in particular support millions of smallholders, and coconut and arecanut provide wide inter-row space that can carry intercrops and so add to food security.

The scope for growth is large. Domestic and export demand for coconut products, rubber, coffee and cocoa is rising; oil palm is being expanded under the National Mission on Edible Oils to cut edible-oil imports; and value addition (virgin coconut oil, coconut water, rubber products, palm oil for biodiesel) is a growing source of income. The main constraints are the long gestation of palms, price volatility, old and senile plantations, and the need for replanting and improved planting material.

Table 4.1 Area and production of the plantation crops of this unit

Crop	Area	Production	Leading states
Coconut	about 2.1-2.3 million ha	about 21,000-22,000 million nuts (2023-24)	Kerala, Karnataka, Tamil Nadu, Andhra Pradesh
Arecanut	949,000 ha (2023-24 final)	1,411,000 t (2023-24 final)	Karnataka, Kerala, Assam, Tamil Nadu
Oil palm	about 0.35-0.40 million ha	about 0.30-0.35 million t fresh fruit bunches	Andhra Pradesh, Telangana, Karnataka, Tamil Nadu
Palmyrah	about 50,000 ha (mostly Tamil Nadu)	not separately estimated nationally	Tamil Nadu, Andhra Pradesh
Rubber	about 0.85 million ha	857,000 t (2023-24)	Kerala, Tripura, Karnataka, Tamil Nadu

Source: NHB, Coconut Development Board, Rubber Board and state estimates, 2023-24. Figures rounded; area and production bases differ between crops.

Export potential of plantation crops

India is the third largest producer of coconut in the world and exports coconut products, coir and activated carbon; it is a leading producer of arecanut and a large exporter of rubber products; and it is the sixth largest producer of natural rubber. The main export products of this group are coir and coir products, coconut-based activated carbon, rubber products and, increasingly, virgin coconut oil. India is a net importer of palm oil, which is why oil palm expansion is a national priority.

Commodity boards and their crops

A commodity board is a statutory or promotional body that supports production, research, marketing and export of a crop. The five statutory commodity boards under the Department of Commerce are the Tea, Coffee, Rubber, Spices and Tobacco Boards. Coconut, cashew and cocoa are supported by boards and directorates under the Ministry of Agriculture and Farmers Welfare.

Table 4.2 Commodity boards and directorates and the crops they support

Board / body	Crop(s)	Headquarters / note
Tea Board	Tea	Kolkata; zonal office at Coonoor, Tamil Nadu
Coffee Board	Coffee	Bengaluru; regional research station at Thandigudi, Tamil Nadu
Rubber Board	Natural rubber	Kottayam, Kerala
Spices Board	Cardamom and 52 spices	Kochi, Kerala
Tobacco Board	Tobacco	Guntur, Andhra Pradesh
Coconut Development Board	Coconut	Kochi, Kerala (Ministry of Agriculture and Farmers Welfare)
Directorate of Cashewnut and Cocoa Development (DCCD)	Cashew, cocoa	Kochi, Kerala
National Horticulture Board (NHB)	Horticulture (all fruit, plantation, spices, flowers)	Gurugram, Haryana
National Medicinal Plants Board	Medicinal and aromatic plants	New Delhi

Source: Ministry of Commerce and Industry and Ministry of Agriculture and Farmers Welfare.

4.2 Common techniques for palms and plantation crops

4.2.1 Climate and soil

Palms and plantation crops are tropical and need a warm, humid climate with a well-distributed rainfall, though rubber and oil palm tolerate a short dry spell. Deep, well-drained soil with good organic matter is needed; waterlogging is fatal to most of them, and drainage is designed before planting. Coconut and arecanut tolerate a range of soils, rubber prefers well-drained lateritic soil, and oil palm needs a deep, friable soil.

4.2.2 Propagation and nursery management

Palms and plantation crops are propagated by seed or vegetatively. Coconut and arecanut are grown from selected seednuts raised in a nursery; oil palm and rubber are grown from selected seed or budded/grafted stock raised in a polybag nursery. A good nursery is essential: seed is selected from high-yielding mother palms, the nursery is raised in a well-drained, shaded site with regular watering, and only vigorous, disease-free seedlings of the right age are planted out. Nursery management includes selection of the mother plant, collection and treatment of seed, sowing, watering, shading, hardening and grading of seedlings.

4.2.3 Planting and planting systems

Palms are planted in square, rectangular or triangular systems. For coconut the common spacings are 7.5 x 7.5 m (square, about 178 palms per hectare) and 7.5 x 7.5 m triangular (about 205 palms per hectare); arecanut is planted at 2.7 x 2.7 m (about 1,370 plants per hectare); oil palm at 9 x 9 x 9 m triangular (about 143 palms per hectare); and rubber at 5 x 5 m (about 400 trees per hectare) for budded stock. Pits are dug to the recommended size and filled with topsoil and farmyard manure before planting at the onset of the monsoon.

Worked example 4.1 Plant population of coconut for a given spacing and area

```
Given: 1 hectare = 10,000 m2.

SQUARE system at 7.5 x 7.5 m:
   Plants/ha = 10,000 / (7.5 x 7.5) = 10,000 / 56.25 = 177.8  ->  about 178 palms

TRIANGULAR (hexagonal) system at 7.5 x 7.5 m:
   Plants/ha = 10,000 / (7.5 x 7.5 x 0.866) = 10,000 / 48.71 = 205.3
   ->  about 205 palms (about 15 per cent more than square)

For a 2-hectare holding: 178 x 2 = 356 palms (square) or 205 x 2 = 410 palms
(triangular).

General rule:  Plants per hectare = 10,000 / (area per plant in m2);
for the triangular system multiply the square number by 1/0.866 (about 1.155).
```

The same method applies to any palm or plantation crop.

4.2.4 Cropping systems, intercropping and multi-tier cropping

Because palms are planted wide, a great deal of space is available between rows for other crops. Intercropping with short-duration crops, banana, pineapple, cocoa, arecanut, pepper, turmeric, ginger or fodder gives extra income until the palm bears, and continues after bearing in the wider alleys. Multi-tier cropping uses three layers - a tall palm canopy, a middle layer of shrubs or small trees, and a ground layer of herbs or climbers - to use light, water and nutrients from the whole soil profile. This is a hallmark of coconut-based farming in the coastal belt.

Diagram 4.1 Coconut palm layout and multi-tier cropping

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COCONUT LAYOUT (triangular system, 7.5 x 7.5 m)
     x       x       x       x       Palms at the corners of equilateral
        x       x       x             triangles; about 205 palms per hectare.
     x       x       x       x       A ring basin (1.8-2.0 m radius) is made
        x       x       x             around each palm for manuring.

MULTI-TIER CROPPING (coconut-based)
   TIER 1 (top)   :  Coconut palm canopy (tall)  ----  o  ----  o  ----
   TIER 2 (middle):  Arecanut / cocoa / banana /  --------o--------o----
                     small trees (partial shade)
   TIER 3 (ground):  Pineapple / turmeric /       ~~~~o~~~~~~o~~~~~~~o~
                     ginger / pepper / fodder
   Each tier uses a different part of the light and root zone, so total
   output per hectare is much higher than a pure stand of palms.

BOUNDARY / FILLER PLANTING
   x - - - - - - - - - - - - - - - - x    Boundary: Casuarina or
   |  o   o   o   o   o   o   o   o  |    Glyricidia as windbreak and
   |  o   o   o   o   o   o   o   o  |    green leaf manure.
   x - - - - - - - - - - - - - - - - x    Filler: banana, papaya.
```

Figure 4.1 Coconut planting layout and the three-tier cropping model.

4.2.5 After care, water, nutrient and weed management

After planting, the seedling is shaded, watered and protected from animals. The palm basin is kept clean and mulched. Manuring follows the recommended dose, applied in a ring 1.5-2.0 m from the trunk, in two splits at the onset of the monsoon and in the post-monsoon period. Green leaf manure from Glyricidia and organic matter are added. Irrigation is by basin or drip, especially in the dry months; drip is now common for coconut and arecanut. Weeds are controlled by clean cultivation, mulching and the cover of intercrops; husk burial and coir-pith application are special practices in coconut.

Worked example 4.2 Fertiliser dose for one coconut palm per year, split into two applications

```
Recommended dose for an adult coconut palm: 500 g N, 320 g P2O5 and 1200 g K2O.

Step 1 - convert nutrient to fertiliser:
   Urea (46 per cent N):   500 / 0.46 = 1,087 g  ->  1.09 kg urea
   SSP (16 per cent P2O5): 320 / 0.16 = 2,000 g  ->  2.00 kg single superphosphate
   MOP (60 per cent K2O): 1200 / 0.60 = 2,000 g  ->  2.00 kg muriate of potash

Step 2 - split the annual dose:
   Application 1 (June-July, with the monsoon):
       half urea (0.55 kg) + all SSP (2.00 kg) + half MOP (1.00 kg)
   Application 2 (September-October, post-monsoon):
       half urea (0.55 kg) + half MOP (1.00 kg)

Step 3 - method: place in a ring 1.5-2.0 m from the trunk, 15-20 cm deep,
cover with soil and irrigate. Add 25-50 kg farmyard manure or green leaf
manure per palm per year. For a hybrid palm the dose is higher.
```

The same three steps apply to arecanut, oil palm and rubber with their own doses.

4.2.6 Mulching and special horticultural practices

Mulching with coconut husk, coir pith, dry leaves or black polythene conserves moisture, suppresses weeds and adds organic matter; burying coconut husk in the basin is a traditional practice that also improves soil moisture. Special practices include the removal of the crown, de-suckering of arecanut and palmyrah, pruning of oil palm fronds and the harvesting of palm sap. Growth regulators are used less than in fruit crops, though auxins are used to check nut drop in coconut.

4.2.7 Pests and diseases

Palms and plantation crops suffer from a characteristic set of pests and diseases. Coconut is attacked by the rhinoceros beetle, the red palm weevil, the black-headed caterpillar and the Eriophyid mite, and by the root (wilt) disease, bud rot and stem bleeding. Arecanut suffers from koleroga (fruit rot), the yellow leaf disease, the spindle bug and the inflorescence caterpillar. Oil palm suffers from the rhinoceros beetle, the bunch moth and Ganoderma basal stem rot. Rubber suffers from abnormal leaf fall (Phytophthora), powdery mildew, Corynespora leaf fall and root disease. Management combines clean cultivation, resistant or tolerant material, biological control and, where necessary, chemical control.

4.2.8 Processing, value addition and maturity

Plantation crops are almost always processed before sale, so the harvest and processing methods matter as much as field management. Coconut is harvested for tender nuts or mature nuts, which are processed into copra, coconut oil, coir, desiccated coconut, coconut milk and virgin coconut oil. Arecanut is processed into dried 'chali' and 'kalipak' and into scented supari. Oil palm gives crude palm oil and palm kernel oil. Palmyrah gives palm sap (neera), palm sugar (jaggery), the tender fruit (nungu) and fibre. Rubber gives sheet rubber, block rubber and latex concentrate. Value addition - virgin coconut oil, coconut water, coconut chips, palm sugar, rubber products - greatly raises the return.

4.3 Coconut (Cocos nucifera)

Origin and importance

Coconut is believed to have originated in the Indo-Pacific region and is grown throughout the tropical coasts. It is called the 'tree of life' because every part is useful: the nut gives food, oil and water; the husk gives coir; the shell gives charcoal and activated carbon; the leaves give thatch and fodder; and the trunk gives timber. Coconut supports more than 12 million people in India and is the source of livelihood for millions of smallholders.

Area and production

India is the third largest producer of coconut in the world, with about 2.1-2.3 million hectares and about 21,000-22,000 million nuts (2023-24). Kerala, Karnataka, Tamil Nadu and Andhra Pradesh together account for about 89 per cent of the area and 91 per cent of production. In Tamil Nadu the main belts are Coimbatore, Tiruppur, Thanjavur, Kanyakumari, Dindigul and Vellore.

Climate and soil

Coconut needs a warm, humid climate with a mean temperature of about 27 deg C and a well-distributed rainfall of 1,000-3,000 mm; it tolerates coastal salinity and wind but not frost or prolonged drought. It grows on a wide range of soils, including sandy coastal soil, laterite and alluvial soil, with pH 5.2-8.0, provided drainage is good. It is tolerant of salinity, which makes it suited to the coastal belt.

Varieties

Coconut varieties are grouped as tall, dwarf and hybrid. The tall types - East Coast Tall (ECT), West Coast Tall (WCT), Tiptur Tall, Kappadam and Laccadive Ordinary - are hardy and long-lived; the dwarf types - Chowghat Orange Dwarf (COD), Chowghat Green Dwarf (CGD) and Malayan Dwarf - bear early and are used as parents; and the hybrids - Kerasankara (WCT x COD), Chandrasankara (COD x WCT), Chandralaksha (Laccadive Ordinary x COD), and the TNAU hybrids VHC 1, VHC 2 and VHC 3 from Veppankulam - give high yields and early bearing. Gangabondam (MYD x WCT) is a popular hybrid of the west coast.

Table 4.3 Important coconut types and their salient features

Type	Examples	Salient features
Tall	East Coast Tall, West Coast Tall, Laccadive Ordinary	Hardy, long-lived, good for copra; bear in 6-8 years
Dwarf	Chowghat Orange Dwarf, Chowghat Green Dwarf, Malayan Dwarf	Early bearing (3-4 years), used as parents for hybrids; lower yield
Hybrid	Kerasankara, Chandrasankara, Chandralaksha, TNAU VHC 1-3, Gangabondam	High yield, early bearing; recommended for commercial planting

Source: Coconut Development Board and TNAU/CPCRI material.

Propagation and nursery management

Coconut is grown from seednuts taken from high-yielding mother palms of a recommended variety. Seednuts are selected from the best bunches, sown in a nursery bed or polybag and raised for 9-12 months, with regular watering and partial shade. Only vigorous seedlings with a good collar girth, early germination and no disease are planted out. Selection of the mother palm and of the seedling is the cheapest way to raise productivity.

Planting density, systems and after care

Coconut is planted at 7.5 x 7.5 m in the square system (about 178 palms per hectare) or the triangular system (about 205 palms per hectare); wider spacing of 8 x 8 m is used in rich soil and closer spacing in poor soil. Pits of 1 x 1 x 1 m are filled with topsoil, farmyard manure and a little common salt, and planting is done at the onset of the monsoon. After care includes shading, watering, and the removal of the lower leaves.

Training, pruning and intercropping

Coconut has no training. The main operations are the removal of dry and diseased leaves and the removal of the male flowers in some cases to improve nut set. Because the palm is planted wide, intercropping and multi-tier cropping are strongly recommended (see Figure 4.1). Common intercrops are banana, pineapple, cocoa, arecanut, pepper, turmeric, ginger, groundnut and fodder grass. Coconut is also grown as a border or filler crop in other orchards.

Nutrient, water and weed management

An adult palm is given about 500 g N, 320 g P2O5 and 1,200 g K2O per year in two splits (see Worked example 4.2), plus 25-50 kg farmyard manure or green leaf manure. Common salt (sodium chloride) is applied in the basin in coastal areas. Irrigation is by basin or drip, especially in the dry months; drip irrigation with fertigation is increasingly used. Weeds are controlled by clean cultivation, mulching and the cover of intercrops; husk burial and coir-pith application are special practices.

Special practices, disorders and pests

Special practices include the burial of coconut husk in the basin, application of coir pith, and irrigation with drip. Disorders include the button shedding (nut drop) from water or nutrient stress, and the 'crown choking' of some palms. The main pests are the rhinoceros beetle, the red palm weevil, the black-headed caterpillar and the Eriophyid mite; the main diseases are the root (wilt) disease of Kerala, bud rot, stem bleeding and leaf blight (Ganoderma). Management combines clean cultivation, removal of affected palms, biological control (Metarhizium for the beetle, baculovirus for the caterpillar) and chemical control.

Maturity, harvest and yield

Tender nuts are harvested at about 6-7 months for coconut water, and mature nuts at about 11-12 months for copra. The palm bears from about the 5th-8th year (tall) or 3rd-4th year (dwarf and hybrid), and yields about 60-100 nuts per palm per year, or higher in well-managed hybrids. Nuts are harvested by climbing or with a pole knife. Post-harvest, the nuts are de-husked and the kernel is dried into copra, from which oil is extracted; the husk gives coir, and the shell gives charcoal and activated carbon. Value-added products are virgin coconut oil, coconut milk, desiccated coconut, coconut water and coconut chips.

Economics note

Coconut gives a steady income for 60-80 years and supports a large processing industry. The return per hectare is greatly increased by intercropping and multi-tier cropping, which make use of the wide inter-row space. The main risks are price volatility of copra and coconut oil, the root (wilt) disease and old and senile plantations. Replanting with hybrids and value addition are the main routes to higher income.

4.4 Arecanut / Betel nut (Areca catechu)

Origin and importance

Arecanut is native to the Malayan region and is grown in India for its seed (the 'nut'), which is chewed with betel leaf and used in religious and social customs. India is the largest producer and consumer of arecanut in the world. The crop is a tall, slender palm with a single trunk and gives a steady income for many years; the stem is used in construction and the husk for fibre.

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Area and production

India grows arecanut on about 949,000 hectares and produces about 1.41 million tonnes (2023-24 final estimate). Karnataka leads with about 10 lakh tonnes from about 6.8 lakh hectares, followed by Kerala, Assam, Tamil Nadu and the north-eastern states. In Tamil Nadu the crop is grown in the Western Ghats districts of Coimbatore, Salem and Kanyakumari.

Climate and soil

Arecanut needs a warm, humid climate with a well-distributed rainfall of 2,000-3,000 mm and a temperature of 14-36 deg C; it is damaged by frost and by drought. It grows on a range of soils but prefers a deep, well-drained loam with pH 5.0-8.0. It needs protection from wind and is often grown under the shade of taller trees.

Varieties

The main varieties are the tall types - South Kanara Local (Kasaragod), Thirthahalli and Sirsi - and the dwarf types - Hirehalli Dwarf and Mohitnagar. The CPCRI releases Mangala, Sumangala, Sreemangala, Swarnamangala and Mohitnagar, which are high-yielding and tolerant of the yellow leaf disease. TNAU and CPCRI material is recommended for new plantings.

Table 4.4 Important arecanut varieties and their salient features

Variety	Type	Salient features
South Kanara Local (Kasaragod)	Tall	Hardy, high yield; leading local type
Thirthahalli, Sirsi	Tall	Popular in Karnataka; good quality
Hirehalli Dwarf	Dwarf	Early bearing, suitable for high-density planting
Mangala, Sumangala, Sreemangala	Tall (CPCRI)	High yield, tolerant of yellow leaf disease
Mohitnagar, Swarnamangala	Tall (CPCRI)	High yield and quality

Source: CPCRI and TNAU material.

Propagation and nursery management

Arecanut is grown from seednuts taken from high-yielding mother palms. Seednuts are selected from the best bunches, sown in a nursery bed or polybag and raised for 12-18 months with regular watering and shade. Only vigorous, disease-free seedlings are planted out. The dwarf types are used for high-density planting.

Planting, spacing and after care

Arecanut is planted at 2.7 x 2.7 m (about 1,370 plants per hectare) in the square system, in pits of 60 cm cube filled with topsoil and farmyard manure, at the onset of the monsoon. After care includes shading of the young palm, watering and protection from animals. The palm is often grown under the partial shade of coconut or other trees.

Training, pruning and intercropping

Arecanut has no training. The main operations are the removal of dry leaves and the de-suckering of the side shoots. Because the palm is planted close, intercropping is limited but is possible with shade-tolerant crops such as cocoa, banana and pineapple in the early years. Arecanut is often grown as a mixed crop in coconut and rubber gardens.

Nutrient, water and weed management

An adult palm is given about 100-150 g N, 40-60 g P2O5 and 150-200 g K2O per year in two splits, plus farmyard manure or green leaf manure. Potassium is important for nut quality. Irrigation is by basin or drip, especially in the dry months; the crop is sensitive to drought. Weeds are controlled by clean cultivation, mulching and the cover of intercrops.

Disorders, pests and diseases

Arecanut suffers from koleroga (fruit rot, caused by Phytophthora), the yellow leaf disease, the spindle bug, the inflorescence caterpillar and the root grub. Koleroga is controlled by timely spraying with Bordeaux mixture and by clean cultivation; the yellow leaf disease is managed by using tolerant varieties such as Mangala and by removing affected palms. The main disorder is the button shedding from water or nutrient stress.

Maturity, harvest and yield

The palm bears from about the 5th-7th year and yields for 30-60 years. Nuts are harvested when they are fully mature - the husk turns orange-yellow and the nut is hard - usually from December to March. Yield is about 2-4 kg of cured nut per palm per year, or 1-3 tonnes per hectare. Post-harvest, the nuts are de-husked, boiled and dried (kalipak) or sun-dried (chali), and graded. Value-added products are scented supari, pan masala and the nut used in traditional medicine.

Economics note

Arecanut is a high-value crop with a strong domestic market and a long productive life. The main risks are the yellow leaf disease and koleroga, and price volatility. The crop is well suited to the humid Western Ghats and is often grown with coconut and pepper as a mixed garden.

4.5 Oil palm (Elaeis guineensis)

Origin and importance

Oil palm is native to West Africa and is the highest-yielding edible-oil crop in the world. It is a perennial palm that gives two oils - crude palm oil from the mesocarp (fruit pulp) and palm kernel oil from the kernel - used in food, soap, cosmetics and, increasingly, biodiesel. India is a very large importer of palm oil, so oil palm expansion is a national priority under the National Mission on Edible Oils - Oil Palm.

Area and production

India grows oil palm on about 0.35-0.40 million hectares and produces about 0.30-0.35 million tonnes of fresh fruit bunches, far below potential. Andhra Pradesh, Telangana, Karnataka, Tamil Nadu, Kerala, Odisha and the north-east are the main states. In Tamil Nadu the crop is grown in the southern and western districts.

Climate and soil

Oil palm needs a hot, humid tropical climate with a rainfall of about 2,000 mm or assured irrigation, a temperature of 22-32 deg C and plenty of sunshine. It is damaged by prolonged drought and by frost. It grows on a deep, well-drained loam with pH 4.0-6.0; good drainage is essential, and it does not tolerate waterlogging.

Varieties and planting material

Oil palm is grown from the Tenera hybrid, which is a cross of the thick-shelled Dura and the shell-less Pisifera. Tenera gives a high proportion of mesocarp and high oil yield. Improved Tenera hybrids and Indian selections are supplied by the ICAR-Indian Institute of Oil Palm Research (formerly the National Research Centre for Oil Palm), Pedavegi, and by state agencies. Planting material is raised in a polybag nursery from selected seed.

Propagation, nursery and planting

Oil palm is grown from seed raised in a polybag nursery for 12-14 months; only vigorous, uniform seedlings are planted. It is planted at 9 x 9 x 9 m in the triangular system (about 143 palms per hectare), in pits of 60 cm cube filled with topsoil and farmyard manure, at the onset of the monsoon. Legume cover crops are grown in the alleys.

Training, pruning and management

The main operations are the removal of the male and female inflorescences in the young palm to build vigour, and the pruning of fronds to maintain the correct leaf area. The palm is kept free of excessive fronds and the crown is maintained for easy harvest. The nutrient dose is high; a bearing palm is given about 1-2 kg N, 0.5-1 kg P2O5 and 1.5-2.5 kg K2O per year in splits, plus magnesium and boron. Irrigation is essential in the dry months; drip fertigation is used. Weeds are controlled by legume cover crops, mulching and clean cultivation.

Pests, diseases, maturity and harvest

The main pests are the rhinoceros beetle, the red palm weevil and the bunch moth; the main disease is Ganoderma basal stem rot. Management combines clean cultivation, biological control and removal of affected palms. The palm bears from about the 3rd-4th year and reaches full bearing by the 8th-10th year. Fruit bunches are harvested when they are ripe - a few loose fruits on the ground and an orange colour - every 7-10 days. Yield is 15-25 tonnes of fresh fruit bunches per hectare per year. The bunches are processed within 24 hours to extract crude palm oil and palm kernel oil.

Economics note

Oil palm is a high-investment, high-return crop with a life of 25-30 years and a steady income from the 4th year. It requires assured irrigation and prompt processing of the fruit bunches, so it is grown under contract with processing mills. Its expansion reduces India's dependence on imported palm oil, and the oil is a feedstock for biodiesel.

4.6 Palmyrah (Borassus flabellifer)

Origin and importance

Palmyrah is native to tropical Asia and is a tall, dioecious palm of the dry coastal belt. Every part is useful: the sap (neera) gives palm sugar (jaggery) and palm wine; the tender fruit (nungu) and the kernel are eaten; the young leaves are used for basketry; and the trunk is used for timber. In Tamil Nadu palmyrah is important in the southern districts and supports a traditional palm-sugar industry.

Area and production

Palmyrah is grown mainly in Tamil Nadu, Andhra Pradesh and Odisha, on about 50,000 hectares, largely as a wild or semi-wild palm; reliable all-India production figures are not separately estimated. The crop is important in the southern districts of Tamil Nadu such as Thoothukudi, Tirunelveli and Ramanathapuram.

Climate and soil

Palmyrah is a palm of the dry tropics and is very drought-tolerant; it grows on poor, sandy, saline and rocky soils and tolerates heat and wind. It needs a warm climate and does not tolerate frost. It grows well on the coastal sandy belt where few other crops succeed.

Types and propagation

There are no named varieties; local types are grown, and the palm is dioecious, so both male and female palms occur. Palmyrah is propagated by seed (the seed is sown in the field or nursery) and the seedling is transplanted. Because the palm is dioecious and slow-growing, only a small proportion of palms are tapped for sap, and selection of good tapping palms is important.

Planting and management

Palmyrah is planted at about 6 x 6 m or wider (about 278 palms per hectare) and needs little care once established. It is largely rainfed. Manuring is with farmyard manure and a light fertiliser dose. The main operation is the tapping of the inflorescence for sap, which is done by a trained tapper; tapping is done in the cool season and the sap is collected twice a day. Palms are pruned by removing dead leaves.

Maturity, harvest and yield

The palm is very slow-growing and begins to yield sap from about the 12th-15th year; the productive life is long. Yield of sap is about 1-2 litres per day per tapped palm, giving palm sugar after boiling. The tender fruit (nungu) is harvested in summer, and the mature fruit gives a kernel. Value-added products are palm sugar (jaggery), palm candy, neera, palm wine and fibre. The palm-sugar industry is an important cottage industry in southern Tamil Nadu.

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4.7 Rubber (Hevea brasiliensis)

Origin and importance

Rubber is native to the Amazon basin of South America and was introduced to India in the nineteenth century. It is grown for its latex, which is processed into natural rubber for tyres, footwear, belts, hoses and thousands of other products. India is the sixth largest producer of natural rubber in the world and the second largest consumer, so domestic production is important for the tyre industry.

Area and production

India grows rubber on about 0.85 million hectares and produced about 857,000 tonnes in 2023-24, a record. Kerala accounts for the largest share, followed by Tripura, Karnataka and Tamil Nadu (Kanyakumari district). The north-east is being expanded under the INROAD project. Consumption is higher than production, so India also imports natural rubber.

Climate and soil

Rubber needs a hot, humid equatorial climate with a rainfall of 2,000-4,000 mm, a temperature of 20-35 deg C and high humidity. It tolerates a short dry spell but not frost. It grows on a deep, well-drained lateritic or alluvial soil with pH 4.5-6.0. Waterlogging is fatal, and the crop needs good drainage.

Clones and planting material

Rubber is grown from budded stock of recommended clones. The leading Indian clone is RRII 105, released by the Rubber Research Institute of India, along with RRII 414, RRII 430, RRIM 600, GT 1, PB 235, PB 260 and PB 311. Clones differ in yield, disease tolerance and adaptation, so the right clone is chosen for the site. Budded stumps or polybag plants are raised in the nursery.

Table 4.5 Important rubber clones and their salient features

Clone	Source	Salient features
RRII 105	Rubber Research Institute of India	Most widely planted Indian clone; high yield, good adaptation
RRII 414, RRII 430	RRII	High-yielding recent releases
RRIM 600	Malaysia (RRIM)	Reliable, widely adapted, good yield
GT 1	Indonesia	Hardy, good for marginal areas
PB 235, PB 260, PB 311	Malaysia (Prang Besar)	High yield; PB 260 popular for new plantings

Source: Rubber Board and Rubber Research Institute of India material.

Nursery, planting and after care

Rubber is raised in a nursery from selected seed (for rootstock) and budded with the chosen clone; budded stumps or polybag plants of 2-3 whorls are planted out. Rubber is planted at 5 x 5 m (about 400 trees per hectare) for budded stock, or wider for seedling stock, in pits of 75 cm cube filled with topsoil and farmyard manure, at the onset of the monsoon. Legume cover crops are grown in the alleys to check erosion and add nitrogen. After care includes weeding, mulching, and the removal of side shoots and lower branches to give a clean trunk.

Training, pruning and management

The tree is trained to a single, clean trunk by removing the side shoots up to the required height; the canopy is allowed to develop. Pruning removes dead wood and crowded branches. The nutrient dose for a mature rubber tree is about 300-600 g N, 200-300 g P2O5 and 300-600 g K2O per year in splits, plus magnesium. The crop is largely rainfed but responds to irrigation in the dry season; it is sensitive to drought. Weeds and cover crops are managed by slashing and by growing legume cover crops.

Tapping, pests and diseases

Rubber is tapped for latex by making a controlled cut (the tapping cut) on the trunk, from which the latex flows into a cup. Tapping begins when the tree reaches the correct girth (about 45-50 cm at 1.2 m height, usually 6-7 years after planting) and is done on a scheduled basis, with the cut renewed each day. The main diseases are abnormal leaf fall (Phytophthora), powdery mildew, Corynespora leaf fall and root disease (Fomes and Ganoderma); the main pests are termites and the cockchafer beetle. Management combines resistant clones, fungicide spraying, clean cultivation and biological control.

Processing and value addition

Latex is processed into sheet rubber (RSS), block rubber and latex concentrate (centrifuged latex). Sheet rubber is made by coagulating the latex, rolling it into sheets and drying them in smokehouses; block rubber is made by crumbing and drying; latex concentrate is used for dipped goods. Value-added products are gloves, footwear, tyres, belts and hoses. India's rubber industry is a major employer and the crop gives a steady income for 25-30 years.

Economics note

Rubber is a long-term crop with a life of 25-30 years and tapping income from the 6th-7th year. The return depends heavily on the world price of natural rubber, which is volatile. The crop is well suited to the warm, humid Western Ghats and the north-east, and the INROAD project is expanding it in the north-east. Rubber is often grown as a mixed crop with coconut, banana and pepper.

Exam focus

Key points to remember

1. The five statutory commodity boards under the Department of Commerce are the Tea, Coffee, Rubber, Spices and Tobacco Boards; coconut, cashew and cocoa are supported by the Coconut Development Board and the DCCD.
2. Coconut plant population: square 7.5 x 7.5 m = 178 palms per hectare; triangular = about 205 palms per hectare.
3. Coconut is the 'tree of life' - the nut gives food, oil and water, the husk gives coir, the shell gives charcoal and the leaves give thatch and fodder.
4. Multi-tier cropping uses three layers - a tall palm canopy, a middle shrub layer and a ground layer - to raise total output per hectare.
5. Arecanut plant population at 2.7 x 2.7 m is about 1,370 plants per hectare; Karnataka is the leading producer.

6. Oil palm is grown from the Tenera hybrid (Dura x Pisifera) and gives crude palm oil and palm kernel oil; it is expanded under the National Mission on Edible Oils.
7. Palmyrah is a slow-growing, dioecious palm of the dry coastal belt; the sap gives palm sugar and the tender fruit is the nungu.
8. Rubber is grown from budded stock of recommended clones; RRII 105 is the leading Indian clone, and tapping begins when the tree reaches the correct girth.
9. The main coconut pests are the rhinoceros beetle, the red palm weevil, the black-headed caterpillar and the Eriophyid mite; the main diseases are root (wilt), bud rot and stem bleeding.
10. Plantation crops are almost always processed before sale, so harvest and processing methods decide the return.

Short-answer questions with model answers

Q1. Calculate the number of coconut palms in one hectare at 7.5 x 7.5 m in the square and triangular systems.

Model answer. Square: 10,000 / (7.5 x 7.5) = 10,000 / 56.25 = about 178 palms. Triangular: 10,000 / (7.5 x 7.5 x 0.866) = about 205 palms.

Q2. What is multi-tier cropping? Give an example with a coconut base.

Model answer. Multi-tier cropping is growing three or more crops of different stature in layers so that each uses a different part of the light and root zone. In a coconut garden, tier 1 is the coconut canopy, tier 2 is arecanut, cocoa or banana, and tier 3 is pineapple, turmeric, ginger or fodder. This raises total output per hectare well above a pure stand of palms.

Q3. Name the leading Indian rubber clone and state when tapping begins.

Model answer. The leading Indian clone is RRII 105, released by the Rubber Research Institute of India. Tapping begins when the tree reaches a girth of about 45-50 cm at 1.2 m height, usually 6-7 years after planting.

Q4. List the commodity boards of India and the crops they support.

Model answer. The five statutory commodity boards under the Department of Commerce are the Tea Board (tea), Coffee Board (coffee), Rubber Board (natural rubber), Spices Board (cardamom and 52 spices) and Tobacco Board (tobacco). Coconut is supported by the Coconut Development Board and cashew and cocoa by the Directorate of Cashewnut and Cocoa Development.

Long-answer questions with model answer outlines

Q1. Discuss the importance, scope and export potential of the plantation crop industry in India.

Outline. Define plantation crops and list them (coconut, arecanut, oil palm, palmyrah, rubber, tea, coffee, cocoa, cashew, spices). Importance: year-round employment, processing industry, foreign exchange, smallholder livelihoods, intercropping and food security. Statistical support: coconut 21,000-22,000 million nuts, arecanut 1.41 million tonnes, rubber 857,000 tonnes. Export: coir and coconut products, rubber products, coffee, tea. Commodity boards and their role. Scope: oil palm expansion under the National Mission on Edible Oils, value addition, replanting with hybrids. Constraints: price volatility, old plantations, diseases.

Q2. Describe the production technology of coconut from nursery to harvest.

Outline. Climate and soil; varieties (tall, dwarf, hybrid - Kerasankara, Chandrasankara, TNAU VHC 1-3); propagation from selected seednuts and nursery management; planting at 7.5 x 7.5 m square (178/ha) or triangular (205/ha); after care; intercropping and multi-tier cropping; nutrient dose 500 g N, 320 g P2O5, 1200 g K2O split into two applications plus farmyard manure and common salt; irrigation and drip; husk burial and mulching; pests (rhinoceros beetle, red palm weevil, black-headed caterpillar, Eriophyid mite) and diseases (root wilt, bud rot, stem bleeding); maturity at 11-12 months for copra and 6-7 months for tender nuts; harvest and yield 60-100 nuts per palm per year; processing into copra, oil, coir; value addition (virgin coconut oil, coconut water).

Q3. Compare the cultivation of arecanut and rubber with respect to climate, planting material, spacing and harvest.

Outline. Arecanut: warm, humid, 2,000-3,000 mm rainfall; grown from selected seednuts raised in a nursery; planted at 2.7 x 2.7 m (about 1,370/ha); bears in 5-7 years; harvested when the husk turns orange-yellow; cured into kalipak or chali. Rubber: hot, humid equatorial climate, 2,000-4,000 mm rainfall; grown from budded stock of clones (RRII 105); planted at 5 x 5 m (about 400/ha); tapped from the 6th-7th year when the girth is 45-50 cm; latex processed into sheet rubber, block rubber and latex concentrate. Bring out differences in propagation, spacing, harvest method and product.

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UNIT V | Production Techniques in Beverage Crops

This unit covers the beverage crops tea, coffee and cocoa, and the nut crop cashew, which is grouped here because it is grown and processed like a plantation crop. The common techniques are set out first, after which each crop is treated in the standard structure of climate and soil, varieties, propagation and nursery, planting and planting systems, training and pruning, nutrient and water management, harvest, processing and value addition.

5.1 Common techniques for beverage crops

Beverage crops are perennial crops grown for a processed product - made tea, coffee beans and cocoa beans - while cashew is grown for its nut. All of them are grown on a plantation scale, mostly in the hills or the coastal belt, and all need a nursery, careful planting, regular pruning or shade management, and prompt processing after harvest. The quality of the final product depends as much on the harvest and processing as on field management.

Table 5.1 Area and production of beverage crops and cashew, India

Crop	Area	Production (2023-24)	Leading states
Tea	about 0.62 million ha	1,382 million kg (made tea)	Assam, West Bengal, Tamil Nadu (Nilgiris)
Coffee	about 0.47 million ha	about 374,000 t (green coffee)	Karnataka, Kerala, Tamil Nadu
Cocoa	109,000 ha	30,000 t (dry beans)	Kerala, Karnataka, Andhra Pradesh, Tamil Nadu
Cashew	1,195,000 ha	795,000 t raw cashew nut	Maharashtra, Andhra Pradesh, Odisha, Tamil Nadu

Source: Tea Board, Coffee Board, DCCD and NHB estimates, 2023-24. Figures rounded.

India is the second largest producer of tea in the world and the second largest exporter after Kenya, exporting about 254 million kg worth about USD 826 million in 2023-24. India is the seventh largest coffee producer and a large exporter of coffee, with exports of about USD 1.29 billion in 2023-24. India is the largest producer and consumer of cashew and a major exporter of cashew kernels, and cocoa is a small but fast-growing crop grown mostly as an intercrop in coconut and arecanut gardens.

The commodity boards - the Tea Board, the Coffee Board and, for cashew and cocoa, the Directorate of Cashewnut and Cocoa Development - support research, extension, quality control and marketing. The Tea Board has a zonal office at Coonoor in Tamil Nadu and the Coffee Board a regional research station at Thandigudi, Tamil Nadu, both of which serve the Nilgiris and the Pulney and Shevaroy hills.

Diagram 5.1 Processing flow for the beverage crops and cashew

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TEA (black tea, orthodox or CTC)
  Plucking (two leaves + a bud) -> Withering (12-18 h) -> Rolling / CTC
  -> Fermentation (oxidation, 2-3 h) -> Drying (firing) -> Sorting & grading
  -> Packing.  Green tea: skip fermentation (steam/pan-fire the leaf).

COFFEE
  (a) WET / washed method (arabica, quality coffee):
      Ripe cherries -> Pulping -> Fermentation (12-24 h) -> Washing
      -> Drying (parchment) -> Hulling -> Grading -> Green beans.
  (b) DRY / natural method (robusta, some arabica):
      Ripe cherries -> Sun-drying (whole cherry) -> Hulling -> Green beans.

COCOA
  Ripe pods -> Opening & removal of beans with pulp -> Fermentation
  (5-7 days) -> Sun-drying -> Roasting -> Winnowing -> Cocoa nibs,
  cocoa mass, cocoa butter and cocoa powder.

CASHEW
  Raw cashew nut (from the apple) -> Sun-drying -> Steam / drum roasting
  -> Shelling -> Kernel drying -> Peeling (testa) -> Grading -> Packing.
  By-products: cashew nut shell liquid (CNSL), cashew apple juice.
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Figure 5.1 Processing flow of tea, coffee, cocoa and cashew.

5.2 Tea (Camellia sinensis)

Origin and importance

Tea is native to the hills of South-East Asia and China and is the most widely consumed beverage in the world. The crop is grown for its young shoots - two leaves and a bud - which are processed into black tea, green tea or oolong tea. India is the second largest producer and the second largest exporter of tea, and the industry is a major employer in the hill districts. In Tamil Nadu, tea is grown mainly in the Nilgiris, the largest tea district in the south.

Area and production

India grows tea on about 0.62 million hectares and produced about 1,382 million kg of made tea in 2023-24. Assam and West Bengal are the largest producers, followed by Tamil Nadu (the Nilgiris), Kerala and the north-eastern states. About 80 per cent of Indian tea is consumed domestically, and the rest is exported, mainly to the CIS countries, Iran, the UAE and the UK.

Climate and soil

Tea needs a warm, humid climate with a rainfall of 1,500-3,000 mm, a temperature of 15-30 deg C and high humidity, and it grows best at 900-2,000 m elevation in the south. It is damaged by frost and by drought. It grows on a deep, well-drained, acidic soil with pH 4.5-5.5; tea soils are typically high in organic matter. Good drainage and a slope for air movement are important.

Varieties and planting material

Tea varieties are grouped as China, Assam and Cambod (Indo-China) types and their hybrids. India grows mostly clonal material and biclonal seed. Important clones are UPASI-9, UPASI-17, UPASI-3, UPASI-27 and the Tocklai (TRI) clones such as TRI-2025, TRI-2026 and TRI-2043. In the Nilgiris the UPASI clones are widely planted. Clonal planting material gives uniform, high-quality leaf and is raised from single-node cuttings in a nursery.

Table 5.2 Important tea clones / types and their salient features

Clone / type	Source	Salient features
UPASI-9	UPASI, Valparai (Tamil Nadu)	Widely grown in the south; good yield and quality
UPASI-17	UPASI	High yield; suited to the Nilgiris
UPASI-3, UPASI-27	UPASI	Popular south Indian clones
TRI-2025, TRI-2026, TRI-2043	Tocklai (TRI), Assam	High-yielding Assam-type clones for the north-east
China type	China	Hardy, good quality, suited to high elevations
Assam type	Assam	Vigorous, high yield, suited to the plains of Assam
Cambod type	Indo-China	Intermediate, adapted to a range of conditions

Source: UPASI, Tocklai and Tea Board material.

Propagation and nursery

Tea is propagated by single-node cuttings of a chosen clone, raised in a nursery bed or sleeve under shade for 9-12 months; biclonal seed is also used. Only vigorous, disease-free plants of the right age are planted. The nursery is kept shaded, watered and weeded, and the plants are hardened before planting.

Planting, spacing and planting systems

Tea is planted at 1.2 x 0.75 m (about 11,000 plants per hectare) or at 1.35 x 0.75 m, in trenches or pits along the contour on slopes. Planting is done at the onset of the rains. On slopes, planting is on the contour with drains and, where necessary, terraces. Green manures and shade trees are grown in the young plantation.

Training and pruning

Tea is trained by 'centering' the young plant and then by a cycle of pruning: the formative prune to build the bush, followed by light, medium and hard prunes at intervals of 2-5 years to renew the bearing surface and keep the bush at the correct height and spread. The pruning cycle depends on elevation and climate. Pruning is done with a pruning knife or machine and is followed by the removal of cut wood and the application of fertiliser.

Nutrient, water and weed management

Tea is a heavy feeder; a mature tea field is given about 100-150 kg N, 30-40 kg P2O5 and 60-80 kg K2O per hectare per year in splits, with zinc and magnesium where needed. Nitrogen is the most important nutrient for leaf yield. Tea is largely rainfed but irrigation in the dry months improves yield. Weeds are controlled by clean weeding, mulching and cover crops; shade trees are managed for light.

Pests, diseases and disorders

The main pests are the tea mosquito bug, the red spider mite, the thrips and the tea looper; the main diseases are blister blight, grey blight, red rust and root diseases. Management combines resistant clones, biological control, pruning and, where necessary, chemical control. The main disorder is the nutrient deficiency of zinc, magnesium and boron, corrected by foliar sprays.

Harvest and yield

Tea is harvested by 'plucking' the tender shoots - two leaves and a bud - at intervals of 7-10 days in the growing season. Fine plucking gives high-quality tea but a lower yield; coarse plucking gives a higher yield but poorer quality. Yield is about 1,700-2,500 kg of made tea per hectare per year in the south, and higher in the best estates. The green leaf is processed within a few hours of plucking.

Processing, value addition and economics

Black tea is made by withering, rolling (or CTC), fermentation, drying and sorting; green tea is made by steaming or pan-firing the leaf to prevent fermentation; oolong tea is partly fermented. Tea is graded by leaf size and quality into leaf, broken, fannings and dust grades. Value-added products are tea bags, flavoured and organic tea and instant tea. Tea is a high-investment, high-employment crop with a life of 40-60 years; returns depend on yield, quality and the world price, and organic and specialty tea fetch a premium.

5.3 Coffee (Coffea arabica and Coffea canephora)

Origin and importance

Coffee is native to Ethiopia and was introduced to India by the saint Baba Budan, who brought seven seeds to the hills of Karnataka in the seventeenth century. The two commercial species are arabica (Coffea arabica), which gives a mild, aromatic coffee, and robusta (Coffea canephora), which is more hardy and gives a stronger coffee. India is the seventh largest producer of coffee in the world and a large exporter, and coffee is grown under shade in the Western and Eastern Ghats, which are biodiversity hotspots.

Area and production

India grows coffee on about 0.47 million hectares and produced about 374,000 tonnes in 2023-24. Karnataka produces about 70 per cent, Kerala about 21 per cent and Tamil Nadu the rest, so the three southern states account for about 96 per cent of production. In Tamil Nadu, coffee is grown in the Shevaroy hills around Yercaud, the Palani (Pulney) hills, the Nilgiris and the Anamalai hills. India exports coffee to more than 50 countries, mainly Italy, Germany and Russia.

Climate and soil

Arabica needs a cool, equable climate with a rainfall of 1,500-2,500 mm and a temperature of 15-25 deg C, and it grows best at 1,000-1,500 m elevation; robusta is more hardy and grows at lower elevations (500-1,000 m) and tolerates higher temperature and rainfall. Both need a distinct dry spell for flowering and a well-drained, deep, acidic soil with pH 5.5-6.5 and high organic matter. Coffee is grown under shade trees to moderate temperature and light.

Varieties

Important arabica varieties are S.795, Kents, Cauvery (Catimor), Sln.5B, Selection 9, Chandragiri and Hemavathi; important robusta varieties are S.274 and CxR (Congensis x Robusta). S.795 is the leading arabica variety; Cauvery and Chandragiri are high-yielding and rust-tolerant; CxR is a high-yielding robusta. The Coffee Board's Central Coffee Research Institute, Balehonnur, releases the varieties.

Table 5.3 Important coffee varieties and their salient features

Species	Variety	Salient features
Arabica	S.795	Leading arabica; good quality and yield
Arabica	Kents	Old, high-quality arabica
Arabica	Cauvery (Catimor)	Rust-tolerant, high yield
Arabica	Sln.5B, Selection 9, Chandragiri, Hemavathi	High-yielding, rust-tolerant selections
Robusta	S.274	Standard robusta; hardy, high yield
Robusta	CxR (Congensis x Robusta)	High-yielding robusta hybrid

Source: Coffee Board and Central Coffee Research Institute material.

Propagation and nursery

Arabica is grown mainly from seed, and robusta from seed or from clonal cuttings. Seed is taken from high-yielding, disease-free mother plants and sown in a nursery bed; seedlings are raised in polybags and planted out at 6-12 months. The nursery is shaded, watered and weeded, and only vigorous seedlings are planted. Shade trees are raised separately and planted before or with the coffee.

Planting, spacing and cropping systems

Arabica is planted at 1.8 x 1.8 m (about 3,000 plants per hectare) or 2 x 2 m, and robusta at 2.7 x 2.7 m (about 1,370 plants per hectare). Coffee is grown as a two-tier or three-tier system with shade trees: the upper tier of tall shade trees (such as silver oak, jack and Ficus) and the lower tier of coffee, with pepper often grown on the shade trees. This multi-tier system moderates light and temperature and adds income. Planting is done at the onset of the monsoon.

Training and pruning

Coffee is trained to a single stem or to a capped (topped) system. Arabica is usually grown on a single stem with a few primary branches, and robusta on a multi-stem or capped system. Pruning removes the dead and weak wood, the suckers and the crowded branches, and renews the bearing wood. Regular pruning is essential because coffee bears on the new wood of the previous season. Shade trees are pruned to regulate light.

Nutrient, water and weed management

Coffee is a heavy feeder; a bearing plant is given about 30-60 g N, 20-40 g P2O5 and 30-60 g K2O per year in splits, plus farmyard manure or compost, and the shade-tree leaf litter adds organic matter. Irrigation is given in the dry months to induce flowering and to fill the berries. Weeds are controlled by clean cultivation and cover crops, and the shade canopy is managed.

Pests, diseases and disorders

The main diseases are the coffee leaf rust (Hemileia vastatrix), which is the most serious, and the black rot of berries; the main pests are the coffee berry borer, the stem borer and the white stem borer. Management combines resistant varieties (Cauvery, Chandragiri), timely fungicide sprays, shade management and the removal of affected parts. The main disorder is the nutrient deficiency of nitrogen, zinc and boron.

Harvest, yield and processing

Coffee is harvested by selectively picking only the ripe red cherries, which gives the best quality; picking is done in several rounds over the season. Yield is about 500-800 kg of clean coffee per hectare for arabica and 800-1,200 kg for robusta. The cherries are processed by the wet (washed) method, which gives 'parchment' coffee of high quality, or by the dry (natural) method, which gives 'cherry' coffee. The parchment or cherry is hulled, graded and cured to give green beans, which are roasted and ground.

Value addition and economics

Value-added products are roasted and ground coffee, instant coffee, flavoured coffee and coffee-based beverages. India also produces a small quantity of the unique 'monsooned' coffee of the Malabar coast. Coffee is a long-term crop with a life of 30-50 years and gives income from the third or fourth year. Returns depend on the world price of coffee, which is volatile, and on the quality of the beans; shade-grown Indian coffee has a good reputation for quality and sustainability.

5.4 Cocoa (Theobroma cacao)

Origin and importance

Cocoa is native to the Amazon basin of South America and is grown for its beans, from which chocolate, cocoa butter and cocoa powder are made. It is a small, shade-loving tree that is grown mainly as an intercrop in coconut and arecanut gardens, so it gives extra income from land already under plantation crops. India is a small producer but a large consumer of cocoa, so there is a large gap between demand and supply.

Area and production

India grows cocoa on about 109,000 hectares and produces about 30,000 tonnes of dry beans. Kerala, Karnataka, Andhra Pradesh and Tamil Nadu are the leading states, and cocoa is grown mostly as an intercrop in coconut gardens. The crop is expanding because of strong domestic demand from the chocolate industry.

Climate and soil

Cocoa needs a warm, humid, tropical climate with a rainfall of 1,500-2,500 mm, a temperature of 20-30 deg C and high humidity. It is shade-loving, especially in the young stage, and is grown under the shade of coconut, arecanut or shade trees. It grows on a deep, well-drained loam with pH 6.0-7.5 and good organic matter; it does not tolerate waterlogging or drought.

Varieties and planting material

Cocoa types are Criollo (fine flavour, low yield), Forastero (bulk cocoa, hardy) and Trinitario (a hybrid of the two). India grows mostly Forastero-type and hybrid material, and the Kerala Agricultural University and the DCCD have released the CCRP hybrids (CCRP-1 to CCRP-9) and other selections with high yield and tolerance to disease. TNAU also recommends cocoa as an intercrop in coconut gardens.

Propagation and nursery

Cocoa is propagated by seed (for seedling plants) and by budding or grafting on a seedling rootstock. Improved hybrids are raised from seed or by grafting. The nursery is shaded and watered, and seedlings are planted out at 6-12 months. Clonal and hybrid material gives uniform, high-yielding plants.

Planting, spacing and cropping systems

Cocoa is planted at 2.7 x 2.7 m or 3 x 3 m (about 1,100-1,370 plants per hectare) as an intercrop in coconut or arecanut gardens, where the palms provide the required shade. It is planted in pits filled with farmyard manure at the onset of the monsoon. The crop is a classic example of intercropping, giving additional income from the same land as the main plantation crop.

Training, pruning and management

Cocoa is trained to a single stem with a 'jorquette' (the whorl of side branches) at about 1-1.5 m, and the side shoots below the jorquette are removed. Pruning removes the dead and crowded branches, the chupons (vertical water sprouts) and excess shade. The crop needs regular watering in the dry months, manuring with farmyard manure and a moderate fertiliser dose, and weed control. It is sensitive to drought and to waterlogging.

Pests, diseases and disorders

The main diseases are the black pod rot (Phytophthora), the swollen shoot and the witches' broom (in some regions); the main pests are the cocoa pod borer, the mealybug and the mirid bug. Management combines resistant varieties, clean cultivation, shade management and timely sprays. The main disorders are the nutrient deficiency of nitrogen, potassium and zinc and the 'cherelle' wilt of young pods.

Harvest, yield and processing

Cocoa pods are harvested when they are ripe - the colour changes from green to yellow or orange - by cutting the pod from the tree. Yield is about 1-2 kg of dry beans per tree per year, or 500-1,000 kg per hectare as an intercrop. The beans and pulp are removed from the pod, fermented for 5-7 days (which develops the chocolate flavour), sun-dried, roasted and winnowed to give cocoa nibs, cocoa mass, cocoa butter and cocoa powder.

Value addition and economics

Value-added products are chocolate, cocoa butter, cocoa powder and cocoa-based beverages. Cocoa as an intercrop gives extra income with little additional land, and the strong domestic demand from the chocolate industry makes it an attractive crop. The main risks are the long gestation (3-4 years), disease pressure and the need for shade and moisture.

5.5 Cashew (Anacardium occidentale)

Origin and importance

Cashew is native to Brazil and was introduced to India by the Portuguese. It is grown for its nut, which is a valuable export crop, and for the cashew apple, which yields juice and the spirit feni. India is the largest producer and consumer of cashew in the world and a major exporter of cashew kernels. The tree is hardy and drought-tolerant and is grown on the coastal belt and the dry uplands.

Area and production

India grows cashew on about 1.2 million hectares and produces about 795,000 tonnes of raw cashew nut. Maharashtra, Andhra Pradesh, Odisha, Karnataka, Tamil Nadu and Kerala are the leading states. In Tamil Nadu, cashew is grown in the coastal and dry districts of Cuddalore, Villupuram, Ariyalur, Pudukkottai and Kanyakumari. India exported about 65,800 tonnes of cashew kernels worth about Rs 2,809 crore in 2023-24.

Climate and soil

Cashew needs a warm, humid tropical climate with a rainfall of 1,000-2,000 mm, a temperature of 20-35 deg C and plenty of sunshine. It is very drought-tolerant and grows on poor, sandy, lateritic and rocky soils with pH 5.0-7.5; it does not tolerate waterlogging or frost. It is one of the best crops for the coastal and dry uplands where few other crops succeed.

Varieties

Important varieties are VRI-1, VRI-2, VRI-3 and VRI-4 (released by TNAU from its Vridhachalam research station), BPP-8, K-22-1, Bhaskara, Priyanka, Ullal-1, Kanaka, Dhana, Anagha and Madakkathara. The VRI varieties are high-yielding and are recommended for Tamil Nadu; BPP-8 and K-22-1 are popular in Andhra Pradesh and Odisha.

Table 5.4 Important cashew varieties and their salient features

Variety	Source	Salient features
VRI-1, VRI-2, VRI-3, VRI-4	TNAU, Vridhachalam	High yield and good kernel quality; recommended for Tamil Nadu
BPP-8	Andhra Pradesh	High yield, bold kernels
K-22-1	Andhra Pradesh	High yield, good quality
Bhaskara, Priyanka	Kerala (KAU)	High-yielding selections
Ullal-1, Kanaka, Dhana	Karnataka (CPCRI)	High yield; Ullal-1 popular in Karnataka
Anagha, Madakkathara	Kerala (KAU)	High-yielding releases

Source: TNAU, KAU and CPCRI material.

Propagation and nursery

Cashew is propagated by seed for rootstock and by softwood grafting, air-layering or budding for planting. Softwood grafting is the standard commercial method and gives true-to-type plants that bear in 3-4 years; seedling plants are variable and bear in 5-8 years. The nursery is raised from selected seed and the grafts are hardened before planting.

Planting, spacing and after care

Cashew is planted at 7 x 7 m or 8 x 8 m (about 156-204 plants per hectare), and high-density planting at 4 x 4 m or 5 x 5 m is used with pruning. Pits of 45-60 cm cube are filled with farmyard manure and planted at the onset of the monsoon. After care includes weeding, mulching, and the removal of the lower branches to give a clean trunk. Intercropping with groundnut, pulses or cashew-based crops is practised in the young orchard.

Worked example 5.1 Plant population of cashew at two spacings

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Given: 1 hectare = 10,000 m2.

Conventional spacing 8 x 8 m:
   Plants/ha = 10,000 / (8 x 8) = 10,000 / 64 = 156.25  ->  about 156 plants

Closer spacing 7 x 7 m:
   Plants/ha = 10,000 / (7 x 7) = 10,000 / 49 = 204.1  ->  about 204 plants

High-density spacing 4 x 4 m (with pruning):
   Plants/ha = 10,000 / (4 x 4) = 10,000 / 16 = 625 plants

For a 2-hectare plot at 7 x 7 m: 204 x 2 = 408 plants.
Note: closer spacing raises the plant number and the early yield, but needs
regular pruning to keep the canopy open and the orchard manageable.
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The same method applies to any spacing and plot size.

Training, pruning and management

Cashew is trained to a modified leader or open centre and kept low for easy harvest. Pruning removes the dead wood, the water sprouts and the crowded branches and keeps the canopy open; it is done after harvest. The crop is largely rainfed but responds to irrigation at flowering and nut development. Manuring is with farmyard manure and a moderate fertiliser dose of about 100-200 g N, 50-100 g P2O5 and 100-200 g K2O per tree per year. Weeds are controlled by clean cultivation, mulching and intercrops.

Pests, diseases and disorders

The main pests are the tea mosquito bug (which is the most serious), the cashew stem and root borer, the leaf miner and the apple borer; the main diseases are the die-back (Colletotrichum), the anthracnose and the powdery mildew. Management combines resistant varieties, timely spraying with the recommended fungicide and insecticide, and the removal of affected parts. The main disorder is the immature nut drop from water or nutrient stress.

Harvest, yield and processing

The cashew nut is the seed attached to the fleshy cashew apple. The nut is harvested when it is mature - the apple turns red or yellow and the nut shell is hard - from February to May. Yield is about 8-12 kg of raw nuts per tree per year, or 1-2 tonnes per hectare. The raw nuts are sun-dried, roasted (steam or drum roasting), shelled and the kernels dried and peeled. The by-products are cashew nut shell liquid (CNSL), used in brake linings and resins, and the cashew apple, used for juice, jam and feni.

Value addition and economics

Value-added products are roasted and salted kernels, cashew kernel splits and pieces, CNSL and cashew apple juice. Cashew is a hardy crop that gives a good return on poor land and a valuable export product. The main risks are the tea mosquito bug, the price of raw nuts and the competition from imported raw nuts for the processing industry. India is the world's largest processor of cashew, importing raw nuts from Africa and processing them for export.

Exam focus

Key points to remember

1. India is the second largest producer and the second largest exporter of tea in the world; the Nilgiris is the leading tea district of south India.
2. Tea is propagated by single-node cuttings of a chosen clone; UPASI-9 and UPASI-17 are leading south Indian clones.
3. Tea is trained by 'centering' and pruned in a cycle of formative, light, medium and hard prunes; plucking is 'two leaves and a bud'.
4. Black tea is made by withering, rolling, fermentation, drying and sorting; green tea is made without fermentation.
5. Coffee has two commercial species - arabica (cool, higher elevation) and robusta (hardy, lower elevation); India is the seventh largest producer.
6. Leading coffee varieties: arabica S.795, Cauvery (Catimor) and Chandragiri; robusta S.274 and CxR.
7. Coffee is grown as a multi-tier system with shade trees; beans are processed by the wet (washed) or dry (natural) method.
8. Cocoa is grown mainly as an intercrop in coconut and arecanut gardens; the beans are fermented for 5-7 days to develop the chocolate flavour.
9. Cashew is propagated by softwood grafting; the VRI varieties of TNAU are recommended for Tamil Nadu, and the tea mosquito bug is the main pest.
10. Cashew processing gives the kernel, and the by-products cashew nut shell liquid (CNSL) and cashew apple products.

Short-answer questions with model answers

Q1. Name two leading clones of tea in south India and one tea district of Tamil Nadu.

Model answer. UPASI-9 and UPASI-17 are leading south Indian clones, released by UPASI, Valparai. The Nilgiris is the leading tea district of Tamil Nadu.

Q2. Distinguish arabica from robusta coffee.

Model answer. Arabica (Coffea arabica) gives a mild, aromatic coffee, needs a cool climate (15-25 deg C) and higher elevation (1,000-1,500 m), and is susceptible to leaf rust. Robusta (Coffea canephora) gives a stronger coffee, is more hardy, grows at lower elevation (500-1,000 m) and tolerates higher temperature and rainfall.

Q3. How is cocoa grown in India and why is it important as an intercrop?

Model answer. Cocoa is grown mainly as an intercrop in coconut and arecanut gardens, which provide the shade it needs. It gives extra income from land already under a plantation crop, with little additional cost, and meets part of the strong domestic demand for cocoa from the chocolate industry.

Q4. Calculate the number of cashew plants in one hectare at 7 x 7 m.

Model answer. Plants per hectare = 10,000 / (7 x 7) = 10,000 / 49 = about 204 plants.

Long-answer questions with model answer outlines

Q1. Describe the production technology of tea from nursery to processing.

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Outline. Climate and soil (warm, humid, 900-2,000 m, acidic soil pH 4.5-5.5); varieties (China, Assam, Cambod types; clones UPASI-9, UPASI-17, TRI clones); propagation by single-node cuttings and nursery management; planting at 1.2 x 0.75 m (about 11,000 plants per hectare) on the contour; training by centering and pruning cycle (formative, light, medium, hard); nutrient dose (100-150 kg N, 30-40 kg P2O5, 60-80 kg K2O per hectare); pests (tea mosquito bug, red spider mite) and diseases (blister blight); harvest by fine plucking of two leaves and a bud; processing into black tea (withering, rolling, fermentation, drying, sorting) or green tea; grading and value addition.

Q2. Compare tea, coffee and cocoa with respect to climate, propagation and processing.

Outline. Tea: warm humid hills, 900-2,000 m, acidic soil; propagated by clonal cuttings; plucking of two leaves and a bud; processed by withering, rolling, fermentation and drying into black tea. Coffee: arabica cool and higher, robusta hardy and lower; propagated by seed (arabica) or clonal (robusta); grown under shade; processed by wet or dry method into green beans. Cocoa: warm humid tropics, grown as an intercrop under shade; propagated by seed or grafting; pods harvested and beans fermented 5-7 days and dried; processed into cocoa nibs, butter and powder. Bring out the differences in propagation, spacing, harvest and product.

Q3. Describe the production technology of cashew with emphasis on propagation, harvest and processing.

Outline. Climate and soil (hardy, drought-tolerant, coastal and dry uplands); varieties (TNAU VRI-1 to VRI-4, BPP-8, K-22-1, Ullal-1); propagation by softwood grafting on seedling rootstock; planting at 7 x 7 m or 8 x 8 m (156-204 plants per hectare) or high density at 4 x 4 m; training and pruning; nutrient dose; pests (tea mosquito bug, stem and root borer) and diseases (die-back, anthracnose); harvest when the apple turns red and the shell is hard; yield 1-2 tonnes per hectare; processing by sun-drying, roasting, shelling and peeling; by-products CNSL and cashew apple products; export and value addition.

APPENDICES | Consolidated Tables, Glossary and References

The appendices bring together, in single tables, the facts that are scattered through the five units: the planting density and spacing of every crop, the rootstock and propagation method of every crop, the maturity index of every crop, and a crop-wise consolidated package of practices. They end with a glossary of technical terms and a list of standard references.

Appendix A Planting density and spacing of the crops in this course

Table A.1 Spacing, plant population and planting system

Crop	Spacing	Plants / trees per ha	System
Mango	10 x 10 m (conventional); 5 x 5 m (HDP); 3 x 2 m (UHDP)	100; 400; 1,333-1,667	Square; rectangular / square
Banana	1.8 x 1.8 m (tall); 1.5 x 1.5 m (dwarf)	3,086; 4,444	Square / paired row
Papaya	1.8 x 1.8 m; 2 x 2 m	3,086; 2,500	Square; raised-bed rows
Guava	6 x 6 m (normal); 3 x 3 m (HDP)	278; 1,111	Square
Sapota	8 x 8 m; 5 x 5 m (closer)	156; 400	Square
Acid lime	4 x 4 m; 5 x 5 m	625; 400	Square
Sweet orange	6 x 6 m	278	Square
Grape	3 x 3 m (bower); 2 x 2 m (trellis)	1,111; 2,500	Bower / trellis rows
Pineapple	60-90 x 30-45 cm (double row)	40,000-55,000	Paired / double row
Pomegranate	3 x 3 m; 4 x 4 m	1,111; 625	Square / bush
Jackfruit	8 x 8 m	156	Square
Apple	6 x 6 m (standard); 3 x 2 m (HDP)	278; 1,667	Square / trellis (HDP)
Pear	5 x 5 m to 7 x 7 m	204-400	Square
Peach	4 x 4 m to 6 x 6 m	278-625	Square
Plum	5 x 5 m to 6 x 6 m	278-400	Square
Strawberry	30 x 30 cm (4 rows per bed)	50,000-100,000	Raised bed (mulched)
Ber	6 x 6 m; 5 x 5 m	278; 400	Square
Custard apple	5 x 5 m; 4 x 4 m	400; 625	Square
Amla	8 x 8 m	156	Square
Dragon fruit	2 x 2 m; 3 x 2 m	2,500; 1,667	Post-and-ring trellis
Almond	6 x 6 m	278	Square
Walnut	8 x 8 m or wider	156	Square
Coconut	7.5 x 7.5 m (square); 7.5 x 7.5 m (triangular)	178; 205	Square / triangular
Arecanut	2.7 x 2.7 m	1,370	Square
Oil palm	9 x 9 x 9 m (triangular)	143	Triangular
Palmyrah	6 x 6 m or wider	278	Square
Rubber	5 x 5 m (budded stock)	400	Square
Tea	1.2 x 0.75 m; 1.35 x 0.75 m	11,000; 9,876	Contour rows
Coffee (arabica)	1.8 x 1.8 m; 2 x 2 m	3,086; 2,500	Square under shade
Coffee (robusta)	2.7 x 2.7 m	1,370	Square under shade

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Crop	Spacing	Plants / trees per ha	System
Cocoa	2.7 x 2.7 m; 3 x 3 m	1,370; 1,111	Intercrop under palms
Cashew	8 x 8 m; 7 x 7 m; 4 x 4 m (HDP)	156; 204; 625	Square / rectangular

Plants per hectare = 10,000 / (area per plant in m2). Figures rounded. HDP = high-density planting; UHDP = ultra-high-density planting.

Appendix B Propagation method and rootstock of the crops in this course

Table B.1 Propagation method and rootstock summary

Crop	Propagation method	Rootstock / note
Mango	Veneer / stone / epicotyl grafting	Seedling of local or polyembryonic type (Olour, Bappakai)
Banana	Sword sucker or tissue culture	No separate rootstock
Papaya	Seed	No rootstock; gynodioecious / hybrid seed preferred
Guava	Air-layering, stooling, grafting	Seedling guava for grafting; own root for layers
Sapota	Softwood grafting, air-layering	Seedling sapota (Kalippatti / local)
Citrus	T / shield budding	Rough lemon, Rangpur lime, Trifoliate orange, Troyer citrange
Grape	Hardwood cuttings	Own root; Dog Ridge / 110-R / 1103-P for problem soils
Pineapple	Suckers, slips, crowns	No rootstock
Pomegranate	Hardwood cuttings	Own root (from certified mother plant)
Jackfruit	Softwood / epicotyl grafting, budding	Seedling jackfruit
Apple	Grafting / budding	M9, M7, MM106, MM111 (dwarfing); seedling stock
Pear	Grafting / budding	Quince A (dwarfing), Kainth (vigorous), seedling pear
Peach	Budding / grafting	Seedling peach, wild peach, almond or plum
Plum	Budding / grafting	Seedling plum, peach or wild plum
Strawberry	Runners (daughter plants)	No rootstock; micropropagated mother plants
Ber	Budding (T / shield)	Seedling wild ber (Ziziphus rotundifolia)
Custard apple	Grafting / budding / air-layering	Seedling of Annona species
Amla	Patch / shield budding, softwood grafting	Seedling amla
Dragon fruit	Stem cuttings	No rootstock
Almond	Budding / grafting	Seedling almond, peach or plum
Pistachio	Budding / grafting	Pistacia integerrima or P. atlantica
Walnut	Budding / grafting	Seedling walnut
Coconut	Seednut raised in nursery	No rootstock (selected mother palm)
Arecanut	Seednut raised in nursery	No rootstock (selected mother palm)
Oil palm	Seed (Tenera hybrid) in polybag nursery	No rootstock (Dura x Pisifera hybrid)
Palmyrah	Seed	No rootstock (dioecious)
Rubber	Budding on seedling rootstock	Seedling stock budded with recommended clone
Tea	Single-node cuttings (clonal); biclonal seed	No rootstock

Crop	Propagation method	Rootstock / note
Coffee	Seed (arabica); seed or clonal (robusta)	No rootstock; shade trees planted separately
Cocoa	Seed; budding / grafting	Seedling cocoa
Cashew	Softwood grafting, air-layering, budding	Seedling cashew

Source: standard texts and TNAU/ICAR/commodity-board material.

Appendix C Maturity indices of the crops in this course

Table C.1 Consolidated maturity indices

Crop	Physical index	Chemical / other index	Days from flowering (approx.)
Mango	Shoulders fill, colour lightens, lenticels prominent	Specific gravity > 1.0-1.02, TSS 12-15 per cent, sap less milky	90-140
Banana	Fingers plump, ridges less angular, colour breaks	TSS 12-15 per cent, pulp:peel ratio	90-150
Papaya	Yellow streak at blossom end, latex watery	TSS 11-14 per cent	120-150
Guava	Skin light green to yellow, smooth	TSS 10-13 per cent, firmness	90-120
Sapota	Brown scurf shed, skin smooth, less latex	TSS 18-22 per cent	120-150
Acid lime / sweet orange	Colour to yellow / orange, glossy surface	Juice > 40-45 per cent, TSS:acid ratio	150-240
Grape	Berries full size and colour, separate from stalk	Brix 16-22 per cent	100-140
Pineapple	Shell colour yellow at base, eyes flat, hollow sound	TSS 12-15 per cent	Fruit 15-20 months after planting
Pomegranate	Rind colour change, metallic sound on tapping	TSS, aril colour	120-150
Jackfruit	Spine colour to yellow-brown, aroma, hollow sound	TSS	120-180
Apple	Skin colour, size, ease of separation	Starch-iodine index, firmness, TSS	Days from full bloom (variety specific)
Pear	Skin colour, firmness	TSS	Days from bloom
Peach	Skin colour, softening	TSS, firmness	Days from bloom
Plum	Skin colour, bloom, softening	TSS	Days from bloom
Strawberry	Full berry colour, firm	TSS, aroma	30-40
Ber	Skin green to golden-yellow	TSS	Days from flowering
Custard apple	Segments widen, colour to light green / yellow, slight softening	TSS	Days from flowering
Amla	Colour green to light green / yellow, size	Vitamin C, TSS	Days from flowering
Dragon fruit	Full red / yellow skin colour, slight softening	TSS	30-50 after flowering
Almond	Hull splits, shell dries	Kernel moisture	Late summer
Walnut	Green husk splits, shell dries	Kernel moisture	Late summer / autumn
Coconut	Nuts mature, husk colour; tender nuts at 6-7 months	Copra content, oil content	11-12 months (copra); 6-7 months (tender)

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Crop	Physical index	Chemical / other index	Days from flowering (approx.)
Arecanut	Husk turns orange-yellow, nut hard	Cured nut weight	December-March (season)
Oil palm	Loose fruits on ground, bunch orange	Oil content, ripeness	Every 7-10 days
Rubber	Tree girth 45-50 cm at 1.2 m (for tapping)	Latex flow	Tapping from 6th-7th year
Tea	Two leaves and a bud, tender shoot	Quality of made tea	Every 7-10 days (plucking)
Coffee	Ripe red cherries	Bean quality	Several rounds in season
Cocoa	Pod colour green to yellow / orange	Bean quality	Pods harvested when ripe
Cashew	Apple red / yellow, shell hard	Kernel quality	February-May (season)

Source: standard post-harvest texts and TNAU/commodity-board material. Days from flowering vary widely with variety and climate.

Appendix D Crop-wise consolidated package of practices

The table below summarises the package of practices for every crop in the course in one place, for quick revision. It should be read together with the detailed crop sections, which give the full explanation and the local recommendations.

Table D.1 Consolidated package of practices (crop-wise summary)

Crop	Propagation / rootstock	Spacing	Training / pruning	Maturity / harvest	Yield (t/ha)
Mango	Veneer/stone graft on polyembryonic stock	10x10 m or 5x5 m HDP	Modified leader; light annual pruning; paclobutrazol	Shoulder fill, SG > 1.0, TSS 12-15 per cent	8-25
Banana	Sucker or tissue culture	1.8x1.8 m	Desuckering; remove male bud; propping	Finger plumpness, colour break	40-60
Papaya	Seed (gynodioecious/hybrid)	1.8x1.8 m or 2x2 m	Remove males and side shoots	Yellow streak, watery latex	40-75
Guava	Air-layer / graft on seedling stock	6x6 m or 3x3 m HDP	Open centre; bahar regulation	Skin yellow, TSS 10-13 per cent	20-40
Sapota	Softwood graft / air-layer	8x8 m	Modified leader; keep low	Scurf shed, skin smooth, TSS 18-22 per cent	15-25
Acid lime	Bud on rough lemon / Rangpur lime	4x4 m or 5x5 m	Open centre; bahar regulation	Colour, juice > 40 per cent	20-30
Sweet orange	Bud on rough lemon / Trifoliate	6x6 m	Open centre	Colour, TSS:acid ratio	15-25
Grape	Hardwood cutting / graft on resistant stock	3x3 m bower; 2x2 m trellis	Two prunings; dormancy-breaker; GA3	Brix 16-22 per cent, colour	20-30
Pineapple	Suckers / slips	60-90 x 30-45 cm	De-suckering; flower induction	Shell colour, flat eyes	40-60
Pomegranate	Hardwood cutting	3x3 m or 4x4 m	Multi-stem bush; bahar regulation	Rind colour, metallic sound	15-25
Jackfruit	Graft / budding	8x8 m	Modified leader; light pruning	Spine colour, aroma	15-30
Apple	Graft on M9 / MM106 / seedling	6x6 m or 3x2 m HDP	Central leader; dormant pruning	Starch index, firmness, colour	15-30
Pear	Graft on quince / Kainth / seedling	5x5-7x7 m	Central leader / open centre	Skin colour, firmness	15-25
Peach	Bud on seedling peach / almond	4x4-6x6 m	Open centre; severe dormant pruning	Skin colour, TSS	10-20
Plum	Bud on seedling plum / peach	5x5-6x6 m	Open centre / modified leader	Skin colour, TSS	10-20
Strawberry	Runners	30x30 cm (beds)	Remove runners and old leaves	Full berry colour	20-30
Ber	Bud on wild ber stock	6x6 m	Open centre; annual heading back	Golden-yellow skin	15-25
Custard apple	Graft / bud on Annona stock	5x5 m	Modified leader; tipping	Segments widen, colour change	8-15

Crop	Propagation / rootstock	Spacing	Training / pruning	Maturity / harvest	Yield (t/ha)
Amla	Bud / graft on seedling stock	8x8 m	Modified leader; heading back	Colour, size	15-25
Dragon fruit	Stem cutting	2x2 m (post-and-ring)	Train on post; remove excess shoots	Full skin colour	20-30
Almond	Bud on seedling almond / peach	6x6 m	Open centre; dormant pruning	Hull splits, shell dries	1-2 (kernel)
Walnut	Graft on seedling walnut	8x8 m	Modified central leader; light pruning	Husk splits, shell dries	2-4
Coconut	Seednut from selected mother palm	7.5x7.5 m (178-205/ha)	Remove dry leaves; intercropping	11-12 months (copra)	60-100 nuts/palm
Arecanut	Seednut from selected mother palm	2.7x2.7 m	Remove dry leaves; de-sucker	Husk orange-yellow, nut hard	1-3
Oil palm	Tenera hybrid seed	9x9x9 m triangular	Remove excess fronds; prune	Loose fruit, orange bunch	15-25 (FFB)
Palmyrah	Seed	6x6 m or wider	Remove dead leaves; tapping	Sap tapping; nungu in summer	1-2 L sap/palm/day
Rubber	Bud on seedling stock	5x5 m	Single clean trunk; prune side shoots	Girth 45-50 cm; tapping	1.5-2.0 (dry rubber)
Tea	Single-node cutting (clonal)	1.2x0.75 m	Centering; pruning cycle	Two leaves and a bud	1.7-2.5 (made tea)
Coffee	Seed (arabica); seed/clonal (robusta)	1.8x1.8 m (A); 2.7x2.7 m (R)	Single stem / capped; shade pruning	Ripe red cherries	0.5-1.2 (clean coffee)
Cocoa	Seed / grafting	2.7x2.7 m (intercrop)	Single stem + jorquette; remove chupons	Pod colour change	0.5-1.0 (dry beans)
Cashew	Softwood grafting	8x8 m or 4x4 m HDP	Modified leader; pruning after harvest	Apple red/yellow, shell hard	1-2 (raw nut)

Summary only; see the crop sections for the full package and local recommendations. HDP = high-density planting; SG = specific gravity; FFB = fresh fruit bunches.

Appendix E Glossary of technical terms

Table E.1 Glossary

Term	Meaning
Air-layering (gootee)	A method of vegetative propagation in which a branch is rooted while still attached to the parent plant
Bahar regulation	Deliberate timing of flowering in a crop (by pruning and water management) to harvest in the most profitable season
Biennial / alternate bearing	A habit in which a heavy crop year is followed by a light crop year
Brix	A measure of the total soluble solids (mainly sugars) in fruit juice; used as a maturity index for grape
Chilling requirement	The number of hours below about 7 deg C that a temperate fruit tree needs to break dormancy and flower
Climacteric fruit	Fruit that shows a rise in respiration and ethylene at ripening and continues to ripen after harvest
Copra	The dried kernel of coconut from which coconut oil is extracted
Curing	Drying and maturing of a product (e.g. arecanut, coffee, rubber) to improve keeping quality
Dead heart	The death of the growing point of a palm or shoot, often from a pest or disease
De-suckering	Removal of unwanted side shoots or suckers (banana, arecanut, palmyrah)
Desuckering (banana)	Removal of all suckers except the follower to divert nutrients to the mother plant
Dioecious	Bearing male and female flowers on separate plants (e.g. palmyrah, papaya)
Drupe	A fleshy fruit with a single hard stone (endocarp), e.g. mango, peach, coconut
Fertigation	Application of soluble fertilisers through the drip irrigation system
Foliar spray	Application of nutrients or growth regulators to the leaves
Gynodioecious	Bearing female and bisexual flowers on separate plants (e.g. some papaya varieties)
Hesperidium	A berry with a leathery rind and oil glands, e.g. citrus
High-density planting (HDP)	Planting more trees per hectare than the traditional spacing to raise early and total yield
Intercropping	Growing a short-duration crop in the space between the rows of a main crop
Koleroga	The fruit-rot disease of arecanut caused by Phytophthora
Leaching	Loss of soluble nutrients from the soil by the downward movement of water
Multi-tier cropping	Growing crops of different stature in layers so that each uses a different part of the light and root zone
Nungu	The tender fruit of palmyrah, eaten in summer
Parchment coffee	Coffee beans in their dry inner covering, produced by the wet (washed) method
Parthenocarpy	Development of fruit without fertilisation, giving seedless fruit
Pseudostem	The false stem of banana, formed by the overlapping leaf sheaths

Term	Meaning
Ratoon	A crop that grows from the stubble or follower of a previous crop (e.g. banana)
Root (wilt) disease	A serious disease of coconut in Kerala causing yellowing, flaccidity and decline
Smudging	Burning of leaves and dung under a mango canopy to raise smoke and ethylene and induce flowering
Sucker	A shoot that arises from the base or root of a plant; used as planting material in banana, pineapple
TSS (total soluble solids)	The soluble solid content of fruit juice, mainly sugars; a maturity index
Testa	The thin brown covering (seed coat) of the cashew kernel, removed during processing
Tenera	The high-yielding oil palm hybrid, a cross of Dura and Pisifera
Training	Building the framework of a young tree by selecting and shaping its branches
Ultra-high-density planting (UHDP)	Very close spacing with a small, controlled canopy
Value addition	Processing a raw crop into a higher-value product (e.g. copra to virgin coconut oil)
Withering	Partial drying of tea leaf before rolling to make it pliable and to develop flavour

Terms are defined as used in this course.

Appendix F References and further reading

The following are standard textbooks and official sources used in the preparation of this material. Students should consult the latest editions and the current crop-wise package of practices.

Textbooks

- Bose, T.K. and Mitra, S.K. (eds). *Fruits of India: Tropical and Subtropical*. Naya Prokash, Kolkata.
- Singh, S.P. (ed). *Commercial Fruits*. Kalyani Publishers, New Delhi.
- Chattopadhyay, T.K. *A Textbook on Pomology* (Volumes). Kalyani Publishers, New Delhi.
- Kumar, N. *Introduction to Horticulture*. Oxford and IBH Publishing, New Delhi.
- Chadha, K.L. *Handbook of Horticulture*. Indian Council of Agricultural Research, New Delhi.
- Chadha, K.L. and Pareek, O.P. *Advances in Horticulture* (Fruit Crops volumes). Malhotra Publishing House, New Delhi.
- Nayar, N.M., Singh, R. and others. *Textbook of Horticulture*.
- Gopalan, C., Rama Sastri, B.V. and Balasubramanian, S.C. *Nutritive Value of Indian Foods*. NIN, Hyderabad.
- Kader, A.A. *Postharvest Technology of Horticultural Crops*. University of California.

Official and institutional sources

- Tamil Nadu Agricultural University, *Crop Production Guide* and the crop-wise *Package of Practices*, Coimbatore.
- ICAR e-Course material on production technology of fruit and plantation crops.
- National Horticulture Board (NHB), *Horticultural Statistics at a Glance* and the annual area and production estimates.

- Department of Agriculture and Farmers Welfare, advance estimates of area and production of horticultural crops.
- Agricultural and Processed Food Products Export Development Authority (APEDA), annual reports and export data.
- Coconut Development Board; Rubber Board; Tea Board; Coffee Board; Spices Board; Directorate of Cashewnut and Cocoa Development.
- ICAR research institutes: IIHR Bengaluru; ICAR-CISH Lucknow; ICAR-CCRI Nagpur; ICAR-DCR (cashew), Puttur; CPCRI (arecanut), Kasaragod; ICAR-IIOPR (oil palm), Pedavegi; Central Coffee Research Institute, Balehonnur.

Note. All statistics in this material are drawn from the standard published sources named above. Where a figure varies between sources or years, a standard range is given. Students should quote the source and year with any figure in an examination answer.

APPENDIX G | Objective Question Bank (Unit-wise)

This bank of objective questions follows the five units in order and is meant for quick revision and self-testing. Each unit has multiple-choice questions, fill-in-the-blanks and true-or-false items, with the answers given immediately.

Unit I Production status of fruit and plantation crops

Multiple-choice questions

1. India's rank in world fruit and vegetable production is
(a) First (b) Second (c) Third (d) Fourth
Ans: (b)

2. India ranks first in the world in the production of
(a) Mango (b) Banana (c) Apple (d) Grape
Ans: (b)

3. The approximate area under fruit crops in India is
(a) 3 million ha (b) 5 million ha (c) 7 million ha (d) 10 million ha
Ans: (c)

4. Fruit production in India in 2023-24 was about
(a) 80 million tonnes (b) 95 million tonnes (c) 113 million tonnes (d) 150 million tonnes
Ans: (c)

5. Which is a climacteric fruit?
(a) Grape (b) Citrus (c) Mango (d) Pineapple
Ans: (c)

6. Which is a non-climacteric fruit?
(a) Banana (b) Papaya (c) Grape (d) Guava
Ans: (c)

7. The richest common source of vitamin C among fruits is
(a) Mango (b) Guava (c) Banana (d) Apple
Ans: (b)

8. Mango is a rich source of
(a) Vitamin C (b) Provitamin A (c) Calcium (d) Iron
Ans: (b)

9. The botanical family of mango and cashew is
(a) Rutaceae (b) Anacardiaceae (c) Rosaceae (d) Myrtaceae
Ans: (b)

10. Citrus fruit belongs to the family
(a) Rutaceae (b) Musaceae (c) Sapotaceae (d) Vitaceae
Ans: (a)

11. A fruit with a leathery rind and oil glands is called a
(a) Drupe (b) Berry (c) Hesperidium (d) Pome
Ans: (c)

12. Apple and pear are examples of
(a) Drupe (b) Pome (c) Aggregate (d) Multiple
Ans: (b)

13. Pineapple and jackfruit are examples of
(a) Simple fruit (b) Aggregate fruit (c) Multiple fruit (d) Nut
Ans: (c)

14. Custard apple is an example of
(a) Multiple fruit (b) Aggregate fruit (c) Berry (d) Drupe
Ans: (b)

15. The average productivity of fruit crops in India is about
(a) 5 t/ha (b) 10 t/ha (c) 15-16 t/ha (d) 30 t/ha
Ans: (c)

16. India's leading fruit-producing state by production is
(a) Maharashtra (b) Andhra Pradesh (c) Uttar Pradesh (d) Tamil Nadu
Ans: (b)

17. Grapes are the leading
(a) Fresh fruit export of India (b) Imported fruit (c) Processed fruit (d) Dry fruit
Ans: (a)

18. India's share of the global horticulture market is about
(a) 1 per cent (b) 5 per cent (c) 10 per cent (d) 20 per cent
Ans: (a)

19. The total horticulture production of India in 2023-24 was about
(a) 250 million tonnes (b) 353 million tonnes (c) 450 million tonnes (d) 150 million tonnes
Ans: (b)

20. Aonla (amla) is a very rich source of
(a) Vitamin A (b) Vitamin C (c) Protein (d) Fat
Ans: (b)

21. Coconut is unusual among fruits because it supplies mainly
(a) Protein (b) Fat (c) Vitamin C (d) Calcium
Ans: (b)

22. The fruit crop classified as temperate is
(a) Banana (b) Apple (c) Mango (d) Papaya
Ans: (b)

23. The fruit crop classified as sub-tropical is
(a) Coconut (b) Citrus (c) Cashew (d) Cocoa
Ans: (b)

24. India is the largest producer and consumer of
(a) Tea (b) Cashew (c) Coffee (d) Rubber
Ans: (b)

25. Which board looks after coconut in India?
(a) Tea Board (b) Coffee Board (c) Coconut Development Board (d) Spices Board
Ans: (c)

Fill in the blanks

1. India is the _____ largest producer of fruits and vegetables in the world. **Ans:** second
2. The rise in respiration and ethylene at ripening is characteristic of _____ fruit. **Ans:** climacteric
3. Mango belongs to the botanical family _____. **Ans:** Anacardiaceae
4. The scientific name of banana is _____. **Ans:** Musa species
5. Papaya is propagated by _____. **Ans:** seed
6. The leading fresh fruit export of India is _____. **Ans:** grape
7. Aonla is the botanical name of _____. **Ans:** Indian gooseberry (Phyllanthus emblica)
8. Fruit and vegetable exports from India in 2023-24 were about USD _____ billion. **Ans:** 1.8
9. The unit used for total soluble solids in fruit is _____. **Ans:** Brix
10. The five statutory commodity boards are tea, coffee, rubber, spices and _____. **Ans:** tobacco

True or false

1. Guava is a rich source of vitamin C. **Ans:** True
2. Non-climacteric fruits ripen after harvest. **Ans:** False (they ripen only on the plant)
3. India is the largest producer of banana in the world. **Ans:** True
4. Coconut is a temperate fruit crop. **Ans:** False
5. Mango is a drupe. **Ans:** True
6. Pineapple is an aggregate fruit. **Ans:** False (it is a multiple/syncarp fruit)
7. Citrus fruit is climacteric. **Ans:** False
8. India is the second largest producer of tea. **Ans:** True
9. Nuts such as almond are low in energy. **Ans:** False (they are energy-dense)
10. Horticulture contributes about one-third of the value of agricultural output. **Ans:** True

Unit II Tropical and sub-tropical fruit crops

Multiple-choice questions

1. Plants per hectare at 5 x 5 m spacing is
(a) 200 (b) 400 (c) 500 (d) 800
Ans: (b)

2. The planting system that gives about 15 per cent more plants than square is
(a) Rectangular (b) Quincunx (c) Hexagonal (d) Contour
Ans: (c)

3. In the quincunx system the number of plants is about
(a) Equal to square (b) Half of square (c) Double of square (d) Three times square
Ans: (c)

4. Urea contains nitrogen at the rate of
(a) 16 per cent (b) 46 per cent (c) 60 per cent (d) 21 per cent
Ans: (b)

5. Muriate of potash contains K2O at the rate of
(a) 46 per cent (b) 16 per cent (c) 60 per cent (d) 30 per cent
Ans: (c)

6. Single superphosphate contains P2O5 at the rate of
(a) 16 per cent (b) 46 per cent (c) 60 per cent (d) 20 per cent
Ans: (a)

7. The training system with a single main stem and tiers of branches is
(a) Open centre (b) Central leader (c) Vase (d) Bush
Ans: (b)

8. Peach and plum are usually trained to
(a) Central leader (b) Open centre (vase) (c) Bush (d) Espalier
Ans: (b)

9. Grape is pruned
(a) Once a year (b) Twice a year (c) Once in two years (d) Not at all
Ans: (b)

10. The dormancy-breaking chemical used in grape is
(a) GA3 (b) NAA (c) Hydrogen cyanamide (d) Ethephon
Ans: (c)

11. Paclobutrazol is used in mango to
(a) Increase fruit size (b) Induce early and regular flowering (c) Check fruit drop (d) Ripen fruit
Ans: (b)

12. NAA is sprayed on mango mainly to
(a) Increase fruit size (b) Check fruit drop (c) Induce flowering (d) Ripen fruit
Ans: (b)

13. TNAU mango varieties include
(a) CO 1 to CO 5 (b) CO 1 to CO 10 (c) PKM 6 (d) None
Ans: (a)

14. Stone grafting in mango is done on seedlings aged
(a) 1 month (b) 8-14 days (c) 6 months (d) 1 year
Ans: (b)

15. Banana plant population at 1.8 x 1.8 m is about
(a) 2,000 (b) 3,086 (c) 5,000 (d) 1,000
Ans: (b)

16. The operation of removing all suckers except the follower in banana is
(a) Pruning (b) Desuckering (c) Denavelling (d) Topping
Ans: (b)

17. The papaya enzyme of commercial value is
(a) Bromelain (b) Papain (c) Pectinase (d) Amylase
Ans: (b)

18. Papaya is a rich source of
(a) Vitamin C and provitamin A (b) Fat (c) Protein (d) Calcium only
Ans: (a)

19. Guava is propagated commercially by
(a) Seed (b) Air-layering (c) Tissue culture (d) Budding
Ans: (b)

20. The leading commercial variety of guava is
(a) Allahabad Safeda (b) Lucknow-49 (Sardar) (c) Banarasi (d) Chittidar
Ans: (b)

21. Sapota is propagated by
(a) Seed (b) Softwood grafting / air-layering (c) Suckers (d) Runners
Ans: (b)

22. The leading rootstock for citrus for vigour and drought tolerance is
(a) Trifoliate orange (b) Rough lemon (c) Troyer citrange (d) Sour orange
Ans: (b)

23. The citrus disease controlled by using Trifoliate orange rootstock is
(a) Canker (b) Phytophthora (gummosis) (c) Greening (d) Tristeza
Ans: (b)

24. Grape maturity is judged mainly by
(a) Skin colour only (b) Brix (c) Firmness (d) Weight
Ans: (b)

25. TNAU acid lime variety is
(a) CO 1 (b) PKM 1 (c) VRI 1 (d) NA 7
Ans: (b)

Fill in the blanks

1. Plants per hectare = 10,000 / _____. **Ans:** area per plant in m2
2. The training system suited to mango is _____. **Ans:** modified leader
3. The application of fertiliser through drip is called _____. **Ans:** fertigation
4. Mango is harvested when specific gravity exceeds about _____. **Ans:** 1.0-1.02
5. Banana is propagated by sword suckers or _____. **Ans:** tissue culture

6. The leading banana export variety is _____. **Ans:** Grand Naine (Robusta)
7. Papaya plants with only female and bisexual flowers are called _____. **Ans:** gynodioecious
8. Guava jelly is rich in _____. **Ans:** pectin
9. Grape is trained on a _____ or trellis. **Ans:** bower
10. The TNAU research station for mango and sapota is at _____. **Ans:** Periyakulam

True or false

1. Hexagonal planting gives fewer plants than square planting. **Ans:** False (more, about 15 per cent)
2. Quincunx planting accommodates filler plants. **Ans:** True
3. Urea is a straight nitrogenous fertiliser. **Ans:** True
4. Mango is a regular bearer in all varieties. **Ans:** False (many are alternate bearers)
5. Banana is a tree. **Ans:** False (it is a giant herb)
6. Papaya bears fruit within a year of planting. **Ans:** True
7. Guava tolerates poor and alkaline soils. **Ans:** True
8. Grape is the leading fresh fruit export of India. **Ans:** True
9. Citrus is very sensitive to waterlogging. **Ans:** True
10. Sapota is propagated mainly by seed for commercial planting. **Ans:** False (by grafting/air-layering)

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Unit III Sub-tropical, temperate crops, minor fruits and nuts

Multiple-choice questions

1. The number of hours below about 7 deg C needed to break dormancy is the
(a) Heat unit (b) Chilling requirement (c) Photoperiod (d) Vernalisation
Ans: (b)

2. Apple is trained to a
(a) Open centre (b) Central leader (c) Bush (d) Espalier
Ans: (b)

3. Peach bears its fruit on
(a) Spurs (b) One-year-old wood (c) Old wood (d) Roots
Ans: (b)

4. The dwarfing rootstock of apple is
(a) M9 (b) Kainth (c) Rough lemon (d) Rangpur lime
Ans: (a)

5. Strawberry is propagated by
(a) Seed (b) Runners (c) Suckers (d) Grafting
Ans: (b)

6. The aggregate fruit among the following is
(a) Mango (b) Custard apple (c) Guava (d) Citrus
Ans: (b)

7. Pomegranate is trained as a
(a) Central leader (b) Multi-stem bush (c) Espalier (d) Cordon
Ans: (b)

8. The leading soft-seeded export variety of pomegranate is
(a) Ganesh (b) Bhagwa (c) Kesar (d) Ruby
Ans: (b)

9. Bahar regulation in pomegranate is done to
(a) Increase tree size (b) Time the crop to the market (c) Reduce pests (d) Increase roots
Ans: (b)

10. Dragon fruit is trained on a
(a) Bower (b) Post with a ring (c) Trellis (d) Wires
Ans: (b)

11. Ber is very responsive to
(a) No pruning (b) Annual heading-back pruning (c) Root pruning (d) Girdling
Ans: (b)

12. The rootstock of ber is
(a) Seedling wild ber (b) Rough lemon (c) Quince (d) M9
Ans: (a)

13. Amla is a very rich source of
(a) Vitamin A (b) Vitamin C (c) Fat (d) Protein
Ans: (b)

14. Most almond varieties are
(a) Self-compatible (b) Self-incompatible (c) Dioecious (d) Parthenocarpic
Ans: (b)

15. The leading walnut variety that flowers late is
(a) Payne (b) Franquette (c) Hartley (d) Chandler
Ans: (b)

16. Pistachio is propagated on the rootstock
(a) Pistacia integerrima (b) M9 (c) Quince (d) Rough lemon
Ans: (a)

17. Low-chill peach varieties include
(a) Elberta (b) Flordasun (c) Redhaven (d) July Elberta
Ans: (b)

18. The leading plum variety is
(a) Santa Rosa (b) Kelsey (c) Frontier (d) Methley
Ans: (a)

19. India is a small producer but a large consumer of
(a) Banana (b) Almond and pistachio (c) Mango (d) Coconut
Ans: (b)

20. Jackfruit is propagated commercially by
(a) Seed (b) Grafting / budding (c) Suckers (d) Runners
Ans: (b)

Fill in the blanks

1. Temperate fruit trees need winter cold, called the _____ requirement. **Ans:** chilling
2. Apple is propagated on dwarfing rootstocks such as _____. **Ans:** M9 (or MM106)
3. Peach is trained to the _____ system. **Ans:** open centre (vase)
4. Strawberry is grown on raised beds covered with _____ mulch. **Ans:** polythene (black/silver-black)
5. Pomegranate is propagated commercially by _____ cuttings. **Ans:** hardwood
6. The leading Indian amla variety is _____. **Ans:** NA-7
7. Dragon fruit belongs to the genus _____. **Ans:** Hylocereus
8. Most almond varieties need _____ varieties for pollination. **Ans:** polliniser
9. Walnut is difficult to graft because it _____. **Ans:** bleeds
10. Ber is propagated by budding on _____ rootstock. **Ans:** wild ber (Ziziphus rotundifolia)

True or false

1. Chilling requirement decides the region suitable for a temperate crop. **Ans:** True
2. Apple bears on one-year-old wood only. **Ans:** False (mostly on spurs)
3. Pomegranate tolerates drought and saline soil. **Ans:** True
4. Dragon fruit is a climbing cactus. **Ans:** True
5. Amla is a rich source of vitamin C. **Ans:** True
6. Almond flowers late and so escapes frost. **Ans:** False (it flowers early and risks frost)
7. India produces enough almond to meet its demand. **Ans:** False

8. Walnut needs a definite winter chilling. **Ans:** True
9. Strawberry is a temperate fruit grown only in the hills. **Ans:** False (low-chill types grow in the plains)
10. Ber gives fruit in winter when few other fruits are available. **Ans:** True

Unit IV Palms and plantation crops

Multiple-choice questions

1. Coconut palms per hectare at 7.5 x 7.5 m square is about
(a) 178 (b) 205 (c) 143 (d) 278
Ans: (a)

2. Coconut palms per hectare in the triangular system at 7.5 x 7.5 m is about
(a) 178 (b) 205 (c) 143 (d) 100
Ans: (b)

3. Coconut is called the
(a) King of fruits (b) Tree of life (c) Poor man's apple (d) Wonder crop
Ans: (b)

4. The leading coconut-producing state in India is
(a) Tamil Nadu (b) Kerala (c) Karnataka (d) Andhra Pradesh
Ans: (b)

5. The TNAU coconut hybrids are
(a) VHC 1-3 (b) CO 1-3 (c) PKM 1-3 (d) VRI 1-3
Ans: (a)

6. Arecanut is planted at a spacing of
(a) 2.7 x 2.7 m (b) 5 x 5 m (c) 7.5 x 7.5 m (d) 9 x 9 m
Ans: (a)

7. The fruit rot disease of arecanut is
(a) Koleroga (b) Bud rot (c) Stem bleeding (d) Leaf blight
Ans: (a)

8. Oil palm is grown from the
(a) Dura type (b) Pisifera type (c) Tenera hybrid (d) Seedling
Ans: (c)

9. Oil palm is planted at
(a) 9 x 9 x 9 m (b) 5 x 5 m (c) 2.7 x 2.7 m (d) 7.5 x 7.5 m
Ans: (a)

10. Palmyrah is
(a) Monoecious (b) Dioecious (c) Hermaphrodite (d) Parthenocarpic
Ans: (b)

11. The tender fruit of palmyrah is called
(a) Neera (b) Nungu (c) Copra (d) Jaggery
Ans: (b)

12. The leading Indian rubber clone is
(a) RRII 105 (b) GT 1 (c) PB 235 (d) RRIM 600
Ans: (a)

13. Rubber tapping begins when the tree girth is about
(a) 20 cm (b) 45-50 cm (c) 80 cm (d) 10 cm
Ans: (b)

14. Rubber is planted at
(a) 5 x 5 m (b) 9 x 9 m (c) 2.7 x 2.7 m (d) 1.8 x 1.8 m
Ans: (a)

15. Multi-tier cropping uses
(a) One crop (b) Two crops (c) Three layers of crops (d) Only palms
Ans: (c)

16. Coconut husk is used for
(a) Oil (b) Coir (c) Copra (d) Toddy
Ans: (b)

17. The Rubber Board is headquartered at
(a) Kochi (b) Kottayam (c) Bengaluru (d) Kolkata
Ans: (b)

18. The Coconut Development Board is at
(a) Kochi (b) Kottayam (c) Chennai (d) Bengaluru
Ans: (a)

19. Cashew and cocoa are supported by the
(a) Tea Board (b) DCCD (c) Coffee Board (d) Rubber Board
Ans: (b)

20. Arecanut is cured mainly into
(a) Copra (b) Kalipak and chali (c) RSS (d) Nibs
Ans: (b)

Fill in the blanks

1. Coconut palms per hectare at 7.5 x 7.5 m square = _____. **Ans:** 178
2. The triangular system gives about _____ per cent more palms than square. **Ans:** 15
3. Arecanut plant population at 2.7 x 2.7 m is about _____. **Ans:** 1,370
4. Oil palm gives crude palm oil and _____ oil. **Ans:** palm kernel
5. Palmyrah sap gives palm sugar and _____. **Ans:** palm wine (neera)
6. The most planted rubber clone in India is _____. **Ans:** RRII 105
7. Coconut is processed into _____ from which oil is extracted. **Ans:** copra
8. The disease of coconut in Kerala is _____ disease. **Ans:** root (wilt)
9. The oil palm research institute is at _____. **Ans:** Pedavegi
10. Growing three crops of different stature in layers is _____ cropping. **Ans:** multi-tier

True or false

1. Coconut is very sensitive to waterlogging. **Ans:** True
2. The triangular system gives fewer palms than the square system. **Ans:** False (more)
3. Arecanut is a tall palm grown for its seed. **Ans:** True
4. Oil palm is a low-yielding oil crop. **Ans:** False (it is the highest-yielding edible oil crop)
5. Palmyrah is a fast-growing palm. **Ans:** False (very slow-growing)
6. Rubber gives a steady income for 25-30 years. **Ans:** True
7. Coconut tolerates coastal salinity. **Ans:** True
8. Karnataka is the leading arecanut producer. **Ans:** True

9. Coir is obtained from the coconut shell. **Ans:** False (from the husk)
10. Multi-tier cropping raises total output per hectare. **Ans:** True

Unit V Beverage crops and cashew

Multiple-choice questions

1. India's rank in world tea production is
(a) First (b) Second (c) Third (d) Fourth
Ans: (b)

2. The leading tea district of south India is
(a) Yercaud (b) The Nilgiris (c) Wayanad (d) Coorg
Ans: (b)

3. A leading south Indian tea clone is
(a) UPASI-9 (b) S.795 (c) CCRP-1 (d) VRI-1
Ans: (a)

4. Tea is harvested by plucking
(a) Mature leaves (b) Two leaves and a bud (c) The whole shoot (d) Only the bud
Ans: (b)

5. Black tea is made by
(a) Drying only (b) Withering, rolling, fermentation, drying (c) Steaming (d) Roasting
Ans: (b)

6. Green tea is made by
(a) Fermenting the leaf (b) Preventing fermentation (c) Roasting the leaf (d) Drying only
Ans: (b)

7. The two commercial species of coffee are
(a) Arabica and robusta (b) Arabica and liberica (c) Robusta and excelsa (d) Arabica and excelsa
Ans: (a)

8. The leading coffee-producing state of India is
(a) Kerala (b) Karnataka (c) Tamil Nadu (d) Andhra Pradesh
Ans: (b)

9. A leading arabica coffee variety is
(a) S.795 (b) S.274 (c) CxR (d) CCRP-1
Ans: (a)

10. Coffee is processed by the
(a) Wet and dry methods (b) Only dry method (c) Only wet method (d) Roasting method
Ans: (a)

11. Cocoa is grown mainly as
(a) A sole crop (b) An intercrop in coconut/arecanut gardens (c) A border crop (d) A dryland crop
Ans: (b)

12. Cocoa beans are fermented to develop
(a) Colour only (b) Chocolate flavour (c) Hardness (d) Size
Ans: (b)

13. India's rank in cashew production in the world is
(a) First (b) Second (c) Third (d) Fourth
Ans: (a)

14. A TNAU cashew variety is
(a) VRI-1 (b) S.795 (c) UPASI-9 (d) CCRP-1
Ans: (a)

15. Cashew is propagated commercially by
(a) Seed (b) Softwood grafting (c) Suckers (d) Runners
Ans: (b)

16. The main pest of cashew is
(a) Tea mosquito bug (b) Red spider mite (c) Berry borer (d) Stem borer
Ans: (a)

17. CNSL is obtained from
(a) Cashew apple (b) Cashew shell (c) Cashew kernel (d) Cashew leaf
Ans: (b)

18. Coffee is grown under
(a) Full sun (b) Shade trees (c) Water (d) Plastic
Ans: (b)

19. The Coffee Board is headquartered at
(a) Bengaluru (b) Kolkata (c) Kochi (d) Kottayam
Ans: (a)

20. The Tea Board zonal office in Tamil Nadu is at
(a) Coonoor (b) Ooty (c) Coimbatore (d) Salem
Ans: (a)

Fill in the blanks

1. India is the _____ largest producer and exporter of tea in the world. **Ans:** second
2. Tea is propagated by single-node _____. **Ans:** cuttings
3. The leading arabica coffee variety is _____. **Ans:** S.795
4. Coffee is grown mainly under _____ trees. **Ans:** shade
5. Cocoa belongs to the genus _____. **Ans:** Theobroma
6. Cocoa beans are fermented for about _____ days. **Ans:** 5-7
7. India is the largest producer and consumer of _____. **Ans:** cashew
8. Cashew nut shell liquid is used in _____. **Ans:** brake linings and resins
9. The main pest of cashew is the _____ bug. **Ans:** tea mosquito
10. The tea clone widely grown in the Nilgiris is _____. **Ans:** UPASI-9

True or false

1. India is the largest tea producer in the world. **Ans:** False (second, after China)
2. Tea grows best on acidic soil. **Ans:** True
3. Robusta coffee is more hardy than arabica. **Ans:** True
4. Coffee is grown in the plains of Tamil Nadu. **Ans:** False (in the hills - Yercaud, Pulneys, Nilgiris)
5. Cocoa gives extra income as an intercrop. **Ans:** True
6. Cashew is a drought-tolerant crop. **Ans:** True
7. Green tea is fermented. **Ans:** False

8. Cashew is processed by roasting and shelling. **Ans:** True
9. India imports raw cashew nuts for processing. **Ans:** True
10. Cocoa is a shade-loving crop. **Ans:** True

APPENDIX H | Solved Numerical Problems

Numerical questions on plant population, fertiliser conversion, high-density economics and cost-and-return are common in examinations. The worked problems below use round figures for clarity; in the examination, substitute the values given in the question and show every step.

H.1 Plant population

Problem 1 Mango at 8 x 8 m in a 2-hectare block

```
Plants per hectare = 10,000 / (8 x 8) = 10,000 / 64 = 156.25  ->  156 plants
For 2 hectares = 156 x 2 = 312 plants.
```

Problem 2 Banana at 2 x 2 m in 1.5 hectares

```
Plants per hectare = 10,000 / (2 x 2) = 10,000 / 4 = 2,500 plants
For 1.5 hectares = 2,500 x 1.5 = 3,750 plants.
```

Problem 3 Coconut at 8 x 8 m in the triangular system, 1 hectare

```
Square number = 10,000 / 64 = 156.25
Triangular = 156.25 x (1 / 0.866) = 156.25 x 1.1547 = 180.4  ->  about 180 palms
```

Problem 4 Guava in the quincunx system at 5 x 5 m, 1 hectare

```
Square grid = 10,000 / 25 = 400 plants
Quincunx adds one centre plant in each square, roughly doubling:
   about 400 x 2 = 800 plants (the centre plants are fillers).
```

Problem 5 Area needed for 1,000 cashew plants at 7 x 7 m

```
Plants per hectare at 7 x 7 m = 10,000 / 49 = 204.1
Area for 1,000 plants = 1,000 / 204.1 = 4.9 hectares (about 4.9 ha).
```

Problem 6 Arecanut at 2.7 x 2.7 m in a 0.5-hectare plot

```
Plants per hectare = 10,000 / (2.7 x 2.7) = 10,000 / 7.29 = 1,371.7  ->  1,371
For 0.5 hectare = 1,371 x 0.5 = about 686 plants.
```

H.2 Fertiliser dose conversion

To convert a nutrient dose into fertiliser, divide the nutrient by the fraction of that nutrient in the fertiliser: urea has 46 per cent N, single superphosphate (SSP) 16 per cent P2O5, and muriate of potash (MOP) 60 per cent K2O. For other fertilisers, use their stated content.

Problem 7 Mango tree: 500 g N, 250 g P2O5, 500 g K2O per year

```
Urea = 500 / 0.46 = 1,087 g = 1.09 kg
SSP  = 250 / 0.16 = 1,562 g = 1.56 kg
MOP  = 500 / 0.60 = 833 g = 0.83 kg
Split: half urea + all SSP + half MOP in June-July;
       half urea + half MOP in September-October.
```

Problem 8 Coconut palm: 500 g N, 320 g P2O5, 1,200 g K2O per year

```
Urea = 500 / 0.46 = 1,087 g = 1.09 kg
SSP  = 320 / 0.16 = 2,000 g = 2.00 kg
MOP  = 1,200 / 0.60 = 2,000 g = 2.00 kg
Apply in two splits in a ring 1.5-2.0 m from the trunk.
```

Problem 9 Tea field: 120 kg N, 40 kg P2O5, 70 kg K2O per hectare per year

```
Urea = 120 / 0.46 = 260.9 kg per hectare
SSP  = 40 / 0.16 = 250.0 kg per hectare
MOP  = 70 / 0.60 = 116.7 kg per hectare
Apply in 3-4 splits during the growing season.
```

Problem 10 Coffee plant: 40 g N, 30 g P2O5, 50 g K2O per year

```
Urea = 40 / 0.46 = 86.9 g per plant
SSP  = 30 / 0.16 = 187.5 g per plant
MOP  = 50 / 0.60 = 83.3 g per plant
Apply in two splits, at the onset of the monsoon and after.
```

Problem 11 Banana crop: 200 g N, 100 g P2O5, 250 g K2O per plant per year

```
Urea = 200 / 0.46 = 434.8 g per plant
SSP  = 100 / 0.16 = 625.0 g per plant
MOP  = 250 / 0.60 = 416.7 g per plant
Apply in 4-6 splits through the season, preferably by fertigation.
```

H.3 High-density planting economics

Problem 12 Compare conventional and high-density guava over the first 5 years

```
Assumptions: fruit price Rs 30 per kg.
                     Conventional (6 x 6 m)   High density (3 x 3 m)
  Plants per hectare        278                      1,111
  Cost of plants (Rs 40)  11,120                    44,440
  Planting and care (5 yr) 1,80,000                3,60,000
  Total cost (5 yr)        1,91,120                4,04,440
  Yield at 5th year (t/ha)   12                       22
  Gross return (Rs 30/kg)  3,60,000                6,60,000
  Net return at 5th year   1,68,880                2,55,560

Conclusion: HDP costs more initially but gives a higher net return
per hectare from the 5th year, and it also recovers the cost earlier
because the trees bear sooner.
```

H.4 Cost and return

Problem 13 Cost and return for one hectare of mango (indicative)

```
Spacing 10 x 10 m  ->  100 plants per hectare; plant cost Rs 80 each.

  ITEM                                        AMOUNT (Rs)
  Cost of plants (100 x 80)                      8,000
  Pits, planting and establishment              40,000
  Annual care, years 1-8 (Rs 60,000/yr)        4,80,000
  Total cost to 8th year                       5,28,000

  Yield at 8th year = 9 t/ha = 9,000 kg.
  Gross return at Rs 40 per kg = 9,000 x 40 = Rs 3,60,000 per year.
  From the 9th year the tree gives 10-12 t/ha, so annual gross return
  rises to about Rs 4,00,000-4,80,000, giving a good net return over the
  25-year life of the orchard.
```

Problem 14 Cost and return for one hectare of papaya (one crop)

Spacing 2 x 2 m -> 2,500 plants per hectare; seedling cost Rs 8 each.

ITEM	AMOUNT (Rs)
Seedlings (2,500 x 8)	20,000
Land preparation, beds and planting	30,000
Manures and fertilisers	40,000
Drip and irrigation (share)	20,000
Plant protection and labour	45,000
Harvesting and packing	25,000
Total cost	1,80,000

Yield = 60 t/ha = 60,000 kg.
Gross return at Rs 12 per kg = 60,000 x 12 = Rs 7,20,000
Net return = 7,20,000 - 1,80,000 = Rs 5,40,000
Benefit-cost ratio = 7,20,000 / 1,80,000 = 4.0

Problem 15 Income from intercropping a young coconut garden

A 1-hectare coconut garden (about 178 palms) has about 60 per cent of the area free between the palms for the first 5-6 years.

Intercrop: banana at 1.8 x 1.8 m on 0.6 ha = about 1,850 plants
Cost of the intercrop Rs 1,20,000
Yield: 40 t on 0.6 ha at Rs 15/kg = Rs 6,00,000
Net income from the intercrop Rs 4,80,000 over the crop period

This additional income offsets the cost of establishing and maintaining the coconut palms until they begin to bear.

Problem 16 Drip emitter requirement for a mango orchard

A 1-hectare mango orchard has 400 plants (5 x 5 m).
Each plant is to get 2 emitters of 4 litres per hour.

Number of emitters = 400 x 2 = 800
Discharge = 800 x 4 = 3,200 litres per hour = 3.2 m3 per hour
If the pump runs for 2 hours a day, water delivered = 6.4 m3 per day, i.e. about 16 litres per plant per day. Adjust the duration to the crop's water requirement and the soil type.

APPENDIX I | Consolidated Nutrient and Plant-Protection Tables

These tables bring together the recommended nutrient doses and the major pests and diseases of every crop in the course, for quick revision.

I.1 Recommended nutrient doses (crop-wise)

Table I.1 Nutrient doses per plant or per hectare per year

Crop	Nitrogen (N)	Phosphorus (P_2O_5)	Potash (K_2O)	Basis
Mango (bearing tree)	500 g	250 g	500 g	Per tree per year
Banana	200 g	100 g	250 g	Per plant per year
Papaya	200-250 g	200-250 g	200-250 g	Per plant per year
Guava (bearing tree)	250-600 g	150-300 g	250-600 g	Per tree per year
Sapota (bearing tree)	500 g	250 g	500 g	Per tree per year
Citrus (bearing tree)	400-600 g	200-300 g	300-500 g	Per tree per year
Grape	200-400 g	200-500 g	200-400 g	Per vine per year
Pineapple	8-12 g	3-4 g	8-12 g	Per plant per year
Pomegranate (bearing)	200-400 g	100-200 g	200-400 g	Per plant per year
Apple (bearing tree)	500-700 g	250-350 g	500-700 g	Per tree per year
Strawberry	5-10 g	3-5 g	5-10 g	Per plant per season
Ber	200-300 g	100-200 g	200-300 g	Per tree per year
Coconut (adult palm)	500 g	320 g	1,200 g	Per palm per year
Arecanut (adult palm)	100-150 g	40-60 g	150-200 g	Per palm per year
Oil palm (bearing)	1-2 kg	0.5-1 kg	1.5-2.5 kg	Per palm per year
Rubber (mature tree)	300-600 g	200-300 g	300-600 g	Per tree per year
Tea	100-150 kg	30-40 kg	60-80 kg	Per hectare per year
Coffee (bearing plant)	30-60 g	20-40 g	30-60 g	Per plant per year
Cashew	100-200 g	50-100 g	100-200 g	Per tree per year

Source: standard package of practices and TNAU/ICAR material. Doses vary with age, soil and yield target; adjust to local recommendations.

I.2 Major pests and diseases (crop-wise)

Table I.2 Important pests and diseases and their management

Crop	Major pests	Major diseases	Key management measures
Mango	Mango hopper, fruit fly, mealybug, stone weevil	Anthracnose, powdery mildew, malformation	Pruning for light, timely sprays, fruit bagging, clean orchard
Banana	Pseudostem weevil, rhizome weevil, aphid (virus vector)	Panama wilt (Fusarium), bunchy top, Sigatoka	Clean tissue-cultured plants, resistant varieties, drainage
Papaya	Aphid (ring spot vector), fruit fly, mealybug	Ring spot virus, root and stem rot	Tolerant varieties, vector control, raised beds, drainage
Guava	Fruit fly, tea mosquito bug, mealybug	Wilt (Fusarium), anthracnose	Resistant rootstock, clean planting material, bagging
Sapota	Leaf webber, bud borer, chiku moth	Sooty mould, leaf spot	Pruning, timely sprays, orchard sanitation
Citrus	Citrus leaf miner, fruit fly, aphid, mites	Canker, gummosis (Phytophthora), greening (HLB)	Tolerant rootstock, clean budwood, drainage, vector control
Grape	Flea beetle, thrips, mealybug, mite	Downy mildew, powdery mildew, anthracnose	Timely fungicide sprays, dry ripening season, canopy management
Pineapple	Mealybug, scale, nematode	Heart rot, root rot	Good drainage, clean suckers, soil treatment
Pomegranate	Fruit borer, aphid, thrips	Bacterial blight, anthracnose	Clean planting material, resistant variety (Bhagwa), bagging
Apple	Codling moth, San Jose scale, woolly aphid	Scab, powdery mildew, canker	Dormant sprays, resistant varieties, orchard sanitation
Strawberry	Mite, aphid, thrips	Grey mould, leaf spot, virus	Mulching, clean plants, drip irrigation, ventilation
Coconut	Rhinoceros beetle, red palm weevil, black-headed caterpillar, Eriophyid mite	Root (wilt), bud rot, stem bleeding, leaf blight	Clean cultivation, biological control (Metarhizium, baculovirus), remove affected palms
Arecanut	Spindle bug, inflorescence caterpillar, root grub	Koleroga (fruit rot), yellow leaf disease	Bordeaux spray, tolerant varieties (Mangala), remove diseased palms
Oil palm	Rhinoceros beetle, red palm weevil, bunch moth	Ganoderma basal stem rot	Clean cultivation, biological control, remove affected palms
Rubber	Termite, cockchafer beetle	Abnormal leaf fall, powdery mildew, Corynespora leaf fall, root disease	Resistant clones, fungicide spray, clean cultivation
Tea	Tea mosquito bug, red spider mite, thrips, looper	Blister blight, grey blight, red rust, root disease	Resistant clones, pruning, biological and chemical control
Coffee	Coffee berry borer, stem borer, white stem borer	Leaf rust, black rot of berries	Resistant varieties, shade management, timely sprays

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Crop	Major pests	Major diseases	Key management measures
Cocoa	Pod borer, mealybug, mirid bug	Black pod rot, swollen shoot, witches' broom	Resistant varieties, shade management, clean cultivation
Cashew	Tea mosquito bug, stem and root borer, leaf miner, apple borer	Die-back, anthracnose, powdery mildew	Resistant varieties, timely sprays, remove affected parts

Source: standard plant-protection texts and TNAU/ICAR/commodity-board material. This is a summary only; use the recommended chemicals and doses for each crop.

APPENDIX J | Post-harvest Technology, Processing and Value Addition

Fruit and plantation crops are perishable, and a large share of the crop is lost between harvest and the consumer. This appendix sets out the general principles of post-harvest handling, ripening, storage, processing and value addition, and brings the crop-wise value-added products together in one place.

J.1 Post-harvest losses and their causes

Post-harvest loss is the loss of quantity and quality of produce after harvest, and in fruit it is large - commonly 15-30 per cent and higher for perishable crops - because fruit continues to respire, transpire and ripen after harvest. Losses are caused by careless harvesting and handling, bruising, exposure to heat, poor grading and packaging, inadequate transport, and the absence of a cold chain. Reducing these losses is often cheaper than raising production, so post-harvest management is as important as field management.

The main causes of loss are mechanical (cuts, bruises, compression), physiological (water loss, shrivelling, chilling injury, over-ripening), pathological (decay by fungi and bacteria) and market-related (glut, poor price, delay). The countermeasures are careful handling, prompt cooling, correct packaging, refrigerated transport and cold storage.

J.2 Harvesting and maturity

Fruit must be harvested at the correct stage of maturity (see Appendix C). The harvest method affects the keeping quality: fruit is harvested by hand or with a clipper, leaving a short stalk, and never by shaking the tree, which bruises the fruit. Harvesting is done in the cool hours of the day, the fruit is collected in clean containers and kept in shade, and it is transported to the packing house without delay.

J.3 Post-harvest operations

Table J.1 Sequence of post-harvest operations and their purpose

Operation	Purpose	Crops where it is important
Cleaning and washing	Remove dust, latex, soil and spray residue	Mango (de-sapping), citrus, grape, banana
Sorting and grading	Remove defective fruit; grade by size, colour and quality	All fruit; essential for export
Waxing	Reduce water loss, improve appearance, extend shelf life	Citrus, apple, mango
Pre-cooling	Remove field heat quickly to slow respiration and decay	Grape, strawberry, mango, banana
Chemical treatment	Control decay (fungicide), delay ripening, degreen	Citrus (degreening), banana (ethylene)
Packaging	Protect the fruit, allow handling and stacking	All fruit; ventilated boxes, trays
Cold storage	Extend shelf life by slowing respiration and decay	Apple, grape, citrus, banana, strawberry
Transport	Move the fruit to the market with minimum damage	Refrigerated or ventilated vehicles for export

Source: standard post-harvest texts (Kader; TNAU material).

J.4 Ripening and degreening

Climacteric fruit (mango, banana, papaya, guava, sapota) can be ripened artificially. Ethylene gas (100-150 ppm) is used in ripening chambers at the correct temperature and humidity; calcium carbide is prohibited because it is harmful. Non-climacteric fruit (citrus, grape, pineapple) do not ripen after harvest, but citrus and some other fruit can be 'degreened' - the green colour is removed with ethylene so that the fruit looks ripe, though it does not become sweeter. Ripening rooms are now used for banana, mango and papaya, and controlled ripening is essential for a uniform market product.

J.5 Storage

Storage extends the season and stabilises price. Ordinary cold storage slows respiration and decay and is used for apple, grape, citrus, banana and strawberry. Controlled-atmosphere (CA) storage, in which oxygen is reduced and carbon dioxide raised, keeps apple for many months. Modified-atmosphere packaging (MAP) is used for small fruit and cut fruit. The storage temperature and humidity are specific to each crop; chilling injury occurs when a tropical fruit is stored too cold.

Table J.2 Approximate storage temperature and life of selected fruit

Crop	Storage temperature	Approximate storage life	Note
Mango	10-13 deg C	2-3 weeks	Chilling injury below 10 deg C
Banana	13-15 deg C	2-4 weeks	Ripen in a chamber before sale
Papaya	10-13 deg C	2-3 weeks	Sensitive to chilling
Guava	8-10 deg C	2-3 weeks	Short life; MAP helps
Sapota	10-15 deg C	2-3 weeks	Ripens quickly
Citrus	5-8 deg C	4-8 weeks	Degreening before storage for colour
Grape	0-2 deg C	2-4 months	SO2 pads for export; fumigation
Apple	0-2 deg C	4-8 months	CA storage extends life
Strawberry	0-2 deg C	5-7 days	Very perishable; pre-cool at once
Coconut (tender nut)	5-8 deg C	2-3 weeks	Keep cool and humid

Source: standard post-harvest texts. Values are indicative and vary with cultivar and handling.

J.6 Processing and value addition

Processing converts a perishable crop into a stable, higher-value product, reduces loss and creates employment. The main methods are drying (amchur, raisins, dried ber, dried amla), canning and bottling (juice, squash, jam, jelly), freezing (pulp, cut fruit), concentration (juice concentrate, pulp), fermentation (wine, vinegar) and extraction (oil, pectin, essential oils). Value addition raises the return and uses fruit that is small, blemished or surplus, which would otherwise be wasted.

Diagram J.1 General flow of fruit processing and value addition

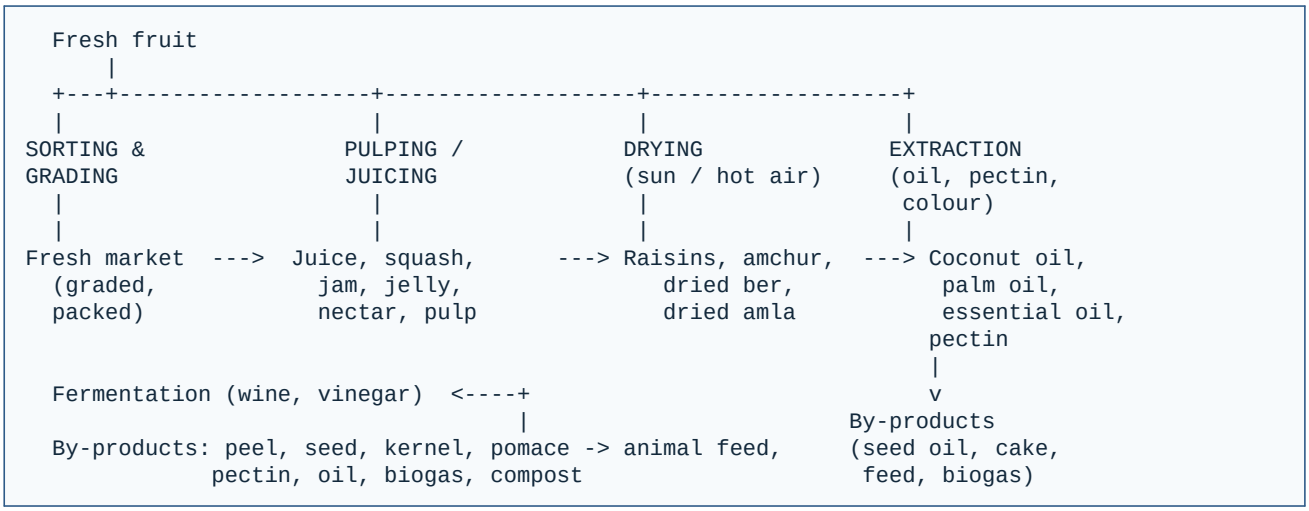


Figure J.1 General flow of fruit processing and value addition.

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Table J.3 Crop-wise major processed and value-added products

Crop	Major value-added products
Mango	Pulp, juice, squash, jam, pickle, chutney, amchur (dried slices), cereal flakes, canned slices
Banana	Chips, puree, pulp, flour, wine, vinegar, fibre from pseudostem, stem and flower as vegetable
Papaya	Papain (enzyme), pulp, juice, jam, tutti-frutti from raw fruit, dried and candied products
Guava	Jelly, jam, cheese, juice, nectar, toffee; source of pectin
Sapota	Milk shake, preserve, jam, dried fruit
Citrus	Juice, squash, cordial, jam, marmalade, peel oil, pectin, candied peel
Grape	Raisins, wine, juice, grape seed oil, vinegar
Pineapple	Canned slices, juice, squash, jam, bromelain, candied fruit
Pomegranate	Juice, syrup, wine, anardana (dried aril), rind tannin
Jackfruit	Chips, canned young fruit, jam, jackfruit flour from seed, tender fruit as vegetable
Apple	Juice, concentrate, jam, sauce, vinegar, cider, dried fruit
Strawberry	Jam, squash, ice cream, freeze-dried fruit, fruit bar
Ber	Candy, preserve, murabba, dried ber
Custard apple	Pulp, ice cream, nectar, custard powder
Amla	Preserve, candy, murabba, dried powder, juice, hair oil, chyawanprash
Dragon fruit	Juice, jam, wine, dried fruit, natural red colour
Coconut	Copra, coconut oil, virgin coconut oil, coconut milk, desiccated coconut, coconut water, coconut chips, coir, activated carbon, shell charcoal
Arecanut	Dried chali and kalipak, scented supari, pan masala
Oil palm	Crude palm oil, palm kernel oil, palm biodiesel, palm wine (toddy)
Palmyrah	Palm sugar (jaggery), palm candy, neera, palm wine, fibre, nungu
Rubber	Sheet rubber (RSS), block rubber, latex concentrate, gloves, footwear, tyres
Tea	Black tea, green tea, oolong tea, tea bags, flavoured and organic tea, instant tea
Coffee	Roasted and ground coffee, instant coffee, monsooned coffee, flavoured coffee
Cocoa	Chocolate, cocoa butter, cocoa powder, cocoa nibs, cocoa-based beverages
Cashew	Roasted and salted kernels, kernel pieces, cashew nut shell liquid (CNSL), cashew apple juice, feni

Source: standard processing texts and commodity-board material.

J.7 Marketing, cold chain and export

Marketing of fruit depends on a chain of intermediaries, and the grower's share of the consumer price is often small. Producer organisations, contract farming, direct marketing and e-marketing (such as the electronic National Agriculture Market) improve the grower's share. For export, a cold chain is essential: harvest, pre-cool, pack in the correct box, and move by refrigerated container. India exports fresh fruit and vegetables worth about USD 1.8 billion, led by grapes, mango, pomegranate, banana and orange, and the main markets are the Netherlands, the UAE, Iraq, Iran and Bangladesh. Sanitary and phytosanitary (SPS) requirements, residue limits and, for some markets, irradiation or vapour-heat treatment must be met.

J.8 Reducing post-harvest losses - a summary

- Harvest at the correct maturity and handle the fruit gently.
- De-sap and clean the fruit promptly; remove field heat by pre-cooling.
- Grade and pack in the correct ventilated or padded container.
- Use the correct storage temperature and humidity for each crop.
- Ripen climacteric fruit in a chamber with ethylene, not calcium carbide.
- Process surplus and low-grade fruit into pulp, juice, jam, chips and dried products.
- Build the cold chain from the farm gate to the market and the port.
- Use value addition and branding to capture a larger share of the consumer price.

APPENDIX K | Orchard Establishment, Farming Systems, Organic and GAP

This appendix covers the practical decisions that come before and around crop production: how to choose a site, lay out an orchard, protect it, irrigate it, and fit it into a farming system. It also covers organic fruit production and Good Agricultural Practices, which are increasingly required by the market.

K.1 Site selection for an orchard

The site decides the fate of the orchard, because fruit crops are perennial and mistakes cannot be corrected easily. The main factors are climate (temperature, rainfall, frost, wind, sunlight), soil (depth, texture, drainage, pH, fertility), water (quantity and quality for irrigation), topography and slope, and the market, labour and transport available nearby. The best site for a tropical crop has deep, well-drained soil, good air drainage and frost-free winters; a temperate crop needs the right elevation and chilling.

- **Climate:** match the crop to the temperature, rainfall and frost pattern of the site.
- **Soil:** deep (at least 1 m), well drained, pH 6.0-7.5 for most crops, free of hard pan.
- **Water:** an assured source of good-quality irrigation water; test for salinity.
- **Topography:** gentle slopes are better than flat, poorly drained land; avoid frost pockets.
- **Market and infrastructure:** proximity to a market, a processing unit, roads, cold storage and labour.
- **Wind:** avoid exposed sites, or plan a windbreak.

K.2 Orchard layout and planting

Before planting, the land is cleared, levelled where necessary, and a layout plan is drawn. The planting positions are marked with a rope and pegs after a base line is fixed along one edge and a right angle is set out; the planting system (square, rectangular, quincunx, hexagonal or contour) is chosen to suit the crop and the site. Pits are dug to the recommended size well before planting, exposed to the sun, and filled with topsoil mixed with farmyard manure, a little rock phosphate and, where needed, a soil amendment. Planting is done at the onset of the monsoon so that the young plant has the whole rainy season to establish.

Diagram K.1 Marking an orchard layout

```
Step 1: fix a base line AB along one edge of the field.
A *---------------------------------* B

Step 2: set a right angle at A (3-4-5 method or a cross staff) and
        run a line AC at 90 degrees.
A *---------------------------------* B
  |
  |
  * C

Step 3: mark the spacing along AB and AC with pegs, then complete the
        grid by running parallel lines.
A *---*---*---*---*---*---*---*---* B
  |   |   |   |   |   |   |   |   |
  *---*---*---*---*---*---*---*---*
  |   |   |   |   |   |   |   |   |
  *---*---*---*---*---*---*---*---*
  |   |   |   |   |   |   |   |   |
  * C
For the contour system on slopes, the lines follow the contour instead
of the straight grid, and a basin is made on the up-slope side.
```

Figure K.1 Marking out an orchard layout.

K.3 Windbreaks and shelterbelts

A windbreak is a row of trees planted to protect the orchard from wind, which otherwise causes lodging, fruit drop, bruising and increased water loss. A good windbreak is a tall, dense, fast-growing, wind-firm species that is not a host of the crop's pests; Casuarina, Glyricidia, Eucalyptus (where permitted), silver oak and bamboo are used. The windbreak is planted at right angles to the prevailing wind, at a distance from the crop, and is pruned to keep it dense to the base. Glyricidia also supplies green leaf manure.

K.4 Irrigation systems

Fruit crops are irrigated by the basin, furrow, sprinkler or drip system. The basin system, in which a ring basin is made around each tree, is traditional and simple; the furrow system is used for row crops such as banana and pineapple. Sprinkler irrigation is used for strawberry and some plantation crops, but it wets the foliage and encourages disease. Drip irrigation is now standard for fruit and plantation crops because it saves water, keeps the soil evenly moist, allows fertigation and keeps the foliage dry. The water requirement depends on the crop, the soil and the climate, and is scheduled by soil moisture, pan evaporation or a tensiometer.

Table K.1 Comparison of irrigation systems

System	Water-use efficiency	Advantages	Limitations
Basin	Low-moderate	Simple, cheap, suits tree crops	High water loss, labour-intensive
Furrow	Moderate	Suits row crops, cheap	Water loss, erosion on slopes
Sprinkler	Moderate-high	Uniform, suits strawberry and fodder	Wets foliage, wind drift, disease
Drip	High (80-90 per cent)	Saves water, allows fertigation, dry foliage, less weed	Initial cost, clogging, needs maintenance

Source: standard irrigation texts and TNAU material.

K.5 Farming systems with fruit and plantation crops

Fruit and plantation crops are rarely grown alone. Intercropping, mixed cropping, multi-tier cropping and agroforestry make fuller use of land, light, water and labour and reduce risk. In the young orchard, short-duration crops (pulses, groundnut, vegetables) give income before the main crop bears. In the mature coconut garden, a three-tier system of coconut, arecanut or cocoa, and pineapple or turmeric is common. Agroforestry combines trees with crops and livestock. These systems are especially important for smallholders, because they spread income and risk.

Table K.2 Farming systems and their uses

System	Description	Example
Intercropping	Short-duration crop between the rows of the main crop	Groundnut or vegetables in a young mango orchard
Mixed cropping	Two or more crops grown together without a fixed row plan	Banana and papaya in a homestead
Multi-tier cropping	Crops of different stature in layers	Coconut + cocoa + pineapple
Agroforestry	Trees with crops and/or livestock	Glyricidia border with fruit and fodder
Boundary / filler planting	Filler or border trees for early income or protection	Papaya as filler; Casuarina as windbreak

Source: standard farming-systems texts.

K.6 Organic fruit production

Organic production avoids synthetic fertilisers and pesticides and relies on organic manures, compost, green manures, biofertilisers, biological pest control and cultural methods. It improves soil health and often fetches a premium price, and it is suited to crops such as coconut, banana, mango and spices where export demand for organic produce is strong. The main practices are the use of farmyard manure and compost, vermicompost, green manuring with Glyricidia and sunhemp, biofertilisers (Azospirillum, Phosphobacteria, Rhizobium, mycorrhiza), biopesticides (neem, Bacillus thuringiensis, Trichoderma, Metarhizium, baculovirus), mulching, and mechanical and cultural weed control. Certification requires a conversion period and record-keeping.

K.7 Good Agricultural Practices and traceability

Good Agricultural Practices (GAP) are a set of standards covering site selection, planting material, soil and water management, plant protection, harvest, post-harvest handling, worker safety and environmental care, designed to produce safe, quality food. GAP certification (such as GLOBALG.A.P. or IndiaGAP) is increasingly required for export, and traceability - the ability to trace a consignment back to the farm and the field - is part of it. Residue limits, hygiene, and the correct use of approved chemicals are the key requirements. Adopting GAP improves market access and price.

K.8 Orchard rejuvenation and replanting

Old, senile and unproductive orchards are common in India. Rejuvenation is the restoration of such an orchard to productivity by heavy pruning, manuring, irrigation, and the removal of dead and diseased wood; it is cheaper than replanting and gives quick results. Where the orchard is beyond recovery or the variety is obsolete, replanting with improved varieties and rootstocks is done. Replanting needs care because old orchard soil may carry pathogens; a rotation with a non-host crop, soil solarisation or fumigation reduces the risk. Rejuvenation and replanting are central to raising the low productivity of Indian orchards.

APPENDIX L | Model Answers to Selected Long-answer Questions

The exam-focus sections give outlines for the long-answer questions of each unit. This appendix gives full model answers to ten important questions, to show the depth and structure expected in the examination. Each answer should be supported by a table or a diagram where the question allows it.

Model answer 1 Importance and scope of the fruit and plantation crop industry in India

The fruit and plantation crop industry is the high-value, high-employment segment of Indian agriculture. India is the second largest producer of fruits and vegetables in the world, after China, and produced about 113 million tonnes of fruit in 2023-24 from about 7.2 million hectares; India ranks first in the world in the production of banana, lime and lemon, and papaya. Horticulture contributes about one-third of the value of agricultural output from about 18 per cent of the cropped area.

The economic importance rests on five pillars. First, value per unit area: a well-managed mango or banana orchard returns several times the income of a cereal crop on the same land. Second, employment: these crops are labour-intensive in planting, pruning, harvesting and processing, and so support rural employment through the year. Third, foreign exchange: India exported fresh fruit and vegetables worth about USD 1.8 billion in 2023-24, and grapes, mango, pomegranate and banana are the leading fresh-fruit earners. Fourth, raw material for industry: pulp, juice, jam, coconut oil, coir, rubber, tea and coffee all depend on these crops. Fifth, nutrition and environment: fruit supplies vitamins, minerals and fibre that cereal diets lack, and perennial orchards check erosion and act as carbon sinks on hill slopes.

The scope for growth is large. Per-capita fruit consumption is below the recommended level, urban demand for quality fruit and processed products is rising, and organised retail, processing and exports are expanding. Average fruit productivity is only about 15-16 tonnes per hectare, and old and seedling orchards, poor planting material and weak post-harvest handling are the main gaps. Plantation crops are the backbone of several hill and coastal economies - tea in the Nilgiris, coffee in Yercaud and the Pulneys, rubber and arecanut in the Western Ghats, cashew in the coastal belt, and coconut in the coastal districts - and their value addition (virgin coconut oil, coconut water, rubber products, palm sugar) is a growing source of income. In short, the industry is important today and has a large, unexhausted scope for the future.

Model answer 2 Classification of fruit crops with examples

Fruit crops are classified in several ways, each serving a purpose. By climatic adaptation, they are tropical (mango, banana, papaya, guava, sapota, pineapple, coconut, cashew), sub-tropical (citrus, grape, pomegranate, custard apple, ber, jackfruit) and temperate (apple, pear, peach, plum, strawberry, almond, walnut, pistachio). This classification decides which crop suits a location.

By botanical family, related crops are grouped together, which predicts pest and cultural similarities: Anacardiaceae (mango, cashew), Musaceae (banana), Rutaceae (citrus), Rosaceae (apple, pear, peach, plum, strawberry, almond), Arecaceae (coconut, arecanut, oil palm, palmyrah) and so on. By fruit type (pomology), fruit is a drupe (mango, peach, plum, coconut), a berry (banana, grape, guava), a hesperidium (citrus), a pome (apple, pear), an aggregate (custard apple, strawberry), a multiple or syncarp (pineapple, jackfruit) or a nut (almond, walnut, pistachio). By ripening behaviour, fruit is climacteric (mango, banana, papaya, guava, sapota, apple, pear, peach, plum, custard apple) or non-climacteric (citrus, grape, pineapple, strawberry, cashew). By life cycle, fruit plants are herbaceous perennials (banana, pineapple, strawberry), woody perennials (mango, guava, sapota), palms (coconut, arecanut, oil palm) or vines (grape). The ripening classification decides harvest maturity, ripening and storage; the pomological classification helps in understanding fruit development and harvest.

Model answer 3 Production technology of mango

Climate and soil. Mango needs a warm, frost-free climate and a dry spell before flowering; deep, well-drained soil with pH 5.5-7.5 is ideal. **Varieties.** Banganapalli, Neelam, Alphonso, Totapuri, Rumani, and the TNAU varieties CO 1 to CO 5 and PKM 1 and PKM 2. **Propagation.** Veneer, stone or epicotyl grafting on a seedling rootstock of a polyembryonic type such as Olour or Bappakai. **Planting.** Conventional spacing 10 x 10 m (about 100 trees per hectare); high-density planting 5 x 5 m (400 trees) and ultra-high-density 3 x 2 m (about 1,667 trees), with canopy control by pruning and paclobutrazol.

Training and pruning. Modified-leader training; annual light pruning after harvest to open the centre, remove water sprouts and dead wood and keep the tree below about 6 m. **Nutrient management.** A bearing tree gets about 500 g N, 250 g P2O5 and 500 g K2O per year, split into two applications, plus farmyard manure and micronutrient sprays. **Water.** Irrigation at flowering and fruit development, with water withheld before flowering to induce the flush; drip fertigation is used. **Growth regulation.** Paclobutrazol for early and regular flowering; NAA to check fruit drop; GA3 to improve size. **Disorders.** Alternate bearing, mango malformation, fruit drop and spongy

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tissue. **Maturity and harvest.** Harvest mature-green when the shoulders fill, the colour lightens, the specific gravity exceeds 1.0-1.02 and the TSS is 12-15 per cent. Yield is 8-10 tonnes per hectare conventionally and 15-25 tonnes in well-managed high-density orchards. **Value addition.** Pulp, juice, pickle, chutney, amchur and dried slices.

Model answer 4 Production technology of banana

Banana is the largest fruit crop of India by production, grown on about 0.99 million hectares with about 37.8 million tonnes. It needs a warm, humid climate (15-35 deg C), rich, well-drained soil with pH 6.0-7.5, and does not tolerate waterlogging or frost. Commercial varieties include Grand Naine (Robusta), Poovan, Nendran, Rasthali, Karpuravalli and the TNAU varieties CO 1, CO 2 and PKM 1 to PKM 5. It is propagated by sword suckers or, increasingly, by tissue-cultured plantlets, which are disease-free and uniform.

Planting is at 1.8 x 1.8 m (about 3,086 plants per hectare) for tall types and closer for dwarf types, in pits filled with farmyard manure. The main cultural operations are desuckering (removal of all suckers except the follower), removal of the male bud (denavelling) to improve bunch size, propping of heavy bunches, and removal of dry leaves. A high-yielding crop is given about 200 g N, 100 g P2O5 and 250 g K2O per plant per year in several splits, plus farmyard manure, applied by fertigation through drip, which is the standard method. The crop needs frequent, generous irrigation.

The bunch is ready about 90-150 days after flowering, judged by the plumpness of the fingers, loss of angularity of the ridges and colour change; fruit is harvested green for distant markets and ripened in a chamber with ethylene. Yield is 40-60 tonnes per hectare. The main disorders are the potassium and magnesium deficiency rosetting and boron deficiency, and the main diseases are Panama wilt (Fusarium), bunchy top and Sigatoka, managed by clean tissue-cultured planting material and resistant varieties. Value-added products are chips, puree, pulp, flour, wine and fibre. The crop gives a net return of about Rs 5 lakh per hectare and a benefit-cost ratio of about 2.9 in a well-managed orchard.

Model answer 5 Production technology of grape

Grape is the leading fresh-fruit export of India, grown on about 0.17 million hectares with about 3.9 million tonnes, mainly in Maharashtra (about 67 per cent), Karnataka, Tamil Nadu and Mizoram. It needs a warm, dry climate with a dry ripening season and a deep, well-drained soil with pH 6.5-8.0. Leading varieties are Thompson Seedless and its selections (Sonaka, Tas-A-Ganesh, Manik Chaman), Sharad Seedless, Flame Seedless, Anab-e-Shahi and Bangalore Blue.

Grape is propagated by hardwood cuttings, or by grafting on resistant rootstocks (Dog Ridge, 110-R, 1103-P) for saline or nematode-infested soils. It is trained on a bower (3 x 3 m) or a trellis (2 x 2 m) and pruned twice a year: forward (foundation) pruning after harvest to build the frame by cutting canes back to 1-2 buds, and back (winter) pruning keeping 4-8 bud canes for the fruiting crop. A dormancy-breaking chemical (hydrogen cyanamide) is applied after pruning to force uniform sprouting.

The crop uses more growth regulators than any other fruit: GA3 to elongate the bunch and increase berry size, cytokinins (CPPU) for size, and girdling to advance maturity. It is given about

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200-400 g N, 200-500 g P2O5 and 200-400 g K2O per vine per year in splits, with magnesium, zinc and boron, and irrigated by drip with water withheld before pruning. The main diseases are downy mildew, powdery mildew and anthracnose, and the main disorder is berry cracking from rain or irregular irrigation. Maturity is judged mainly by Brix (16-22 per cent) and berry colour, and the crop is timed to ripen in the dry season. Yield is 20-30 tonnes per hectare, and the crop is high-investment and export-oriented; value-added products are raisins, wine and juice.

Model answer 6 Importance of plantation crops and commodity boards

Plantation crops are perennial crops grown on a plantation scale and processed before sale; in India the group includes coconut, arecanut, oil palm, palmyrah, rubber, tea, coffee, cocoa, cashew and spices. They are grown mainly in the coastal and hill belts of the south and the north-east, provide year-round employment, support a large processing industry and earn foreign exchange. Coconut, arecanut and rubber in particular support millions of smallholders, and coconut and arecanut provide wide inter-row space that can carry intercrops and add to food security.

India is the third largest producer of coconut in the world (about 21,000-22,000 million nuts), the largest producer and consumer of arecanut (about 1.41 million tonnes) and cashew, and the sixth largest producer of natural rubber (about 857,000 tonnes in 2023-24). The scope for growth is large: oil palm is being expanded under the National Mission on Edible Oils to cut edible-oil imports, and value addition (virgin coconut oil, coconut water, rubber products, palm sugar) is a growing source of income.

A commodity board is a statutory or promotional body that supports production, research, marketing and export of a crop. The five statutory commodity boards under the Department of Commerce are the Tea Board (tea, Kolkata, with a zonal office at Coonoor), the Coffee Board (coffee, Bengaluru, with a research station at Thandigudi), the Rubber Board (natural rubber, Kottayam), the Spices Board (cardamom and 52 spices, Kochi) and the Tobacco Board (tobacco, Guntur). Coconut is supported by the Coconut Development Board, and cashew and cocoa by the Directorate of Cashewnut and Cocoa Development. The National Horticulture Board supports horticulture as a whole. These boards fund research, extension, quality control and market promotion, and so are central to the development of the plantation sector.

Model answer 7 Production technology of coconut

Coconut is called the tree of life because every part is useful. India is the third largest producer in the world, with about 2.1-2.3 million hectares and about 21,000-22,000 million nuts; Kerala, Karnataka, Tamil Nadu and Andhra Pradesh account for about 89 per cent of the area. The palm needs a warm, humid climate with a mean temperature of about 27 deg C and 1,000-3,000 mm rainfall, and grows on a wide range of well-drained soils with pH 5.2-8.0; it tolerates coastal salinity.

Coconut varieties are tall (East Coast Tall, West Coast Tall, Laccadive Ordinary), dwarf (Chowghat Orange Dwarf, Chowghat Green Dwarf, Malayan Dwarf) and hybrid (Kerasankara, Chandrasankara, Chandralaksha, and the TNAU hybrids VHC 1, VHC 2 and VHC 3). It is grown from seednuts taken from high-yielding mother palms and raised in a nursery for 9-12 months, and planted at 7.5 x 7.5 m in the square system (about 178 palms per hectare) or the triangular system (about 205 palms per hectare).

An adult palm is given about 500 g N, 320 g P2O5 and 1,200 g K2O per year in two splits, plus farmyard manure or green leaf manure and common salt in coastal areas; irrigation is by basin or drip. Because the palm is planted wide, intercropping and multi-tier cropping (coconut + arecanut or cocoa + pineapple or turmeric) greatly raise the return per hectare. Special practices are husk burial and coir-pith application. The main pests are the rhinoceros beetle, the red palm weevil, the black-headed caterpillar and the Eriophyid mite, and the main diseases are root (wilt), bud rot and stem bleeding. Nuts are harvested at 6-7 months for tender nuts and 11-12 months for copra; yield is 60-100 nuts per palm per year. Processing gives copra, coconut oil, coir, desiccated coconut and activated carbon, and value addition gives virgin coconut oil and coconut water.

Model answer 8 Production technology of tea

India is the second largest producer and the second largest exporter of tea in the world, producing about 1,382 million kg in 2023-24 from about 0.62 million hectares; Assam, West Bengal and Tamil Nadu (the Nilgiris) are the leading states. Tea needs a warm, humid climate with 1,500-3,000 mm rainfall, a temperature of 15-30 deg C, and grows best at 900-2,000 m elevation on a deep, well-drained, acidic soil with pH 4.5-5.5.

Tea varieties are the China, Assam and Cambod types and their hybrids; India grows mostly clonal material, and important clones are UPASI-9, UPASI-17, UPASI-3 and the Tocklai clones TRI-2025, TRI-2026 and TRI-2043. It is propagated by single-node cuttings raised in a shaded nursery for 9-12 months and planted at 1.2 x 0.75 m (about 11,000 plants per hectare) on the contour. The plant is trained by centering and pruned in a cycle of formative, light, medium and hard prunes at intervals of 2-5 years to renew the bearing surface. A mature field is given about 100-150 kg N, 30-40 kg P2O5 and 60-80 kg K2O per hectare per year in splits, with zinc and magnesium where needed.

Tea is harvested by plucking the tender shoots - two leaves and a bud - at intervals of 7-10 days; fine plucking gives high quality but a lower yield. The main pests are the tea mosquito bug, the red spider mite and the thrips, and the main diseases are blister blight, grey blight and red rust. Yield is about 1,700-2,500 kg of made tea per hectare per year. The green leaf is processed into black tea by withering, rolling, fermentation, drying and sorting; green tea is made by preventing fermentation; oolong tea is partly fermented. Value-added products are tea bags, flavoured and organic tea and instant tea.

Model answer 9 Production technology of cashew

Cashew is native to Brazil and is grown for its nut and its apple. India is the largest producer and consumer of cashew in the world, with about 1.2 million hectares and about 795,000 tonnes of raw cashew nut; Maharashtra, Andhra Pradesh, Odisha, Karnataka, Tamil Nadu and Kerala are the leading states, and in Tamil Nadu the coastal districts of Cuddalore, Villupuram, Ariyalur and Kanyakumari are important. The tree needs a warm, humid tropical climate with 1,000-2,000 mm rainfall and is very drought-tolerant, growing on poor, sandy and lateritic soils with pH 5.0-7.5; it does not tolerate waterlogging or frost.

Important varieties are VRI-1, VRI-2, VRI-3 and VRI-4 (released by TNAU from Vridhachalam), BPP-8, K-22-1, Bhaskara, Priyanka, Ullal-1, Kanaka, Dhana, Anagha and Madakkathara. Cashew is propagated by softwood grafting on a seedling rootstock, which gives true-to-type plants that bear in 3-4 years. It is planted at 7 x 7 m or 8 x 8 m (about 156-204 plants per hectare), or at 4 x 4

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m for high-density planting with pruning, in pits filled with farmyard manure. The tree is trained to a modified leader and pruned after harvest to keep the canopy open.

Cashew is largely rainfed but responds to irrigation at flowering and nut development; a moderate fertiliser dose of about 100-200 g N, 50-100 g P2O5 and 100-200 g K2O per tree is applied. The main pest is the tea mosquito bug, followed by the stem and root borer and the leaf miner, and the main diseases are die-back, anthracnose and powdery mildew. The nut is harvested when the apple turns red or yellow and the shell is hard, from February to May; yield is 1-2 tonnes per hectare. The raw nuts are sun-dried, roasted, shelled and the kernels dried and peeled; the by-products are cashew nut shell liquid (CNSL), used in brake linings and resins, and the cashew apple, used for juice and feni. India is the world's largest processor of cashew and a major exporter of kernels.

Model answer 10 High-density planting - concept, advantages and management

High-density planting (HDP) means planting more trees per hectare than the traditional spacing, to use land, light, labour and inputs more efficiently and to raise early and total yield. Ultra-high-density planting (UHDP) uses very close spacing with a small, controlled canopy. In mango, the traditional spacing of 10 x 10 m gives about 100 trees per hectare; high-density planting at 5 x 5 m gives 400 trees and ultra-high-density planting at 3 x 2 m gives about 1,667 trees. Similar principles apply to guava (6 x 6 m to 3 x 3 m) and apple (6 x 6 m to 3 x 2 m on dwarfing rootstocks).

The advantages are earlier bearing and return, higher yield per hectare, better use of land and inputs, easier harvest and spraying because the trees are small, and better fruit quality through improved light. The main requirements are a compact or dwarfing variety or rootstock, a training and pruning system that keeps the canopy small and open, growth regulators (paclobutrazol in mango) to control size and induce regular flowering, precise water and nutrient management by fertigation, and adequate plant protection.

The disadvantages are the higher initial cost of plants and establishment, the greater need for skilled pruning and management, the risk of overcrowding and poor light if the canopy is not controlled, and the need to remove or thin trees if they outgrow the spacing. A worked comparison shows that although HDP costs more up front, it recovers the extra cost from the 7th-8th year and gives a higher cumulative return over the life of the orchard, which makes it especially suited to smallholdings and to early-return varieties. In practice, HDP is now standard for guava, papaya, banana, pineapple, strawberry and apple on dwarfing rootstocks, and is increasingly used for mango and citrus.

APPENDIX M | Short Notes and One-line Facts

Short notes are a common form of examination question. The notes below cover the technical terms and practices most often asked, and can be used for quick revision.

Bahar regulation. The deliberate timing of flowering in a fruit crop by controlling pruning and irrigation, so that the crop is harvested in the most profitable season. It is widely used in acid lime, guava and pomegranate, and lets the grower target a market window or an export season.

High-density planting (HDP). Planting more trees per hectare than the traditional spacing to raise early and total yield. It needs a compact canopy, controlled by dwarfing rootstock, training, pruning and growth regulators, and precise water and nutrient management. Example: mango at 5 x 5 m (400 trees per hectare) instead of 10 x 10 m (100 trees).

Fertigation. The application of water-soluble fertilisers through the drip irrigation system. It places nutrients in the root zone in small, frequent doses, so nutrient-use efficiency rises and labour falls. The injection is followed by a clear-water flush to prevent clogging.

Multi-tier cropping. Growing crops of different stature in layers so that each uses a different part of the light and root zone. A coconut garden may have coconut in the top tier, arecanut or cocoa in the middle tier and pineapple or turmeric in the ground tier, which raises total output per hectare.

Intercropping. Growing a short-duration crop in the space between the rows of the main crop, usually in the young orchard, to give early income and suppress weeds. The intercrop must not shade or compete with the main crop or share its pests.

Mulching. Covering the soil surface with organic matter (straw, husk, coir pith) or polythene to conserve moisture, moderate soil temperature, suppress weeds and add organic matter. In strawberry it also keeps the fruit clean.

Training and pruning. Training builds the framework of the young tree by selecting and shaping its branches; pruning removes or shortens wood on the bearing tree to admit light, control size, remove dead and diseased wood and regulate fruiting.

Rootstock. The seedling or clonal plant on which the scion is grafted or budded. The rootstock controls tree vigour, size, precocity, fruit quality and tolerance of soil, drought, salinity and disease. Example: rough lemon gives vigour and drought tolerance in citrus.

Climacteric and non-climacteric fruit. Climacteric fruit (mango, banana, papaya, guava, sapota, apple) show a rise in respiration and ethylene at ripening and continue to ripen after harvest. Non-climacteric fruit (citrus, grape, pineapple, strawberry) do not ripen after harvest and must be picked fully ripe.

Chilling requirement. The number of hours below about 7 deg C that a temperate fruit tree needs to break dormancy and flower. It decides the elevation and region suitable for a crop; low-chill varieties extend apple, peach and plum to warmer areas.

Polyembryony. The occurrence of more than one embryo in a seed. In mango, polyembryonic types (such as Olour and Bappakai) produce seedlings that are true to type and are therefore used as rootstocks.

Fruit drop. The shedding of flowers and young fruit, most commonly the 'June drop' of mango, caused by water or nutrient stress, pests or disease. It is checked by NAA sprays, balanced nutrition and timely irrigation.

Alternate bearing. The habit in which a heavy crop year is followed by a light one, common in many mango varieties. It is managed by thinning, pruning and the use of paclobutrazol to induce regular flowering.

Smudging. Burning leaves and dung under a mango canopy to raise smoke and ethylene and induce off-season flowering. It is a traditional technique for staggering the crop.

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Girdling (ringing). Removal of a ring of bark from a branch or trunk to block the downward movement of food, which increases fruit set and size and advances maturity. It is used in grape and some other crops, but weakens the tree if overdone.

Paclobutrazol. A growth retardant applied to the soil or as a foliar spray that dwarfs the tree and induces early, regular flowering. It is central to high-density planting in mango.

Ethephon. A growth regulator that releases ethylene and is used to hasten ripening and colour development and to regulate flowering. Calcium carbide, formerly used for ripening, is now prohibited because it is harmful.

GA3 (gibberellic acid). A growth regulator used to increase fruit size and berry size, to elongate bunches in grape, to improve fruit set and to delay maturity.

NAA (naphthalene acetic acid). An auxin used as a foliar spray to check fruit drop and to improve fruit set and size. It is used in mango, citrus and other crops.

Commodity board. A statutory or promotional body that supports production, research, marketing and export of a crop. The five statutory boards under the Department of Commerce are the Tea, Coffee, Rubber, Spices and Tobacco Boards.

Good Agricultural Practices (GAP). A set of standards covering site, planting material, soil and water, plant protection, harvest, post-harvest handling, worker safety and the environment, designed to produce safe, quality food. GAP certification is increasingly required for export.

Cold chain. The unbroken series of refrigerated steps - pre-cooling, cold storage, refrigerated transport - that keeps perishable produce cool from the farm to the consumer. It is essential for fruit quality and for export.

Value addition. Processing a raw crop into a higher-value product, such as copra into virgin coconut oil, or mango into pulp, pickle and amchur. It reduces post-harvest loss and raises the grower's return.

Organic farming. Production without synthetic fertilisers and pesticides, relying on organic manures, biofertilisers, biopesticides and cultural methods. It improves soil health and often fetches a premium price.

Rejuvenation. The restoration of an old, senile orchard to productivity by heavy pruning, manuring, irrigation and the removal of dead and diseased wood. It is cheaper than replanting and gives quick results.

Root (wilt) disease. A serious disease of coconut in Kerala causing yellowing, flaccidity of the leaflets and decline of the palm. It is managed by removing affected palms, growing resistant material and balanced nutrition.

Koleroga. The fruit-rot disease of arecanut caused by Phytophthora, in which the nuts rot on the bunch. It is controlled by spraying Bordeaux mixture and by clean cultivation.

Brix. A measure of the total soluble solids, mainly sugars, in fruit juice. It is the standard maturity index for grape (16-22 per cent) and is also used for citrus and other crops.

Tenera. The high-yielding oil palm hybrid obtained by crossing the thick-shelled Dura with the shell-less Pisifera. It gives a high proportion of mesocarp and high oil yield, and is the commercial

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planting material.

Nungu. The tender fruit of palmyrah, eaten in summer. Palmyrah also gives palm sap (neera), from which palm sugar and palm wine are made.

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APPENDIX N | Consolidated List of Recommended Varieties

This table lists the recommended varieties of each crop for quick revision. TNAU varieties are named where they exist; for other crops the ICAR or commodity-board releases are given. Variety descriptions are brief and indicative.

Table N.1 Recommended varieties of the crops in this course

Crop	TNAU varieties	Other recommended varieties	Notes
Mango	CO 1, CO 2, CO 3, CO 4, CO 5; PKM 1, PKM 2	Banganapalli, Neelam, Alphonso, Totapuri, Rumani, Kesar, Dashehari, Langra	Banganapalli for quality; Neelam for regular bearing
Banana	CO 1, CO 2; PKM 1 to PKM 5	Grand Naine, Poovan, Nendran, Rasthali, Karpuravalli, Monthan, Red Banana	Grand Naine (tissue culture) for export
Papaya	CO 1 to CO 7; PKM 1, PKM 2	Ranchi, Pusa Dwarf, Pusa Nanha, Red Lady, Coorg Honey Dew	Gynodioecious/hybrid varieties preferred
Guava	CO 1, CO 2, CO 3; PKM series	Allahabad Safeda, Lucknow-49 (Sardar), Banarasi, Apple Colour, Arka Mridula	Lucknow-49 leading commercial variety
Sapota	CO 1, CO 2; PKM 1 to PKM 5	Kalippatti, Cricket Ball, Oval, Dwarapudi, Kirthabarthi	PKM 3 noted for yield
Acid lime	PKM 1, PKM 2	Kagzi lime, Pramalini, Vikram, Sai Sarbati, Balaji, Chakradhar	PKM 1 high-yielding for Tamil Nadu
Sweet orange	Selections	Sathgudi, Batavian (Mosambi), Blood Red, Jaffa, Pineapple, Hamlin	Sathgudi main southern variety
Grape	CO 1; PKM selections	Thompson Seedless, Sonaka, Tas-A-Ganesh, Sharad Seedless, Flame Seedless, Anab-e-Shahi, Bangalore Blue	Seedless types for export
Pineapple	TNAU selections	Kew, Queen, Mauritius, Jaldhup, Amritha	Kew for canning; Queen for fresh
Pomegranate	CO 1 and selections	Ganesh, Bhagwa, Mridula, Ruby, Arakta, Kesar, Phule Arakta	Bhagwa for export (soft seed)
Jackfruit	TNAU selections	Singapore Jack, Black Gold, Coorg Honey Dew, Muttom Varikka, Palur	Varikka (firm flesh) types preferred
Apple	Low-chill types trialled	Royal Delicious, Red Delicious, Golden Delicious, McIntosh, Ambri, Scarlet Spur; low-chill Anna, Dorsett Golden	Delicious group dominates
Pear	-	Bartlett, Flemish Beauty, Bosc, Anjou; sub-tropical Patharnakh, Gola, Le Conte, Punjab Beauty	Patharnakh and Gola for the plains
Peach	-	Flordasun, Flordaprince, Shan-e-Punjab, Elberta, Redhaven, Sharbati, Prabhat, Partap	Low-chill types for the sub-tropics
Plum	-	Santa Rosa, Satluj Purple, Kala Amritsari, Kelsey, Frontier, Methley	Santa Rosa leading variety
Strawberry	-	Chandler, Tioga, Selva, Sweet Charlie, Winter Dawn, Camarosa, Festival, Nabila	Winter Dawn low-chill for the plains
Ber	-	Umran, Gola, Seb, Kaithali, Katha, Mehrun, ZG-2, Goma Kirti, Banarasi Karaka, Illaichi	Umran leading commercial variety
Custard apple	-	Balanagar, Mammoth, Washington, Red Sitaphal, Purple, Arka Sahan	Balanagar and Mammoth leading
Amla	TNAU selections	NA-7, NA-10, Chakaiya, Kanchan, Krishna, BSR-1, Anand-1, Anand-2	NA-7 high-yielding

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Crop	TNAU varieties	Other recommended varieties	Notes
Dragon fruit	-	Hylocereus undatus (white), H. polyrhizus (red), Selenicereus megalanthus (yellow)	Red and yellow types fetch a premium
Almond	-	Nonpareil, Ne Plus Ultra, Mission, Drake, IXL, Waris, Merced, Parbat	Most varieties self-incompatible
Pistachio	-	Kerman (with Peters as polliniser), Chico	Rootstock P. integerrima
Walnut	-	Lake English, Opex Culch, Sulaiman, Gobind, Hamdan, JKSK; Payne, Franquette, Eureka, Hartley, Chandler, Howard	Franquette/Payne late-flowering
Coconut	VHC 1, VHC 2, VHC 3 (hybrids)	East Coast Tall, West Coast Tall, Laccadive Ordinary (tall); Chowghat Orange Dwarf, Chowghat Green Dwarf (dwarf); Kerasankara, Chandrasankara, Chandralaksha, Gangabondam (hybrid)	Hybrids for high yield and early bearing
Arecanut	TNAU material	South Kanara Local, Thirthahalli, Sirsi, Hirehalli Dwarf; CPCRI: Mangala, Sumangala, Sreemangala, Mohitnagar, Swarnamangala	Mangala tolerant of yellow leaf disease
Oil palm	TNAU hybrids	Tenera hybrid (Dura x Pisifera); ICAR-IIOPR material	Tenera is the commercial type
Palmyrah	Local types	Local selections (dioecious)	No named varieties
Rubber	-	RRII 105, RRII 414, RRII 430, RRIM 600, GT 1, PB 235, PB 260, PB 311	RRII 105 leading Indian clone
Tea	-	UPASI-9, UPASI-17, UPASI-3, UPASI-27; TRI-2025, TRI-2026, TRI-2043	UPASI clones for the south
Coffee	-	Arabica: S.795, Kents, Cauvery, Sln.5B, Selection 9, Chandragiri, Hemavathi; Robusta: S.274, CxR	Cauvery and Chandragiri rust-tolerant
Cocoa	-	Criollo, Forastero, Trinitario; CCRP hybrids 1-9	Forastero and hybrids grown in India
Cashew	VRI-1, VRI-2, VRI-3, VRI-4	BPP-8, K-22-1, Bhaskara, Priyanka, Ullal-1, Kanaka, Dhana, Anagha, Madakkathara	VRI series for Tamil Nadu

Source: TNAU, ICAR, commodity-board and state agricultural university material. Variety names are given for revision; always confirm current recommendations.

APPENDIX O | Model Question Papers

Two model question papers are given below, one for a full-semester theory examination and one for a mid-semester test. The pattern follows the usual university format of objective, short-answer and long-answer questions; the marks are indicative.

Model Question Paper 1 (Theory - 100 marks)

Part A - Answer all questions (20 x 1 = 20 marks)

- India's rank in world fruit production.
- Define a climacteric fruit with one example.
- Scientific name of banana.
- Plants per hectare at 5 x 5 m spacing.
- Nutrient content of urea.
- Name one dwarfing rootstock of apple.
- Full form of TSS.
- Name one TNAU mango variety.
- The two commercial species of coffee.
- Name the leading rubber clone of India.
- What is fertigation?
- The fruit-rot disease of arecanut.
- Name one soft-seeded pomegranate variety.
- The training system used for peach.
- What is bahar regulation?
- The leading fresh-fruit export of India.
- Name the board that supports coconut.
- One rich source of vitamin C among fruits.
- What is multi-tier cropping?
- The main pest of cashew.

Part B - Answer any five (5 x 6 = 30 marks)

- Write short notes on (a) high-density planting and (b) fertigation.
- Describe the systems of planting with diagrams.
- Explain the classification of fruit crops with examples.
- Describe the propagation and rootstock of citrus.
- Explain the production technology of banana.
- Write short notes on (a) commodity boards and (b) value addition.

Part C - Answer any two (2 x 25 = 50 marks)

- Describe the production technology of mango from planting to harvest, with a note on maturity indices and value addition.

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- Discuss the importance, scope and export potential of the plantation crop industry in India, and describe the production technology of coconut.
- Describe the production technology of tea and coffee, bringing out the differences in climate, propagation and processing.

Model Question Paper 2 (Mid-semester - 50 marks)

Part A - Answer all questions (10 x 1 = 10 marks)

- Define a rootstock.
- Plants per hectare at 7.5 x 7.5 m.
- Nutrient content of muriate of potash.
- Name one TNAU cashew variety.
- The aggregate fruit among custard apple and mango.
- Full form of GAP.
- Name one low-chill peach variety.
- What is desuckering?
- The leading tea district of Tamil Nadu.
- The chilling requirement is measured in what unit?

Part B - Answer any three (3 x 5 = 15 marks)

- Explain the training systems used in fruit crops with diagrams.
- Describe the maturity indices of mango and banana.
- Write short notes on (a) paclobutrazol and (b) GA3.
- Describe the propagation and planting of coconut.

Part C - Answer any one (1 x 25 = 25 marks)

- Describe the production technology of any one tropical fruit crop in detail.
- Discuss the production technology of cashew with emphasis on propagation, harvest and processing.

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APPENDIX P | Abbreviations, Units and Conversion Factors

A quick reference of the abbreviations and units used throughout this material, and the conversion factors commonly needed in horticultural calculations.

Table P.1 Common abbreviations

Abbreviation	Full form	Abbreviation	Full form
TNAU	Tamil Nadu Agricultural University	ICAR	Indian Council of Agricultural Research
NHB	National Horticulture Board	APEDA	Agricultural and Processed Food Products Export Development Authority
MIDH	Mission for Integrated Development of Horticulture	GAP	Good Agricultural Practices
HDP	High-density planting	UHDP	Ultra-high-density planting
TSS	Total soluble solids	SG	Specific gravity
NPK	Nitrogen, phosphorus, potassium	FYM	Farmyard manure
SSP	Single superphosphate	MOP	Muriate of potash
DAP	Di-ammonium phosphate	PGR	Plant growth regulator
GA3	Gibberellic acid	NAA	Naphthalene acetic acid
IBA	Indole butyric acid	CPPU	Forchlorfenuron (a cytokinin)
RSS	Ribbed smoked sheet (rubber)	CNSL	Cashew nut shell liquid
VHC	Veppankulam Hybrid Coconut	CPCRI	Central Plantation Crops Research Institute
IIHR	Indian Institute of Horticultural Research	CISH	Central Institute for Subtropical Horticulture
CCRI	Central Citrus Research Institute	DCCD	Directorate of Cashewnut and Cocoa Development
IIOPR	Indian Institute of Oil Palm Research	KAU	Kerala Agricultural University
UPASI	United Planters' Association of Southern India	TRI	Tocklai Tea Research Institute
RRII	Rubber Research Institute of India	HLB	Huanglongbing (citrus greening disease)
CA storage	Controlled-atmosphere storage	MAP	Modified-atmosphere packaging
MT	Million tonnes	ha	Hectare
ppm	Parts per million	Brix	A measure of total soluble solids

Abbreviations are expanded on first use in the text as well.

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Table P.2 Useful units and conversion factors

Item	Value / conversion
1 hectare	10,000 square metres; 2.471 acres
1 acre	0.405 hectare; about 4,047 square metres
1 tonne	1,000 kilograms
1 quintal	100 kilograms
1 lakh	100,000 (one hundred thousand)
1 crore	10 million (one hundred lakh)
Plants per hectare	10,000 / area per plant in square metres
Hexagonal plant number	Square number x 1.1547 (i.e. 1 / 0.866)
Urea	46 per cent N
Single superphosphate	16 per cent P2O5
Muriate of potash	60 per cent K2O
Di-ammonium phosphate	18 per cent N, 46 per cent P2O5
Nutrient to fertiliser	Nutrient / nutrient fraction in the fertiliser
1 mm of water over 1 hectare	10,000 litres (10 cubic metres)

Use these conversions when working out plant population, fertiliser doses and irrigation.